

电子电路与系统基础(A2)---动态电路

第8讲：阻抗匹配

李国林

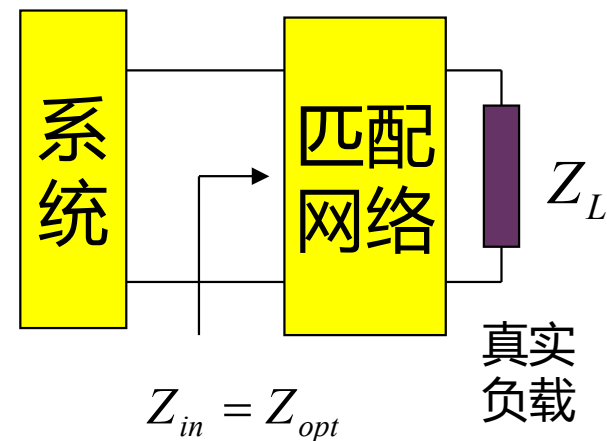
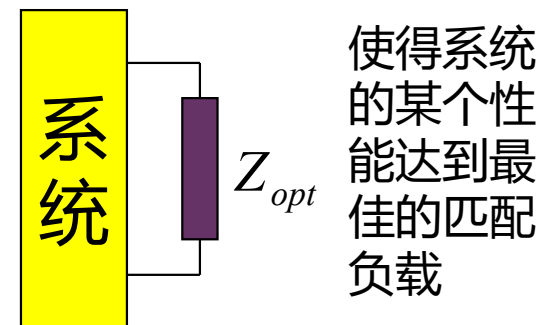
清华大学电子工程系

A班课程内容安排

第一学期：电阻	序号	第二学期：动态
电路定律	1	电容电感
电阻电源	2	一阶时域
电路定理	3	正弦稳态
方程列写	4	一阶时频
网络分析	5	二阶时域
二极管	6	二阶时频
晶体管	7	有源滤波
MOSFET	8	阻抗匹配
BJT	9	寄生效应
反相电路	10	频率特性
数字门	11	负反馈
小信号放大	12	正反馈
三种组态	13	振荡器
大信号放大	14	期末复习
期末复习	15	电路抽象

阻抗匹配是为了获得某种性能最佳

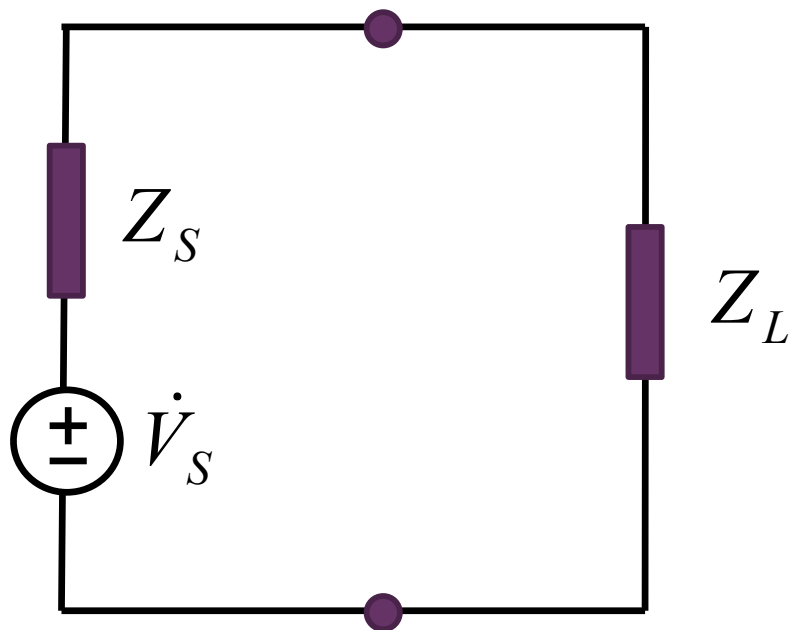
- 为了使得系统的某个性能达到最优，希望负载阻抗为某个特定的和系统参数有关的一个最佳值 Z_{opt}
 - 最佳性能：最大功率传输，最小噪声系数，最大功率输出，某种定义下的滤波传输特性，...
- 但实际的负载 Z_L 往往和最佳阻抗相差甚远
- 于是我们希望有一个匹配网络，将负载阻抗 Z_L 变换为 Z_{opt}
 - 匹配网络为一个二端口阻抗变换器，一端接了负载 Z_L 后从另一端看进去的阻抗 Z_{in} 应该等于 Z_{opt}
- 在系统和负载之间加上一个匹配网络（阻抗变换网络），可以使系统的某个性能达到最佳
 - 匹配网络也称为阻抗变换网络



以最大功率传输为例的 阻抗匹配 内容

- 阻抗匹配网络，多是利用双向无损网络实现的阻抗变换网络
 - 双向无损网络
 - 阻性：理想变压器、理想回旋器、理想环行器、…
 - 高度抽象的理想元件
 - 动态：并臂/串臂电容/电感，二端口电感（互感变压器）/二端口电容，传输线，…
 - 实际采用的网络
- 利用谐振实现阻抗变换和匹配
 - L型LC匹配网络：并大串小Q相等
 - 互感变压器：阻抗变换和带通滤波特性

一、最大功率传输匹配



电阻电路的最大功率传输匹配

$$R_L = R_S$$

$$P_L = P_{L,\max} = \frac{V_{S,rms}^2}{4R_S} = P_{S,\max}$$

$$Z_S = R_S + jX_S \quad Z_L = R_L + jX_L$$

$$\dot{I} = \frac{\dot{V}_S}{Z_S + Z_L} = \frac{\dot{V}_S}{(R_S + R_L) + j(X_S + X_L)}$$

$$= I_p e^{j\varphi_I} = \sqrt{2} I_{rms} e^{j\varphi_I}$$

$$P_L = I_{rms}^2 R_L = \frac{V_{S,rms}^2}{(R_S + R_L)^2 + (X_S + X_L)^2} R_L$$

共轭匹配

$$P_L = \frac{R_L}{(R_S + R_L)^2 + (X_S + X_L)^2} V_{S,rms}^2$$

$$X_L = -X_S$$

$$P_L = \frac{R_L}{(R_S + R_L)^2} V_{S,rms}^2$$

$$R_L = R_S$$

共轭匹配

$$Z_L = Z_S^*$$

$$P_L = \frac{R_L}{(R_S + R_L)^2} V_{S,rms}^2 \leq \frac{V_{S,rms}^2}{4R_S} = P_{Smax}$$

共轭匹配内蕴谐振

- 电感的电抗是正的，电容的电抗是负的
 - 电感的电纳是负的，电容的电纳是正的
- 要想抵偿电感电抗，则需电容
- 要想抵偿电容电抗，则需电感

$$R_L = R_S \quad \text{相等：匹配}$$

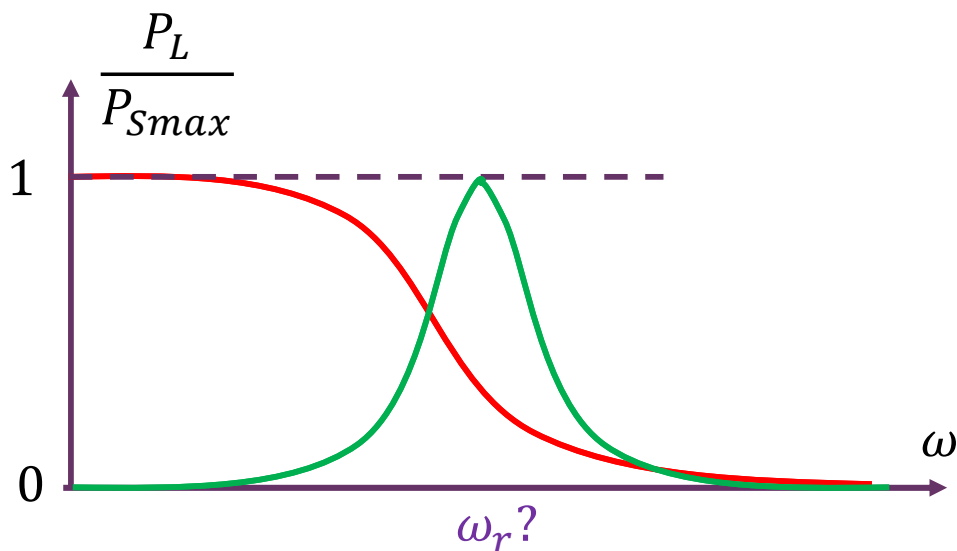
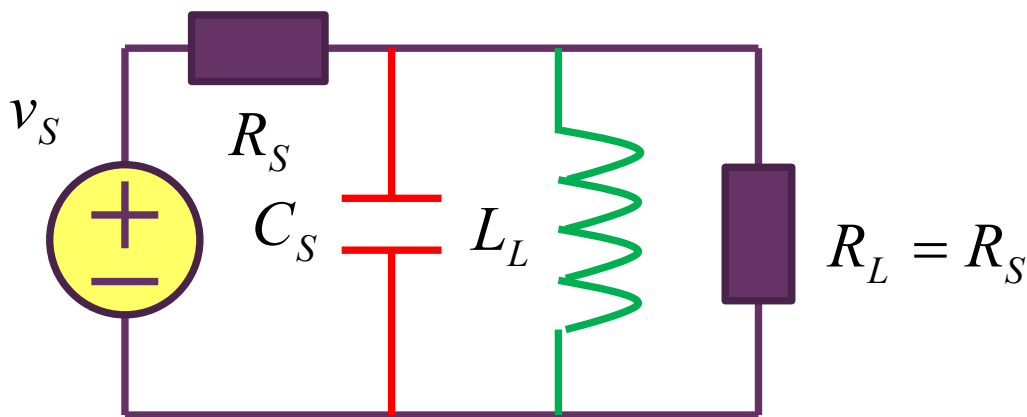
$$\text{共轭匹配} \quad Z_L = Z_S^*$$

$$X_L + X_S = 0 \quad \text{谐振：回路中的电抗正负抵消为0}$$

$$P_L = P_{L,\max} = \frac{V_{S,rms}^2}{4R_S} = P_{S,\max}$$

例1 用谐振实现特定频点的匹配

- 已知某信源输出有寄生电容，如何使得负载获得最大功率传输匹配？



$$Z_L = Z_S^*$$

$$Y_L = Y_S^*$$

共轭匹配

$$G_L = G_S$$

$$B_L + B_S = 0$$

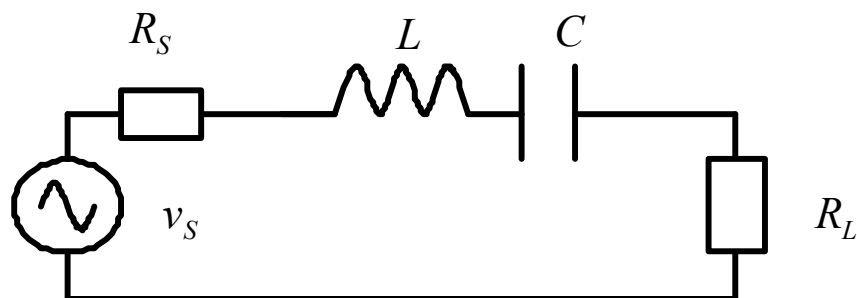
$$-\frac{1}{\omega_r L_L} + \omega_r C_S = 0$$

$$L_L = \frac{1}{\omega_r^2 C_S}$$

用感性负载抵偿容性负载，可在特定频点可实现共轭匹配：带通匹配特性

谐振匹配就存在匹配频点问题

LC带通传输网络不具阻抗匹配作用



$$|H|^2 = 4 \frac{R_S}{R_L} \frac{V_{L,rms}^2}{V_{S,rms}^2} = \frac{\frac{V_{L,rms}^2}{R_L}}{\frac{V_{S,rms}^2}{4R_S}} = \frac{P_L}{P_{S,max}}$$

定义：基于功率传输的电压传递函数

$$H(s) = 2 \sqrt{\frac{R_S}{R_L}} \frac{V_L(s)}{V_S(s)} = 2 \sqrt{\frac{R_S}{R_L}} \frac{R_L}{R_S + sL + \frac{1}{sC} + R_L} = H_0 \frac{2\xi\omega_0 s}{s^2 + 2\xi\omega_0 s + \omega_0^2}$$

$$H_0 = \frac{2\sqrt{R_S R_L}}{R_S + R_L} \quad \omega_0 = \frac{1}{\sqrt{LC}} \quad \xi = \frac{R}{2Z_0} = \frac{R_S + R_L}{2\sqrt{\frac{L}{C}}}$$

$$H_0 \leq 1 \quad H_0 = 1 (R_L = R_S) \quad \mathbf{H_0^2} \text{谐振频点的功率增益}$$

如果 $R_L \neq R_S$ ，如何让负载获得最大功率？

■ LC带通网络

- 传递函数最大值位于谐振频点，该点功率传递系数 $\frac{4R_LR_S}{(R_L+R_S)^2}$ 只有在 $R_L = R_S$ 时获得最大值100%
 - 串联LC和并联LC谐振腔本身不具阻抗变换作用，仅提供带通选频特性

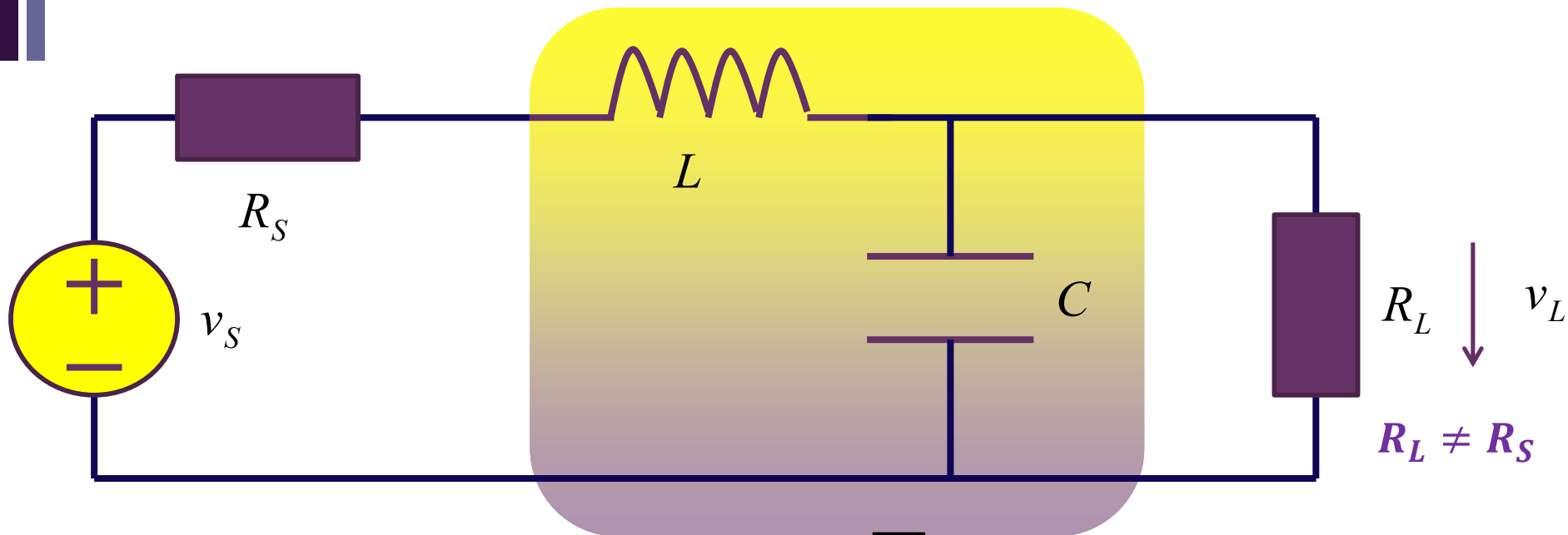
■ LC低通网络

- 中心频点0频点上串臂电感短路，并臂电容开路，源直连负载，中心频点零频点的功率传递系数 $\frac{4R_LR_S}{(R_L+R_S)^2}$ 只有在 $R_L = R_S$ 时获得最大值100%
- 但是，如果系统阻尼系数 $\xi < 0.707$ ，传递函数存在谐振峰频点，是否存在在谐振峰频点上达到功率传输100%的可能性？即使 $R_L \neq R_S$

■ LC高通网络

- 中心频点 ∞ 频点上串臂电容短路，并臂电感开路，源直连负载，中心频点无穷频点的功率传递系数 $\frac{4R_LR_S}{(R_L+R_S)^2}$ 只有在 $R_L = R_S$ 时获得最大值100%
- 但是，如果系统阻尼系数 $\xi < 0.707$ ，传递函数存在谐振峰频点，是否存在在谐振峰频点上达到功率传输100%的可能性？即使 $R_L \neq R_S$

直观思路：谐振峰频点上强制匹配



- 1、求基于功率的传递函数
$$H(s) = 2 \sqrt{\frac{R_s V_L(s)}{R_L V_S(s)}} = \dots = H_0 \frac{\omega_0^2}{s^2 + 2\xi\omega_0 s + \omega_0^2}$$

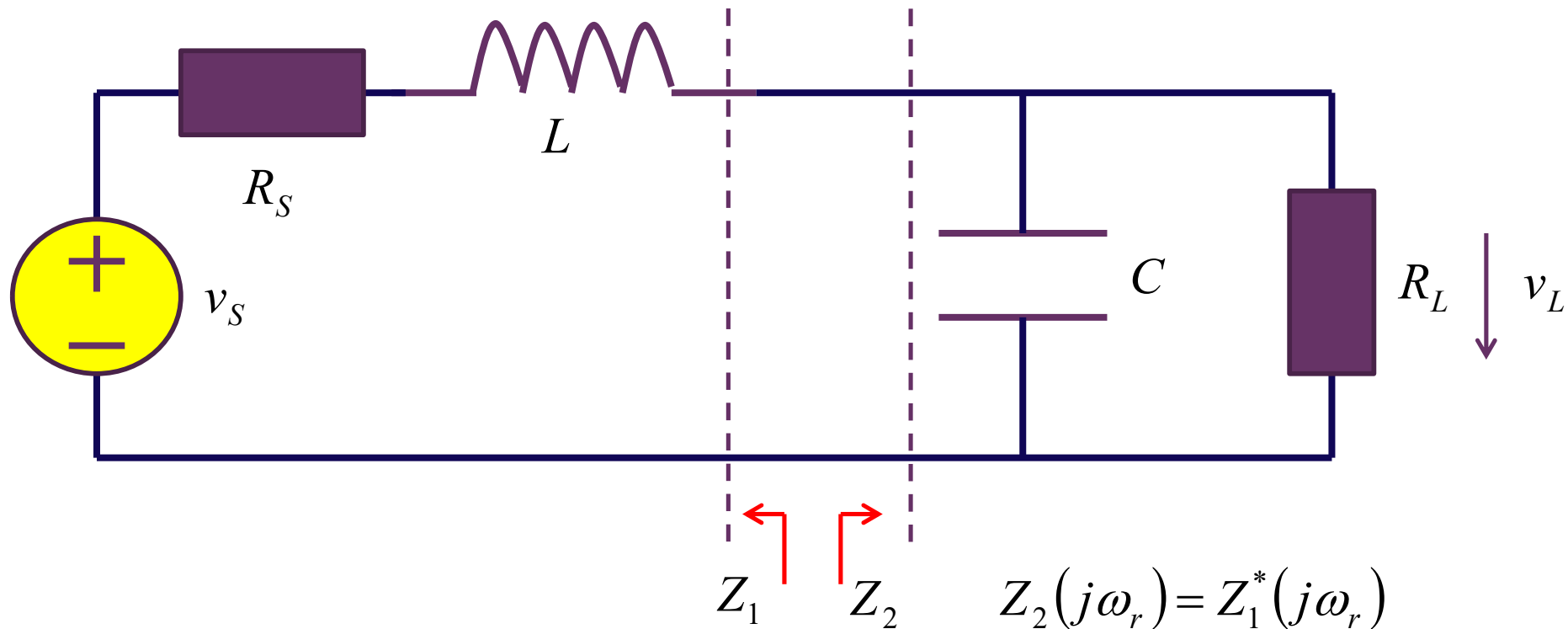
确定系统阻尼系数 ξ 、自由振荡频率 ω_0 、中心频点传递系数 H_0 和电路参量 R_s 、 R_L (已知)、 L 、 C (未知)之间的关系

- 2、令谐振峰频点 $\omega_e = \sqrt{1 - 2\xi^2}\omega_0$ 为匹配频点 ω_r (已知)，令谐振峰值为1

$$A(\omega_e) = H_0 \frac{1}{2\xi\sqrt{1 - \xi^2}} = 1 \quad \text{由这两个方程求两个未知量 } L、C，$$

最终获得低通**LC**匹配网络，由于谐振峰频点上功率增益为1，故而在该频点上负载获得的是信源的额定功率：**LC**本身无损，结论无异议。

实现最大功率传输匹配的物理要求 必须共轭匹配：基本思路



$$Z_1(j\omega_r) = R_S + j\omega_r L \quad Z_2(j\omega_r) = \frac{R_L}{1 + j\omega_r R_L C} = \frac{R_L}{1 + (\omega_r R_L C)^2} - \frac{j\omega_r R_L^2 C}{1 + (\omega_r R_L C)^2}$$

共轭匹配

$$Z_1(j\omega_r) = R_S + j\omega_r L \quad Z_2(j\omega_r) = \frac{R_L}{1 + j\omega_r R_L C} = \frac{R_L}{1 + (\omega_r R_L C)^2} - \frac{j\omega_r R_L^2 C}{1 + (\omega_r R_L C)^2}$$

$$Z_2(j\omega_r) = Z_1^*(j\omega_r) \quad \frac{R_L}{1 + (\omega_r R_L C)^2} - \frac{j\omega_r R_L^2 C}{1 + (\omega_r R_L C)^2} = R_S - j\omega_r L$$

$$\frac{R_L}{1 + (\omega_r R_L C)^2} = R_S$$

$$\frac{R_L^2 C}{1 + (\omega_r R_L C)^2} = L$$



$$C = \frac{1}{\omega_r R_L} \sqrt{\frac{R_L}{R_S} - 1}$$

$$L = \frac{R_S}{\omega_r} \sqrt{\frac{R_L}{R_S} - 1}$$

只要**L**和**C**如是取值，
即可在 ω_r 频点获得
最大功率传输匹配

只解决了 **$R_L > R_S$** 情况，
如果 **$R_L < R_S$** 呢？

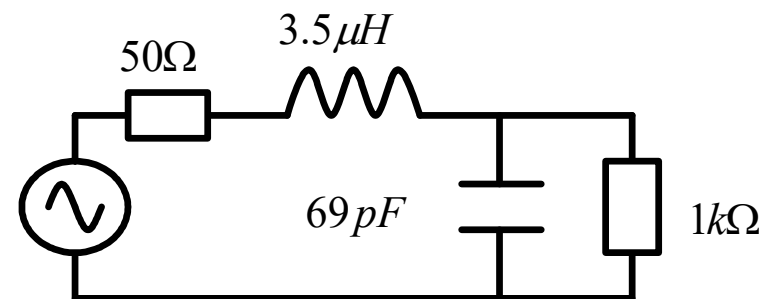
例 用LC网络实现特定频点的匹配传输

- 已知 $R_S=50\Omega$, $R_L=1k\Omega$, 请设计一个LC低通匹配网络, 在 $f_r=10\text{MHz}$ 频点上实现最大功率传输匹配

L型低通网络

$$L_1 = \frac{R_S}{\omega_r} \sqrt{\frac{R_L}{R_S} - 1} = \frac{50}{2\pi \times 10M} \sqrt{\frac{1k}{50} - 1} = 3.469\mu H$$

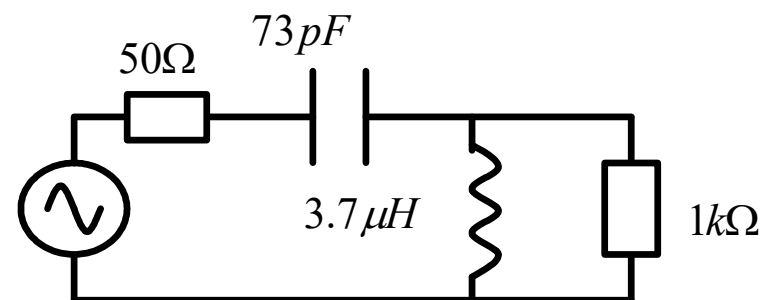
$$C_2 = \frac{1}{\omega_r R_L} \sqrt{\frac{R_L}{R_S} - 1} = \frac{1}{2\pi \times 10M \times 1k} \sqrt{\frac{1k}{50} - 1} = 69.37\text{ pF}$$



L型高通网络：用相同的思路，自行推导获得此公式

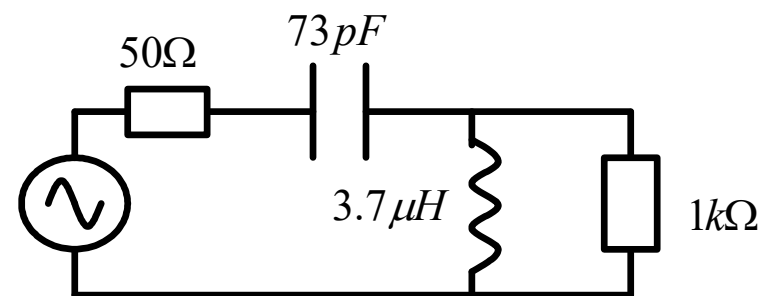
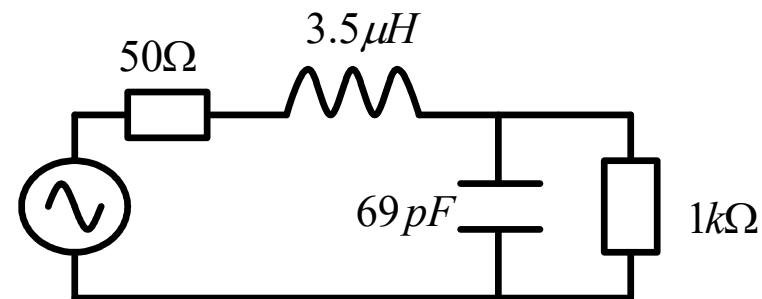
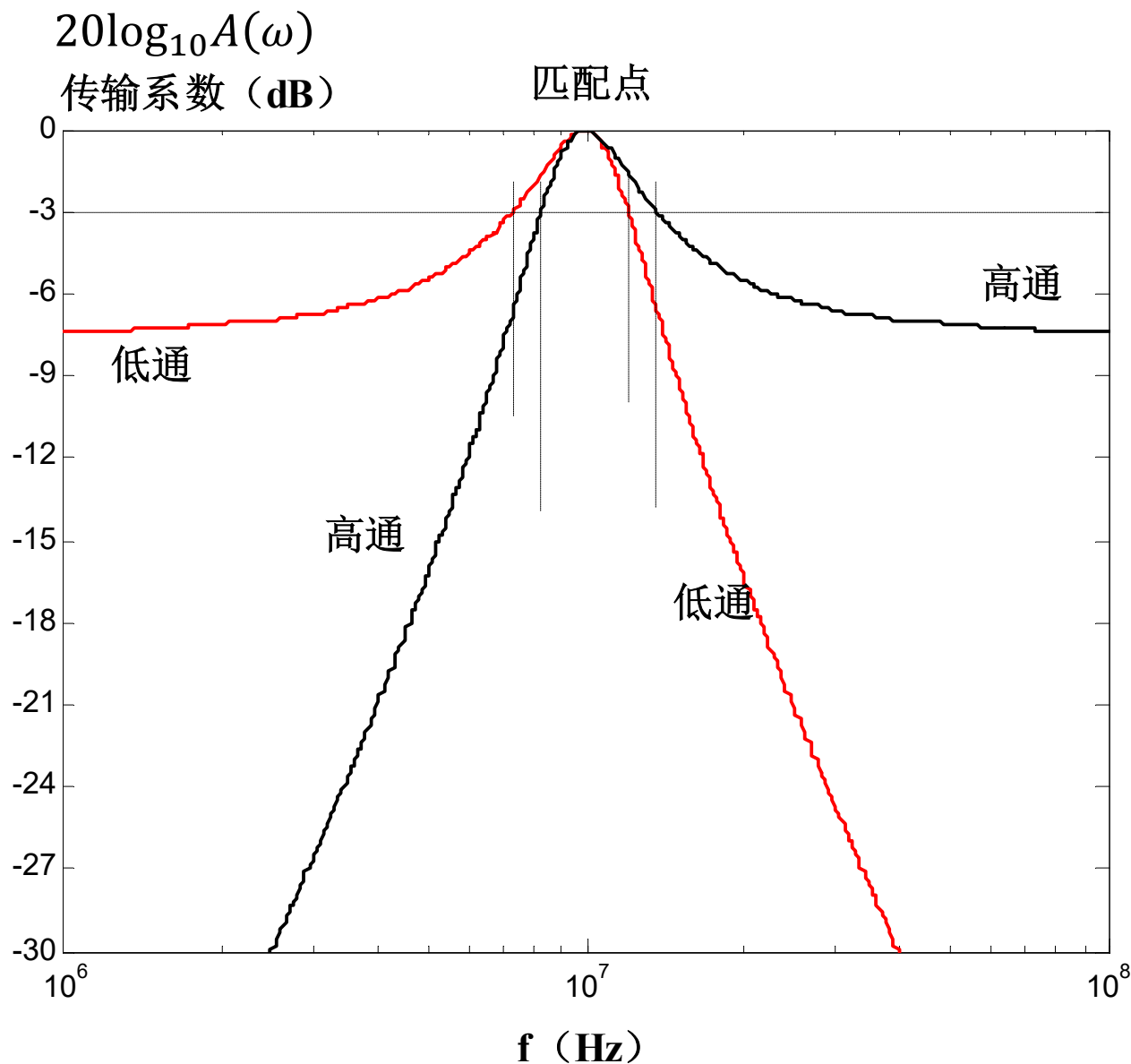
$$C_1 = \frac{1}{\omega_r R_S} \sqrt{\frac{R_L}{R_S} - 1} = \frac{1}{2\pi \times 10M \times 50} \sqrt{\frac{1k}{50} - 1} = 73.03\text{ pF}$$

$$L_2 = \frac{R_L}{\omega_r} \sqrt{\frac{R_L}{R_S} - 1} = \frac{1k}{2\pi \times 10M} \sqrt{\frac{1k}{50} - 1} = 3.651\mu H$$

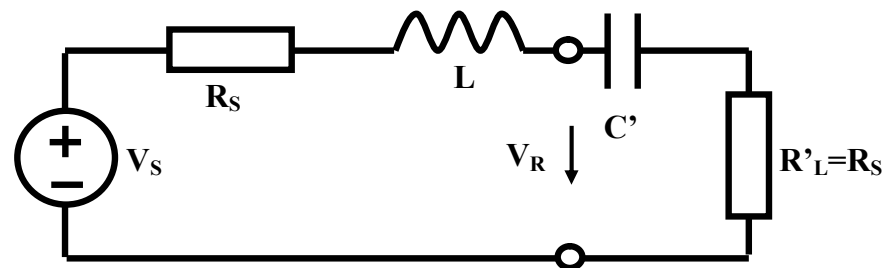
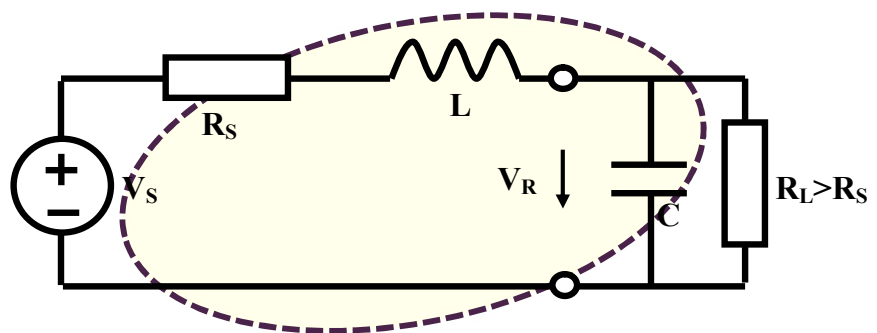


最大功率传输匹配

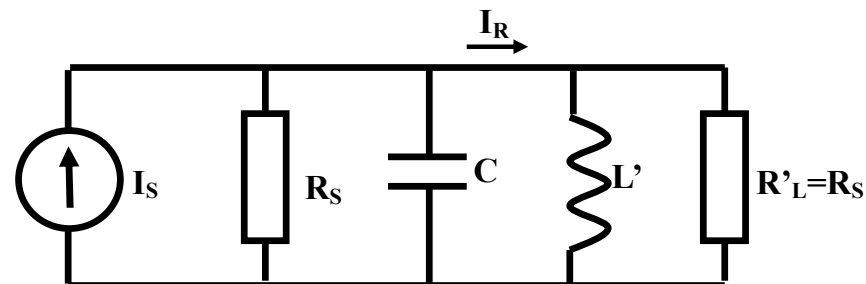
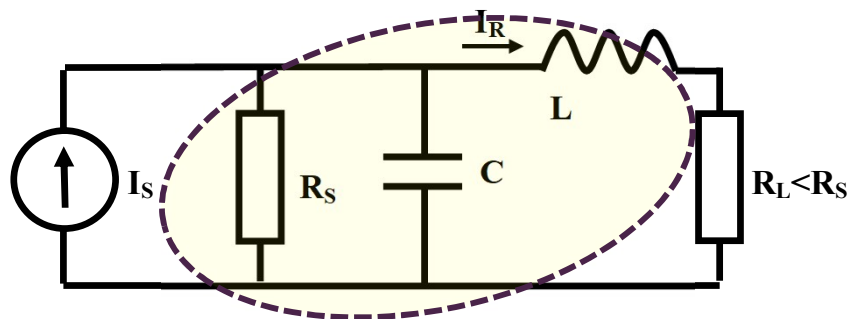
$$H(j\omega) = 2 \sqrt{\frac{R_S}{R_L}} \frac{V_L(j\omega)}{V_S(j\omega)} = A(\omega)e^{j\varphi(\omega)}$$



从谐振角度进行理解 and 设计

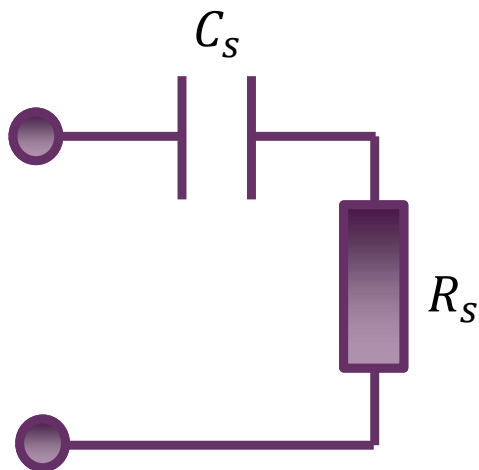


串联谐振是电压谐振



并联谐振是电流谐振

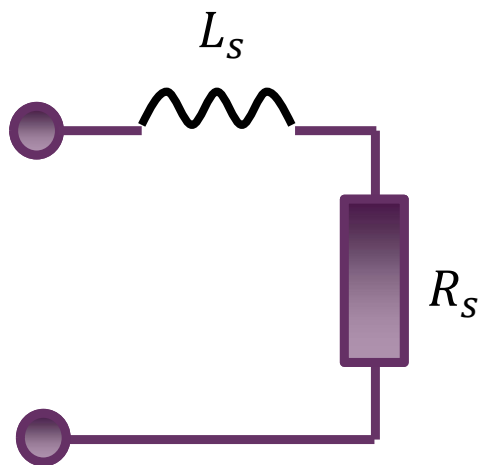
设计口诀：并大串小Q相等



$$Q = \frac{\text{串联电抗}}{\text{串联电阻}} = \frac{1}{\omega C_s} = \frac{1}{\omega R_s C_s}$$

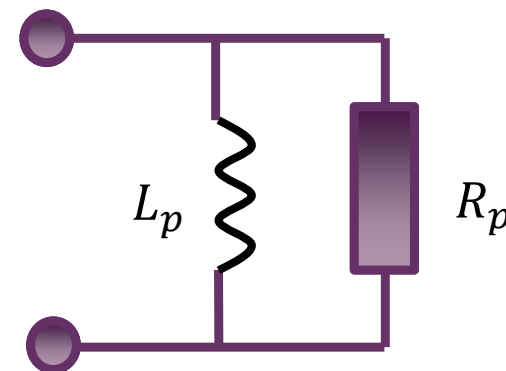
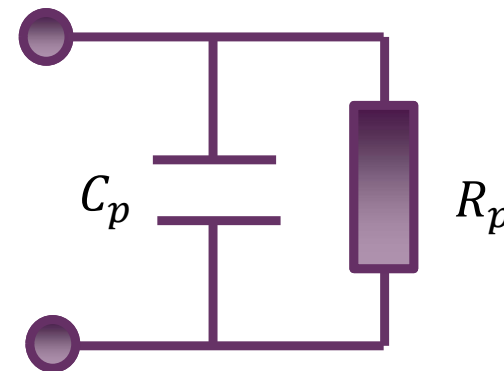
$$Q = \frac{\text{并联电纳}}{\text{并联电导}} = \frac{\omega C_p}{G_p} = \omega R_p C_p$$

$$Q = \sqrt{\frac{R_p}{R_s} - 1}$$

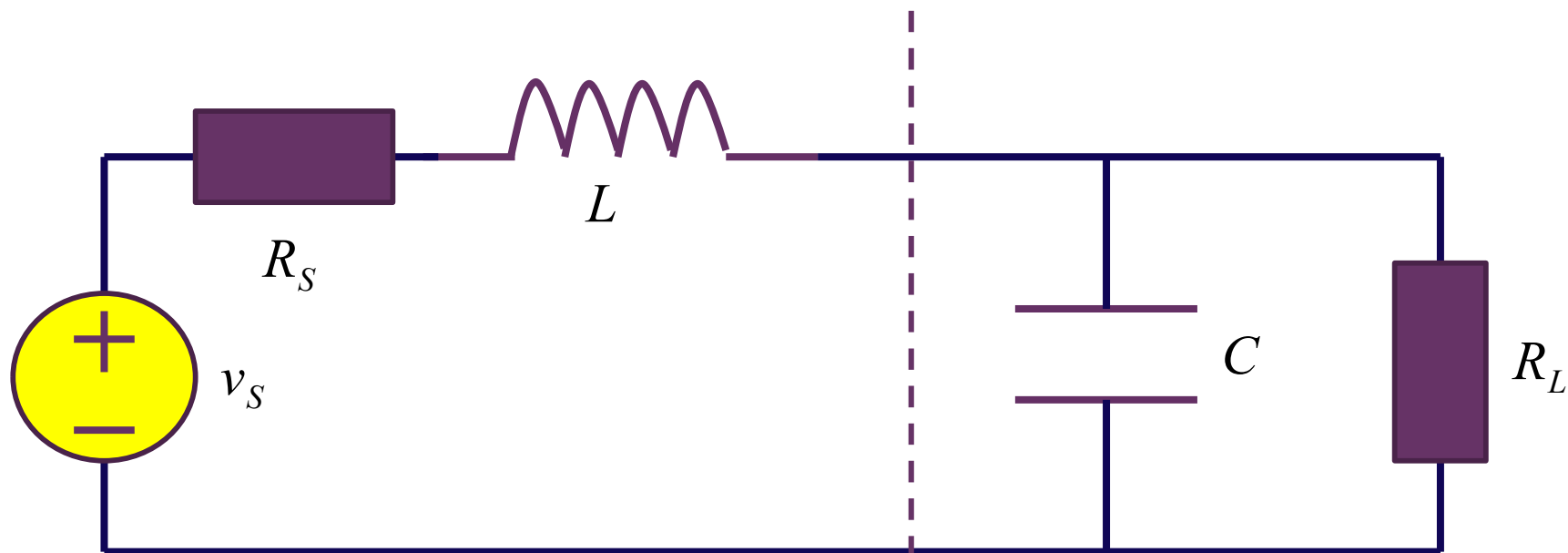


$$Q = \frac{\text{串联电抗}}{\text{串联电阻}} = \frac{\omega L_s}{R_s}$$

$$Q = \frac{\text{并联电纳}}{\text{并联电导}} = \frac{1}{\omega L_p} = \frac{R_p}{\omega L_p}$$



用设计口诀进行匹配网络设计



$$Q = \frac{\omega_r L}{R_S} \quad Q = \sqrt{\frac{R_L}{R_S} - 1}$$



$$L = \frac{R_S}{\omega_r} Q = \frac{R_S}{\omega_r} \sqrt{\frac{R_L}{R_S} - 1}$$



$$Q = \omega_r R_L C$$

$$C = \frac{1}{\omega_r R_L} Q = \frac{1}{\omega_r R_L} \sqrt{\frac{R_L}{R_S} - 1}$$

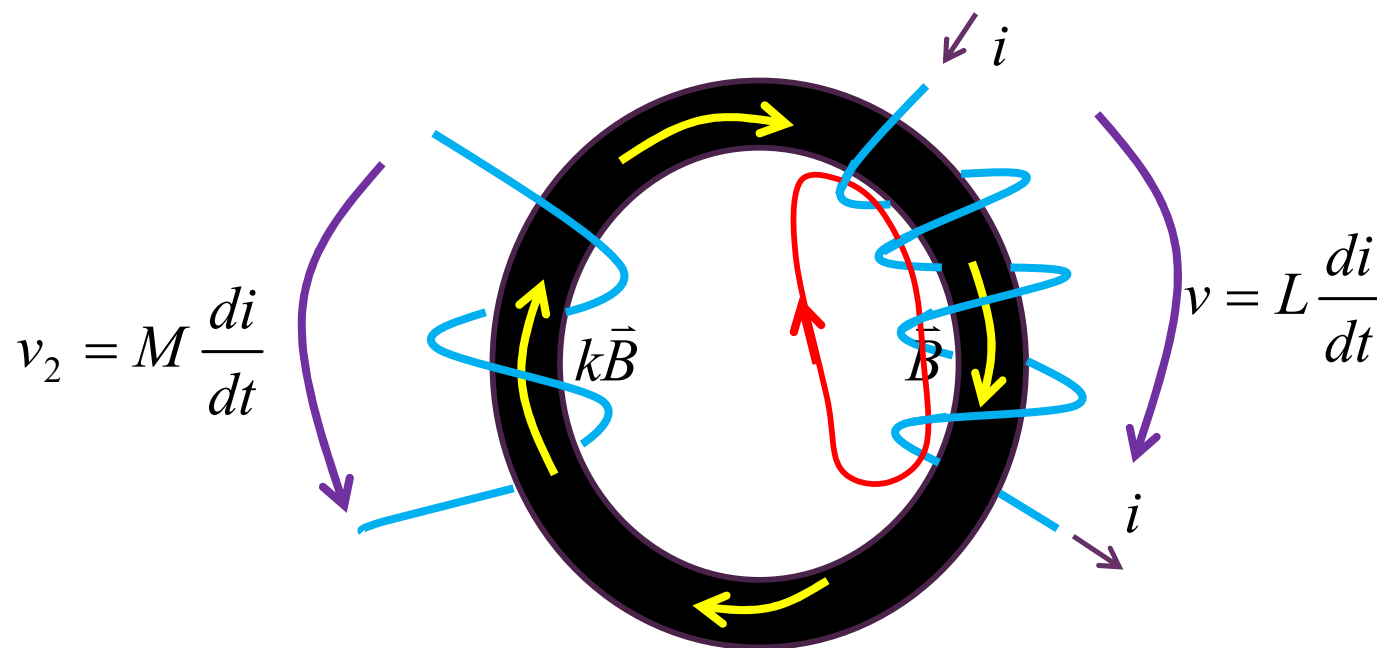
二、互感变压器 二端口电感网络

μ : 磁导率

S : 环横截面面积

N : 总匝数

p : 环周长



$$B = \mu \frac{N}{p} i$$

$$\Phi_0 = B \cdot S = \mu \frac{N}{p} i \cdot S$$

$$\Xi = \mu \frac{S}{p}$$

Ξ : 磁环磁导

$$M = kN_2(N\Xi)$$

$$\Phi_0 = N\Xi i$$

N_2 : 第二线圈总匝数

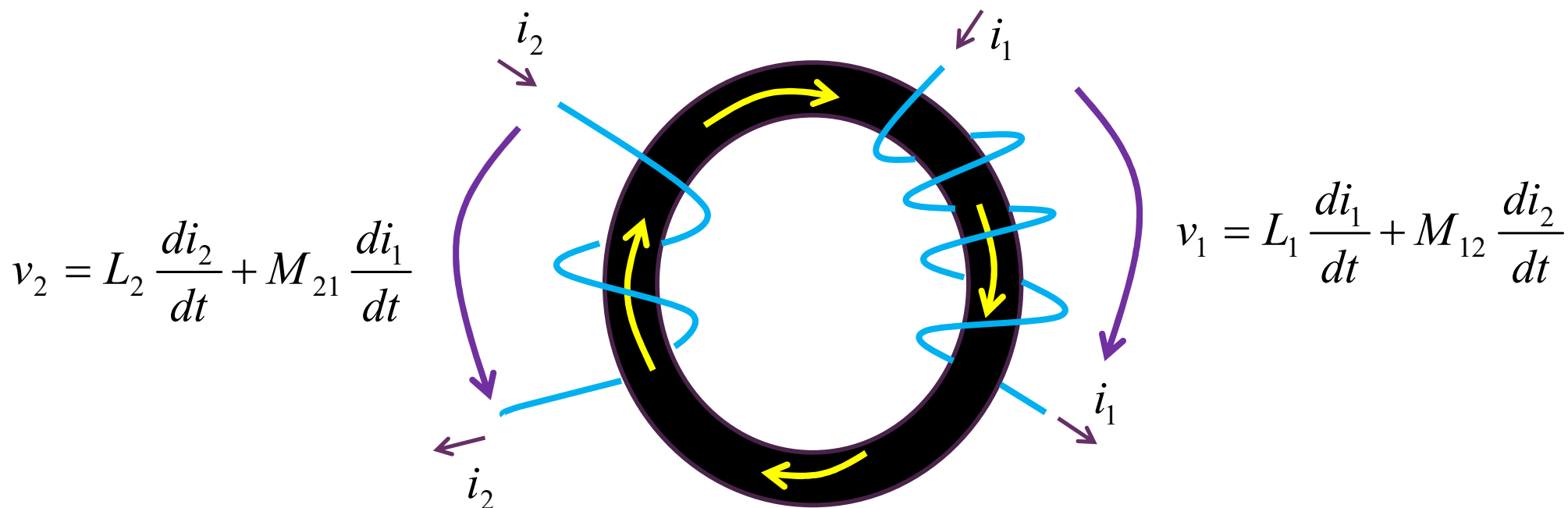
k : 第二线圈链接的第一线圈产生磁通的百分比

互感: Mutual Inductance 自感: Self Inductance

$$\Phi = N\Phi_0 = N^2\Xi i$$

$$L = \frac{\Phi}{i} = N^2\Xi = N(N\Xi)$$

二端口网络描述



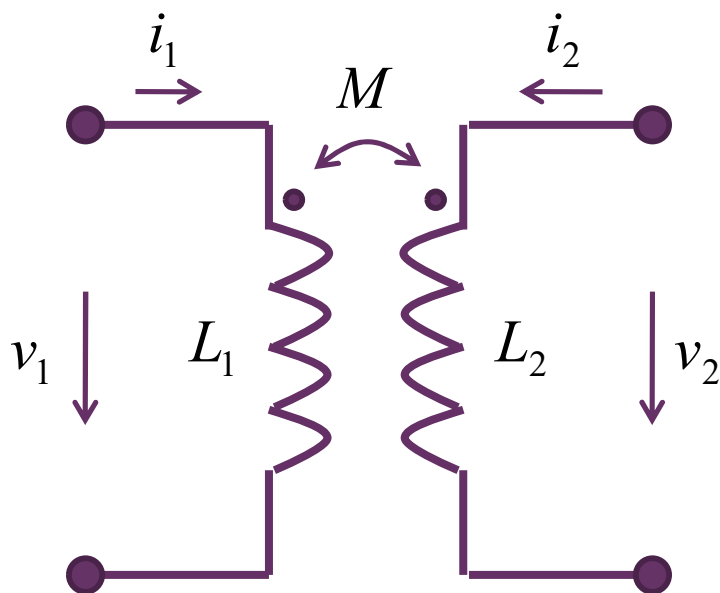
$$L_1 = N_1^2 \Xi$$

$$M_{12} = M_{21} = M = k N_1 N_2 \Xi$$

$$L_2 = N_2^2 \Xi$$

互易网络

端口约束方程：z参量描述



$$L_1 = N_1^2 \Xi \quad L_2 = N_2^2 \Xi$$

$$M = k N_1 N_2 \Xi = k \sqrt{L_1 L_2} = k M_0$$

$$k = \frac{M}{\sqrt{L_1 L_2}} = \frac{M}{M_0}$$

耦合系数：磁通链接百分比

$$0 \leq k \leq 1$$

$$\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} L_1 & M \\ M & L_2 \end{bmatrix} \frac{d}{dt} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix}$$

$$\begin{bmatrix} \dot{V}_1 \\ \dot{V}_2 \end{bmatrix} = \begin{bmatrix} j\omega L_1 & j\omega M \\ j\omega M & j\omega L_2 \end{bmatrix} \begin{bmatrix} \dot{I}_1 \\ \dot{I}_2 \end{bmatrix}$$

$$\dot{\mathbf{V}} = \mathbf{Z} \dot{\mathbf{I}}$$

二端口电感

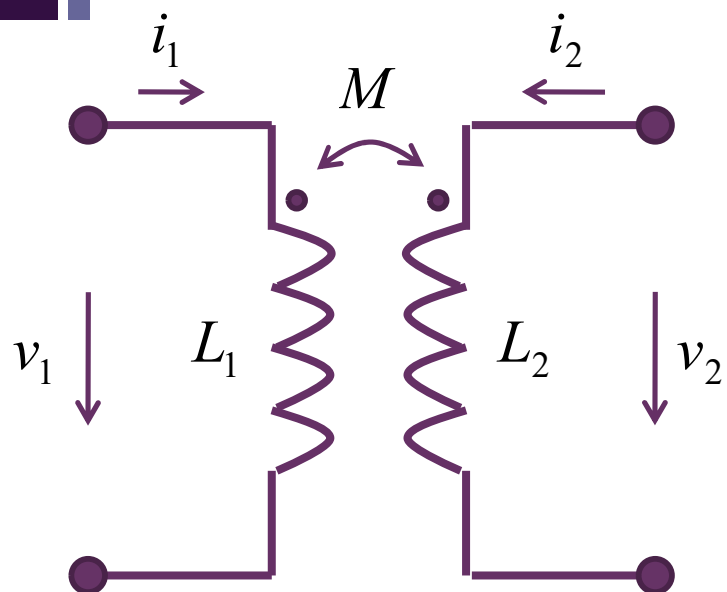
$$v = L \frac{di}{dt}$$

单端口电感

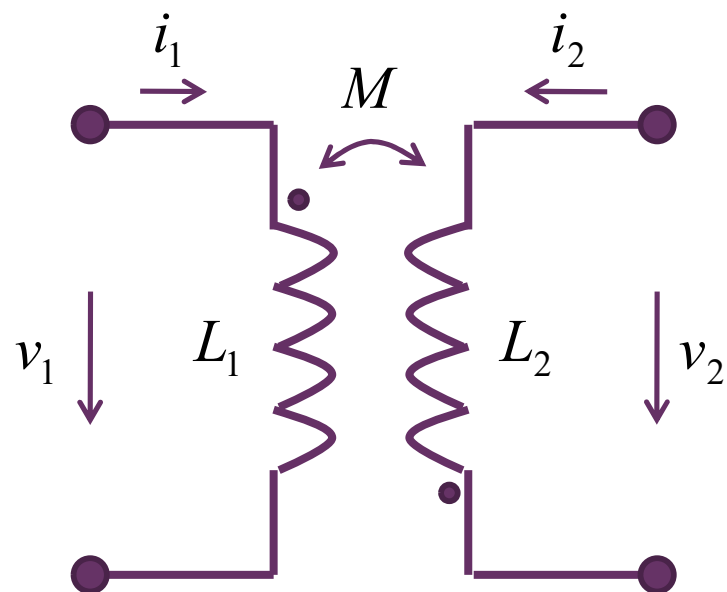
$$\dot{V} = j\omega L \dot{I}$$

$$\dot{V} = Z \dot{I}$$

同名端



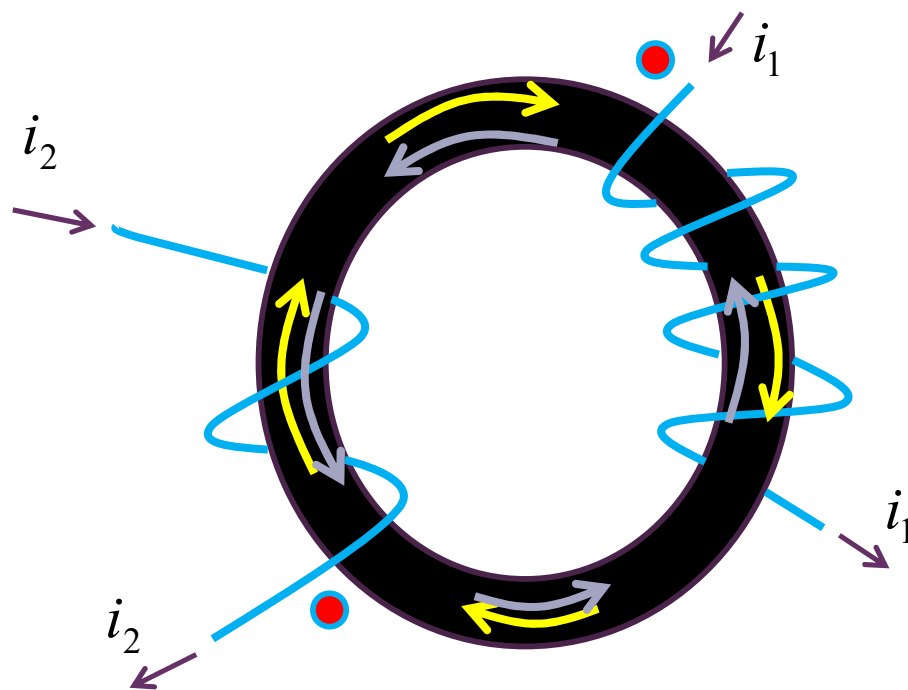
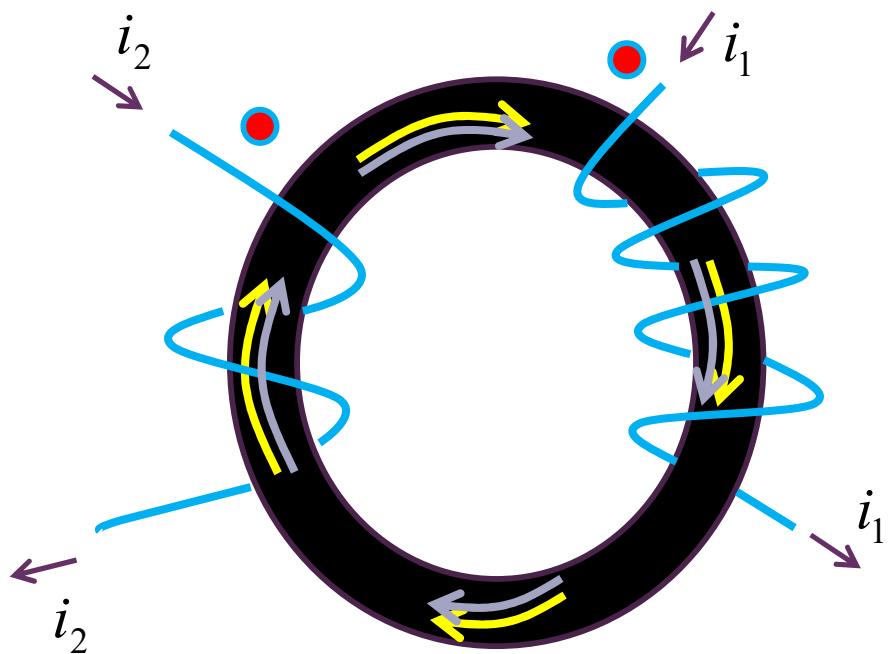
$$\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} L_1 & M \\ M & L_2 \end{bmatrix} \frac{d}{dt} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix}$$



$$\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} L_1 & -M \\ -M & L_2 \end{bmatrix} \frac{d}{dt} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix}$$

$$\begin{bmatrix} v_1 \\ -v_2 \end{bmatrix} = \begin{bmatrix} L_1 & M \\ M & L_2 \end{bmatrix} \frac{d}{dt} \begin{bmatrix} i_1 \\ -i_2 \end{bmatrix}$$

同名端判定



流入电流使得磁通加强的两个端点是同名端

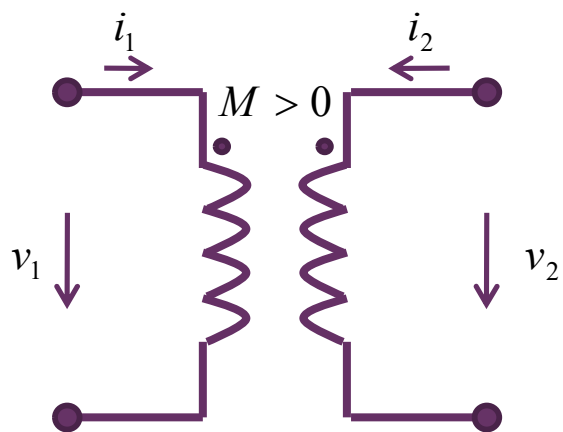
$$\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} L_1 & M \\ M & L_2 \end{bmatrix} \frac{d}{dt} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix}$$

磁通加强，感生电动势加强

$$\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} L_1 & -M \\ -M & L_2 \end{bmatrix} \frac{d}{dt} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix}$$

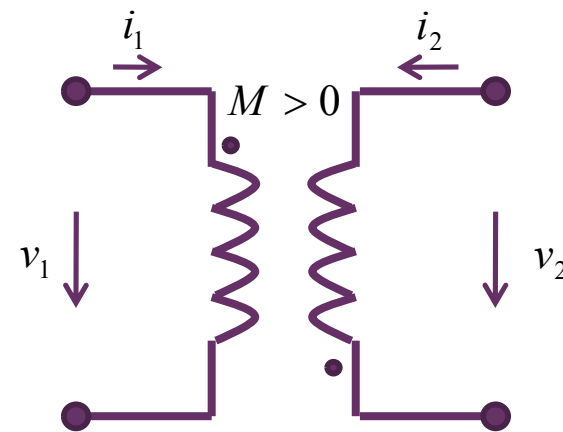
磁通相抵，感生电动势相减

描述方程统一



$$\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} L_1 & M \\ M & L_2 \end{bmatrix} \frac{d}{dt} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix}$$

$$M > 0$$



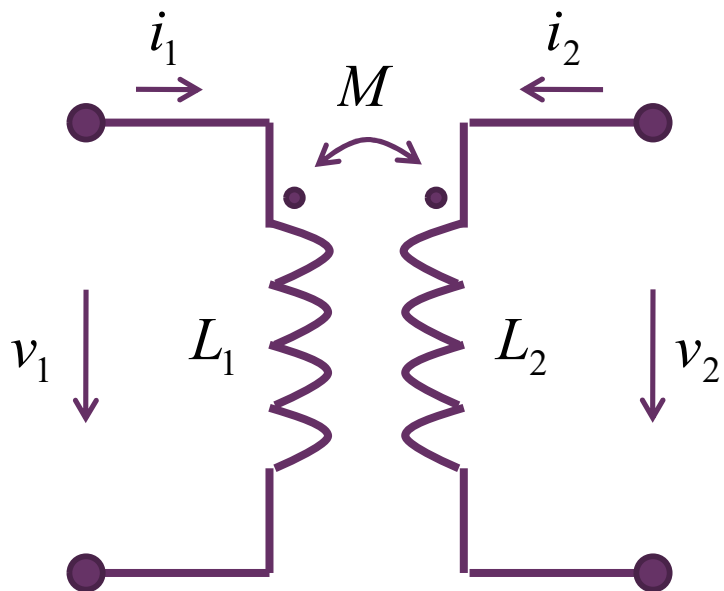
$$\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} L_1 & -M \\ -M & L_2 \end{bmatrix} \frac{d}{dt} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix}$$

$$M > 0$$

$$M > 0 \quad \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} L_1 & M \\ M & L_2 \end{bmatrix} \frac{d}{dt} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix} \quad M < 0$$

$$-1 \leq k = \frac{M}{\sqrt{L_1 L_2}} \leq 1$$

储能关系



如果两端电流实际流入导致磁通加强， $\mathbf{M}i_1i_2>0$ ，储能累加形态

如果两端电流实际流入导致磁通抵消， $\mathbf{M}i_1i_2<0$ ：一个端口进能量，一个端口出能量（一个端口电流自同名端入，另一个端口电流自同名端出：一端接源，一端接负载，做传输系统时的情形）

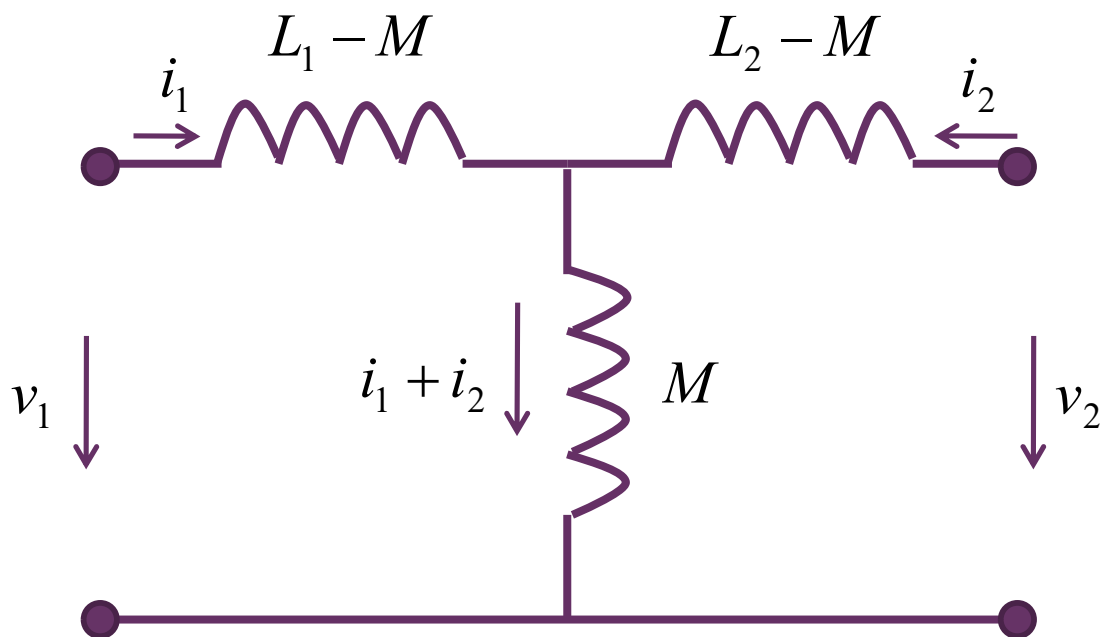
$$\begin{aligned}
 p(t) &= v_1(t)i_1(t) + v_2(t)i_2(t) \\
 &= \left(L_1 \frac{di_1(t)}{dt} + M \frac{di_2(t)}{dt} \right) i_1(t) \\
 &\quad + \left(L_2 \frac{di_2(t)}{dt} + M \frac{di_1(t)}{dt} \right) i_2(t) \\
 &= \frac{1}{2} L_1 \frac{d}{dt} i_1^2(t) + M \frac{d}{dt} (i_1(t)i_2(t)) + \frac{1}{2} L_2 \frac{d}{dt} i_2^2(t)
 \end{aligned}$$

$$E(t) = \int_{-\infty}^t p(v) dv$$

$$= \frac{1}{2} L_1 i_1^2(t) + M i_1(t) i_2(t) + \frac{1}{2} L_2 i_2^2(t)$$

假设： $E(-\infty) = 0$

T型等效电路：互易网络z参量等效电路

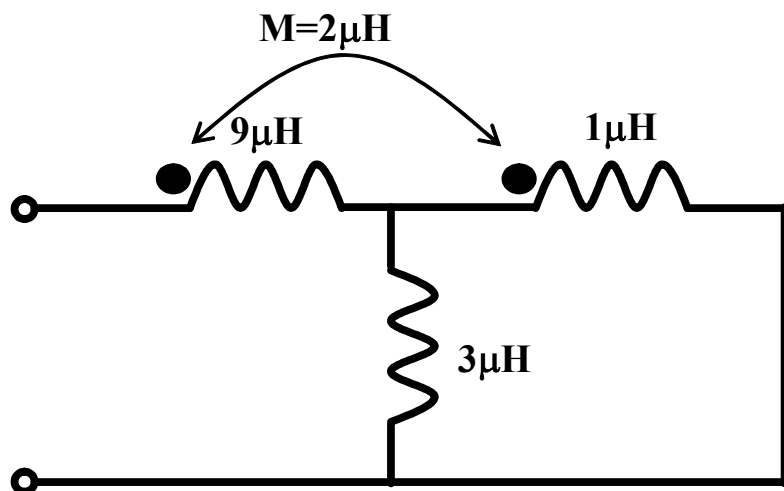


$$v_1 = (L_1 - M) \frac{di_1}{dt} + M \frac{d}{dt}(i_1 + i_2) = L_1 \frac{di_1}{dt} + M \frac{di_2}{dt}$$

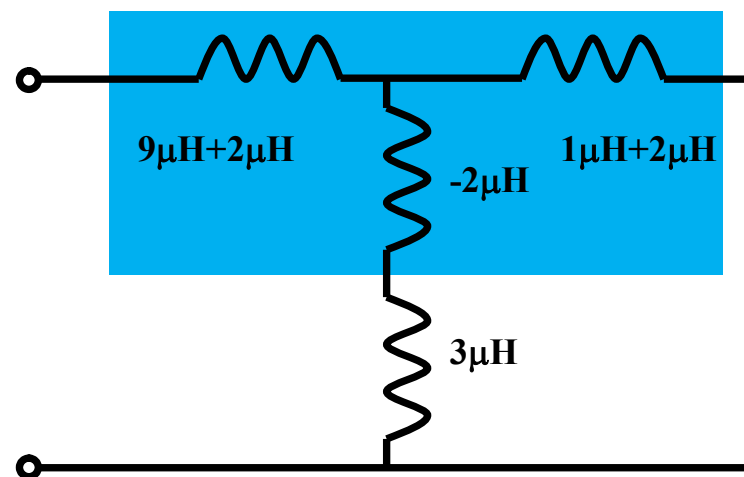
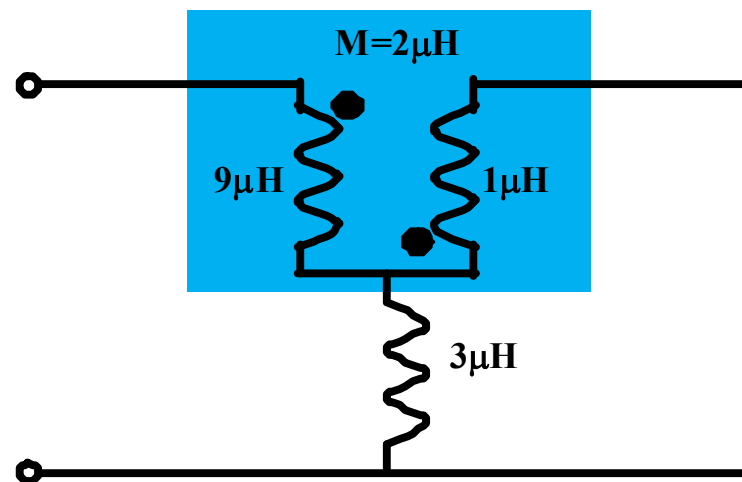
$$v_2 = (L_2 - M) \frac{di_2}{dt} + M \frac{d}{dt}(i_1 + i_2) = L_2 \frac{di_2}{dt} + M \frac{di_1}{dt}$$

具有完全一致的端口方程，故而是等效电路

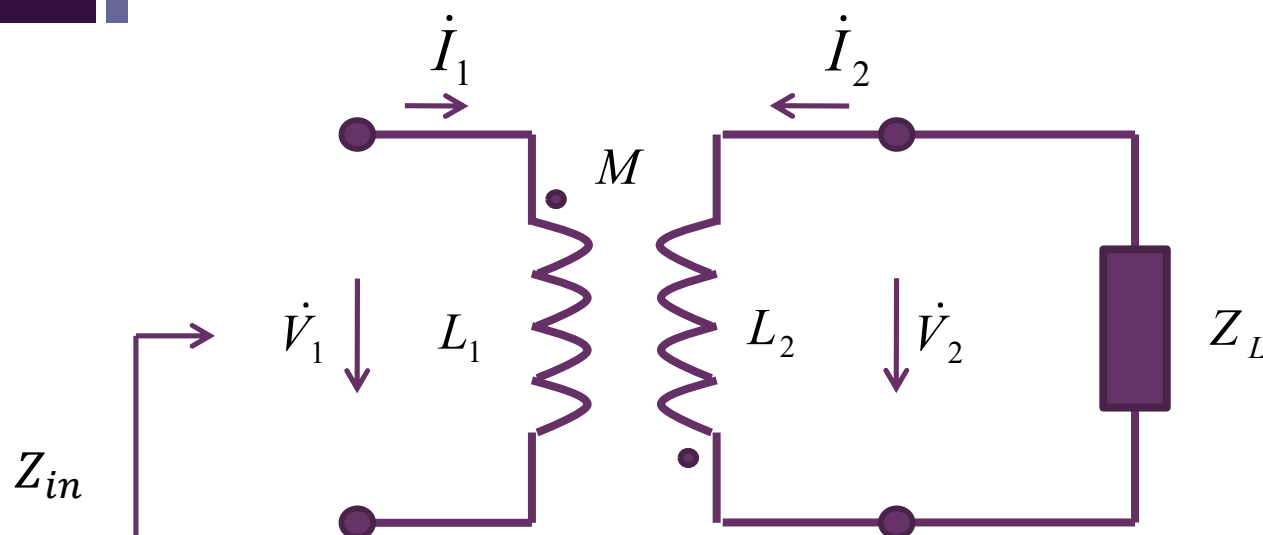
等效电路应用例



$$\begin{aligned}
 L &= (L_1 + M) \text{串} ((L_3 - M) \text{并} (L_2 + M)) \\
 &= 11\mu H \text{串} (1\mu H \text{并} 3\mu H) \\
 &= 11\mu H + \frac{3 \cdot 1}{3 + 1} \mu H = 11.75\mu H
 \end{aligned}$$



互感的阻抗变换作用



$$\begin{bmatrix} \dot{V}_1 \\ \dot{V}_2 \end{bmatrix} = \begin{bmatrix} j\omega L_1 & j\omega M \\ j\omega M & j\omega L_2 \end{bmatrix} \begin{bmatrix} \dot{I}_1 \\ \dot{I}_2 \end{bmatrix}$$

$$Z_{in} = Z_{11} - \frac{Z_{12}Z_{21}}{Z_{22} + Z_L}$$

$$= j\omega L_1 + \frac{(\omega M)^2}{j\omega L_2 + Z_L}$$

$$Z_{in, Z_L=R_L} = j\omega L_1 + \frac{(\omega M)^2}{j\omega L_2 + R_L} = \frac{(\omega M)^2}{R_L^2 + (\omega L_2)^2} R_L + j\omega L_1 \left(1 - \frac{L_2}{L_1} \frac{(\omega M)^2}{R_L^2 + (\omega L_2)^2} \right)$$

$$= \frac{k^2}{1 + Q_L^2} n^2 R_L + j\omega \left(1 - \frac{k^2}{1 + Q_L^2} \right) L_1 \quad \begin{matrix} k=1, Q_L=0 \\ \cong \\ n^2 R_L \end{matrix}$$

$$n = \sqrt{\frac{L_1}{L_2}} = \frac{N_1}{N_2} \quad k = \frac{M}{\sqrt{L_1 L_2}} \quad Q_L = \frac{R_L}{\omega L_2}$$

理想变压器抽象

理想变压器是互感变压器的理想抽象模型

$$M_0 = \sqrt{L_1 L_2} \rightarrow \infty, k = \frac{M}{M_0} \rightarrow 1 \quad \text{理想抽象条件}$$

$$\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} L_1 & M \\ M & L_2 \end{bmatrix} \frac{d}{dt} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = M_0 \begin{bmatrix} n & k \\ k & \frac{1}{n} \end{bmatrix} \frac{d}{dt} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix}$$

互感变压器是二阶元件

$$n = \frac{N_1}{N_2} = \sqrt{\frac{L_1}{L_2}}$$

匝数比

$k = 1$
全耦合

↓ 二阶微分方程退化为一阶微分方程

$v_1 = n v_2$ 全耦合互感变压器是一阶元件

$$v_2 = M_0 \frac{d}{dt} \left(i_1 + \frac{i_2}{n} \right)$$

$M_0 \rightarrow \infty$
极大电感量

↓

$$\frac{d}{dt} \left(i_1 + \frac{i_2}{n} \right) \rightarrow 0 \quad i_1 + \frac{i_2}{n} = I_0$$

直流短路，短路电流无关系
扣除直流后，两回路电流线性关系

→

$$\begin{aligned} v_1 &= n v_2 \\ i_1 &= -\frac{1}{n} i_2 \end{aligned}$$

理想变压器是零阶元件

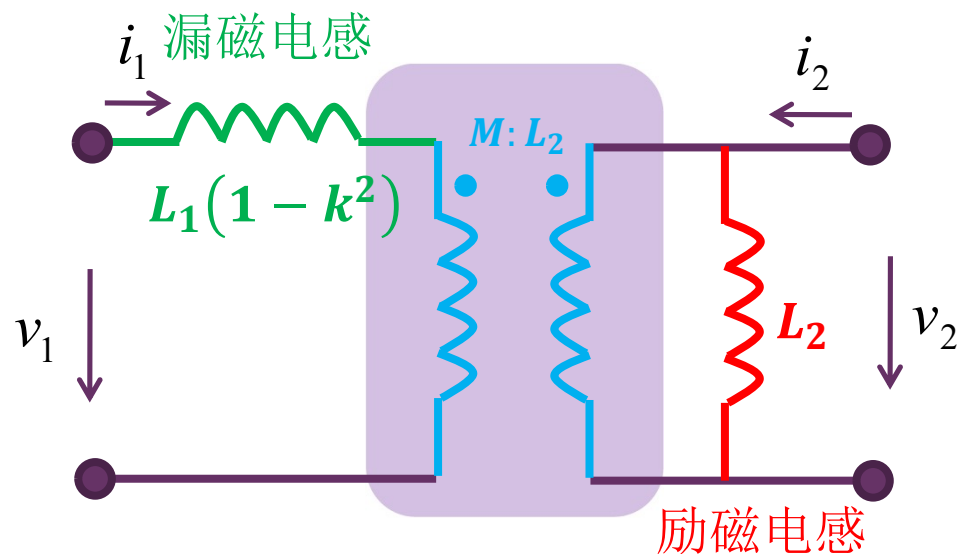
互感变压器的h参量模型

- T型网络等效属z参量模型，其h参量、g参量电路模型如图所示

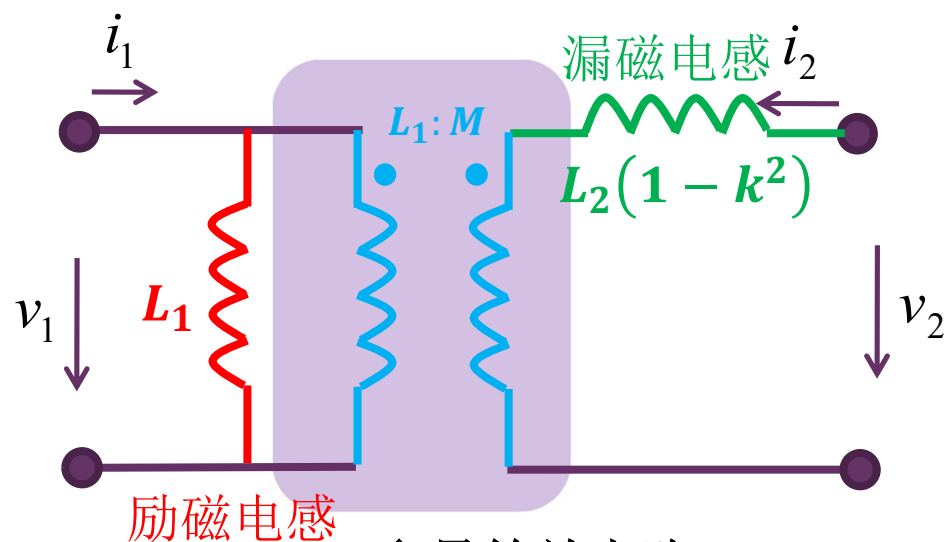
- 作业证明

- 上学期作业：纯阻、纯容、纯感互易网络中蕴含理想变压器

- 全耦合变压器是一阶系统：没有漏磁，只有励磁



h参量等效电路



g参量等效电路

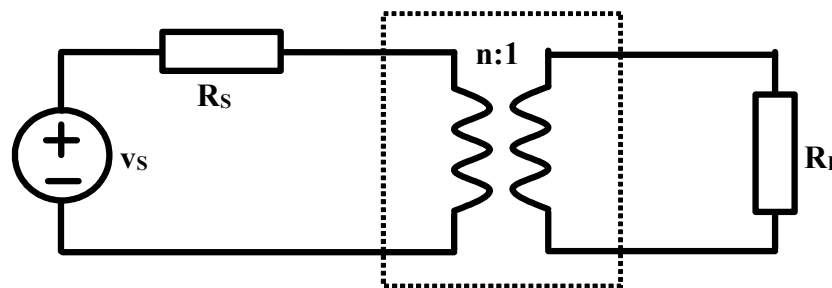
互感变压器的作用

- 直流隔离：两个回路之间没有直流通路，可以不共同一个地，可以完全独立

- 信号和电能通过互感双向传递

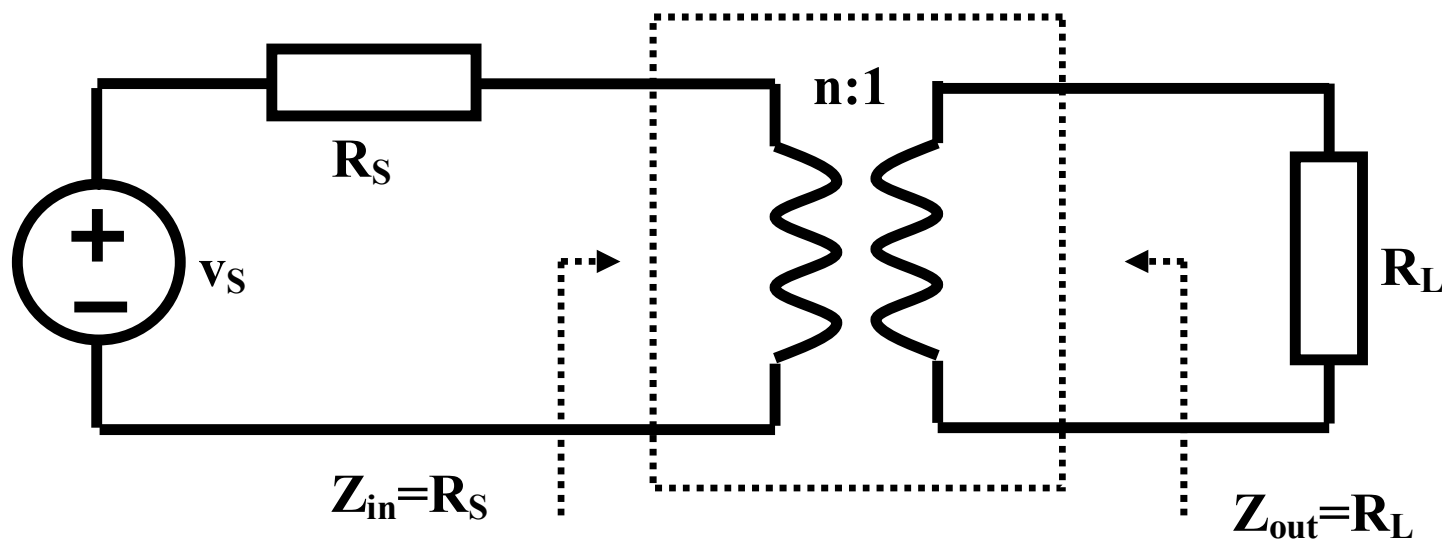
- 双向网络具有阻抗变换能力

- $$Z_{in} = Z_{11} - \frac{Z_{12}Z_{21}}{Z_{22} + Z_L}$$



- 本身具有感性，可以和电容谐振形成带通滤波特性

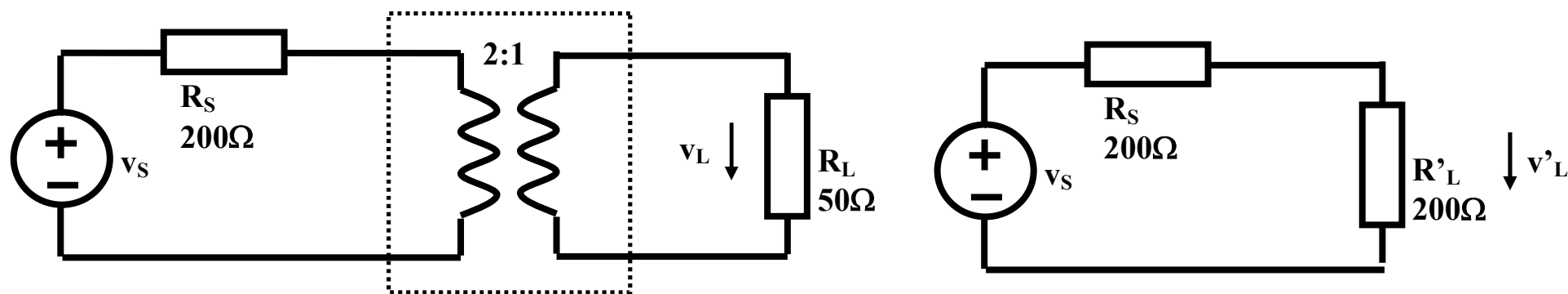
理想变压器



$$n = \sqrt{\frac{R_S}{R_L}} = \sqrt{\frac{200}{50}} = 2$$

除了直流频点之外，任意频点均可实现阻抗匹配：最大功率匹配传输

理想变压器传递函数



理想变压器抽象不包括零频点

$$\begin{aligned}
 H &= 2 \sqrt{\frac{R_S}{R_L}} \frac{\dot{V}_L}{\dot{V}_S} = 2 \sqrt{\frac{R_S}{R_L}} \frac{\dot{V}'_L \cdot \frac{1}{n}}{\dot{V}_S} = 2 \frac{\dot{V}'_L}{\dot{V}_S} \\
 &= 2 \frac{R'_L}{R_S + R'_L} = 2 \frac{n^2 R_L}{R_S + n^2 R_L} = 2 \frac{200}{200 + 200} = 1
 \end{aligned}$$

$$\eta = |H|^2 = \frac{P_L}{P_{S,\max}} = 1 = 100\%$$

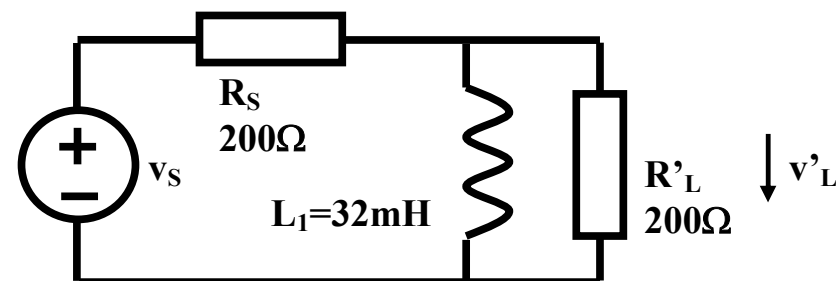
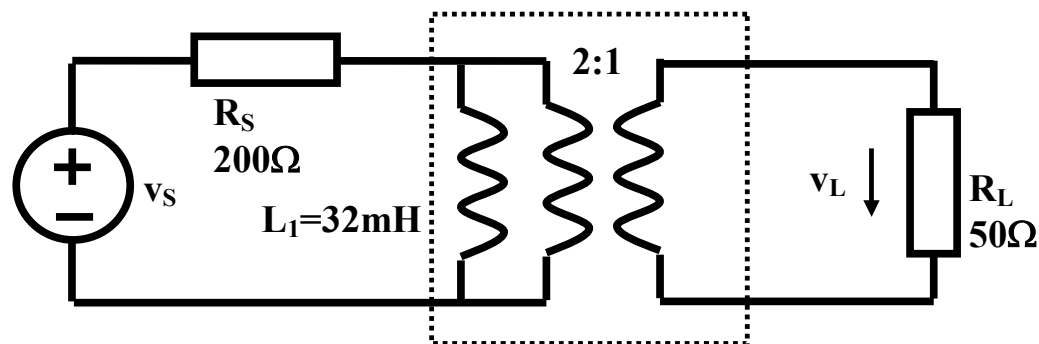
全耦合变压器

$$k = 1$$

$$L_1 = 32mH$$

$$L_2 = 8mH$$

$$M = 16mH$$



$$H = 2 \sqrt{\frac{R_S}{R_L}} \frac{\dot{V}_L}{\dot{V}_S} = 2 \sqrt{\frac{R_S}{R_L}} \frac{\dot{V}'_L \cdot \frac{1}{n}}{\dot{V}_S} = 2 \frac{\dot{V}'_L}{\dot{V}_S}$$

$$= 2 \frac{R'_L \parallel j\omega L_1}{R_S + R'_L \parallel j\omega L_1} = 2 \frac{j\omega L_1 G_S}{1 + j\omega L_1 (G_S + G'_L)} = \frac{j\omega \tau}{1 + j\omega \tau} = \frac{s \tau}{1 + s \tau} = \frac{s}{s + \omega_0}$$

$$\tau = L_1 (G_S + G'_L) = 2L_1 G_S$$

$$= 2 \times 32mH \times \frac{1}{200\Omega} = 0.32ms$$

$$\tau = 2G_S L_1$$

$$f_{3dB} = \frac{\omega_0}{2\pi} = \frac{1}{2\pi\tau} = 497Hz$$

具有一阶高通滤波特性：在高频实现了阻抗匹配

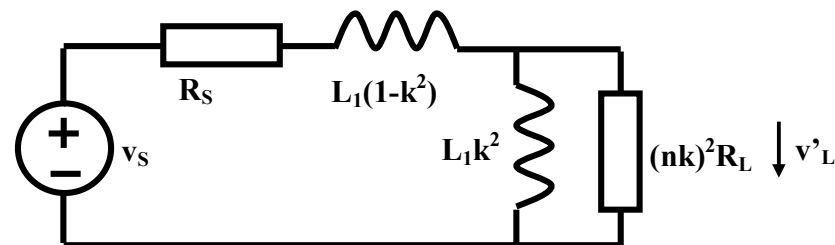
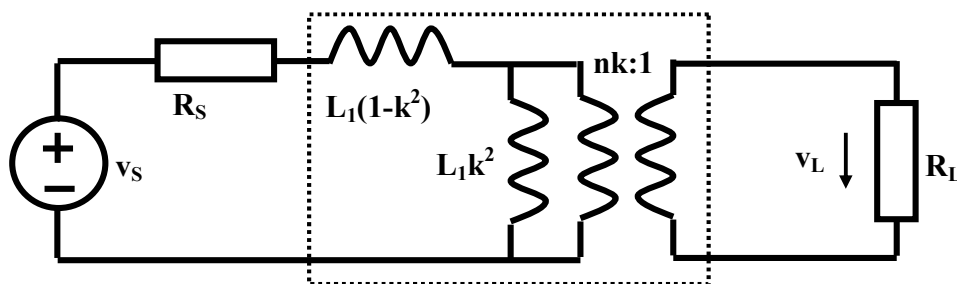
互感变压器

$$k < 1$$

$$L_1 = 32mH$$

$$L_2 = 8mH$$

$$M = k \times 16mH$$



$$H = 2 \sqrt{\frac{R_S}{R_L}} \frac{\dot{V}_L}{\dot{V}_S} = 2 \sqrt{\frac{R_S}{R_L}} \frac{\dot{V}'_L \cdot \frac{1}{nk}}{\dot{V}_S} = 2 \frac{1}{k} \frac{\dot{V}'_L}{\dot{V}_S}$$

$$= \frac{2}{k} \frac{(nk)^2 R_L \parallel j\omega L_1 k^2}{R_S + j\omega L_1 (1-k^2) + (nk)^2 R_L \parallel j\omega L_1 k^2} = \frac{2}{k} \frac{k^2 \frac{sL_1 R_S}{sL_1 + R_S}}{R_S + sL_1 (1-k^2) + k^2 \frac{sL_1 R_S}{sL_1 + R_S}}$$

$$= \frac{2ksL_1 R_S}{(R_S + sL_1 (1-k^2))(sL_1 + R_S) + k^2 sL_1 R_S} = \frac{2ksL_1 R_S}{s^2 L_1^2 (1-k^2) + 2sL_1 R_S + R_S^2}$$

$$= k \frac{2 \frac{R_S}{L_1 (1-k^2)} s}{s^2 + 2 \frac{R_S}{L_1 (1-k^2)} s + \frac{R_S^2}{L_1^2 (1-k^2)}} = k \frac{2\xi\omega_0 s}{s^2 + 2\xi\omega_0 s + \omega_0^2}$$

$$\omega_0 = \frac{R_S}{L_1 \sqrt{1-k^2}}$$

$$\xi = \frac{1}{\sqrt{1-k^2}} > 1$$

典型的二阶带通滤波特性

互感变压器：带通滤波特性

$$H = 2\sqrt{\frac{R_S}{R_L}} \frac{\dot{V}_L}{\dot{V}_S} = k \frac{2\xi\omega_0 s}{s^2 + 2\xi\omega_0 s + \omega_0^2}$$

$$\omega_0 = \frac{R_S}{L_1 \sqrt{1-k^2}}$$

$$\xi = \frac{1}{\sqrt{1-k^2}} > 1$$

过阻尼情况下的二阶带通滤波器

$$f_0 = \frac{\omega_0}{2\pi} = \frac{1}{2\pi G_S L_1 \sqrt{1-k^2}} = \frac{995}{\sqrt{1-k^2}} \text{ Hz}$$

$$\eta = \frac{P_L}{P_{S,\max}} = |H(j\omega_0)|^2 = k^2 \times 100\%$$

$$BW_{3dB} = \frac{f_0}{Q} = 2\xi f_0 = \frac{2}{2\pi G_S L_1 (1-k^2)} = \frac{1989}{1-k^2} \text{ Hz}$$

k=0.95

$$f_0 = \frac{995}{\sqrt{1-k^2}} \text{ Hz} = 3.19 \text{ kHz}$$

$$BW_{3dB} = \frac{1989}{1-k^2} \text{ Hz} = 20.4 \text{ kHz}$$

$$\eta_{\max} = k^2 = 90.25\% = -0.45 \text{ dB}$$

k=0.50

$$f_0 = \frac{995}{\sqrt{1-k^2}} \text{ Hz} = 1.15 \text{ kHz}$$

$$BW_{3dB} = \frac{1989}{1-k^2} \text{ Hz} = 2.65 \text{ kHz}$$

$$\eta_{\max} = k^2 = 25\% = -6 \text{ dB}$$

只有谐振才能最大功率传输

■ 互感变压器

- 有两个等效电感，没有谐振电容，无法最大功率传输匹配

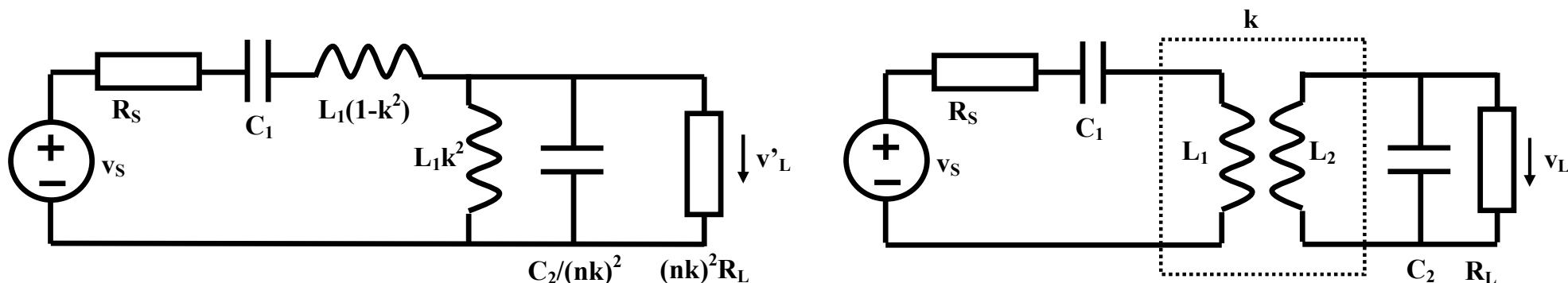
■ 全耦合变压器

- 一个等效电感，可视为存在一个并联谐振零电容，故而谐振频率为无穷（高通）

■ 理想变压器

- 可视为无穷大电感和零电容的并联谐振，谐振频率无穷，带宽无穷：除了零频外，所有频点都可匹配

通过简单谐振实现匹配



电容 $\mathbf{C_1}$ 和漏磁电感 $\mathbf{L_1(1-k^2)}$ 串联谐振于选定频点 ω_r ，该频点上串联 $\mathbf{LC}$ 相当于短路

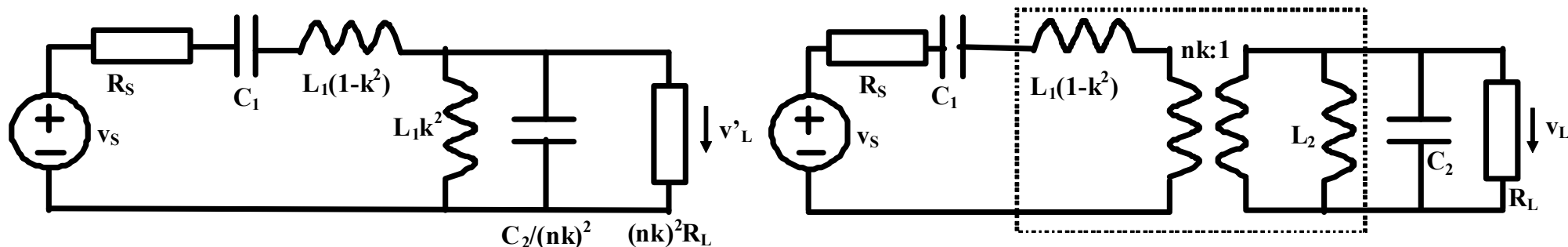
端口2并接 $\mathbf{C_2}$ 电容和励磁电感 $\mathbf{L_2}$ 并联谐振于同一个选定频点 ω_r ，该频点上并联 $\mathbf{LC}$ 相当于开路

在 ω_r 频点上， $\mathbf{(nk)^2R_L}$ 的等效负载电阻和信源内阻 $\mathbf{R_s}$ 直连，只要两者相等，则可在该频点上实现最大功率传输匹配

$$R_s = (nk)^2 R_L$$

$$n = \frac{1}{k} \sqrt{\frac{R_s}{R_L}} \stackrel{k=0.5}{=} \frac{1}{0.5} \sqrt{\frac{200}{50}} = 4$$

k=0.5时的简单匹配电路



$$n = \frac{1}{k} \sqrt{\frac{R_S}{R_L}} \stackrel{k=0.5}{=} \frac{1}{0.5} \sqrt{\frac{200}{50}} = 4$$

$$L_1 = 32mH, L_2 = \frac{1}{n^2} L_1 = 2mH$$

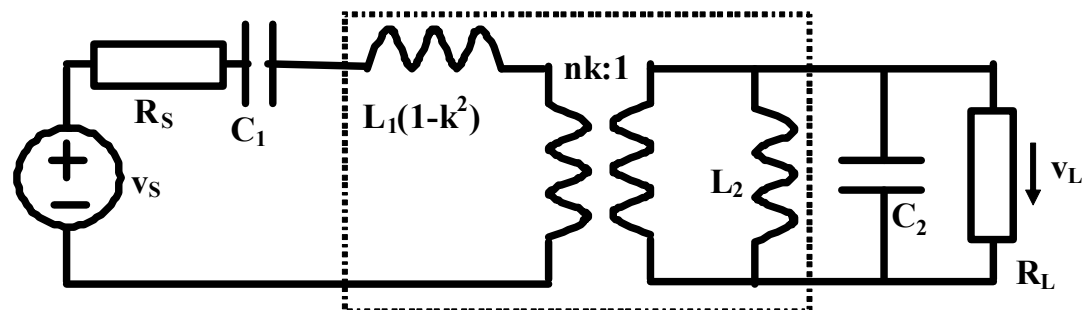
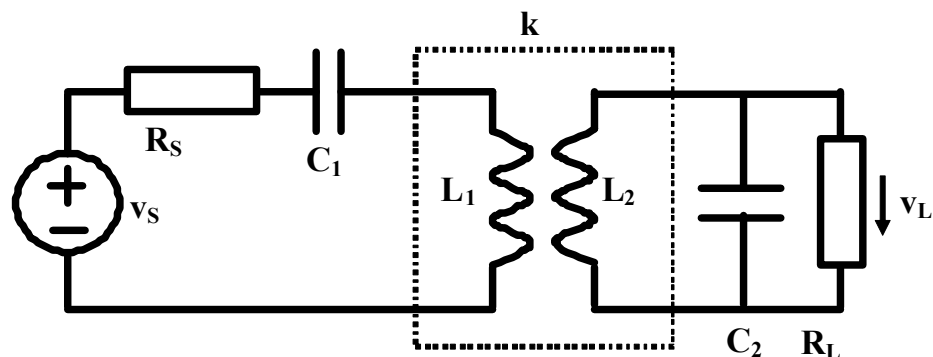
$$M = k\sqrt{L_1 L_2} = 4mH$$

$$f_r = 1kHz$$

$$C_1 = \frac{1}{(2\pi f_r)^2 L_1 (1-k^2)} = \frac{1}{(2\pi \times 1000)^2 \times 32 \times 10^{-3} \times (1-0.5^2)} = 1.06\mu F$$

$$C_2 = \frac{1}{(2\pi f_r)^2 L_2} = \frac{1}{(2\pi \times 1000)^2 \times 2 \times 10^{-3}} = 12.7\mu F$$

传递函数

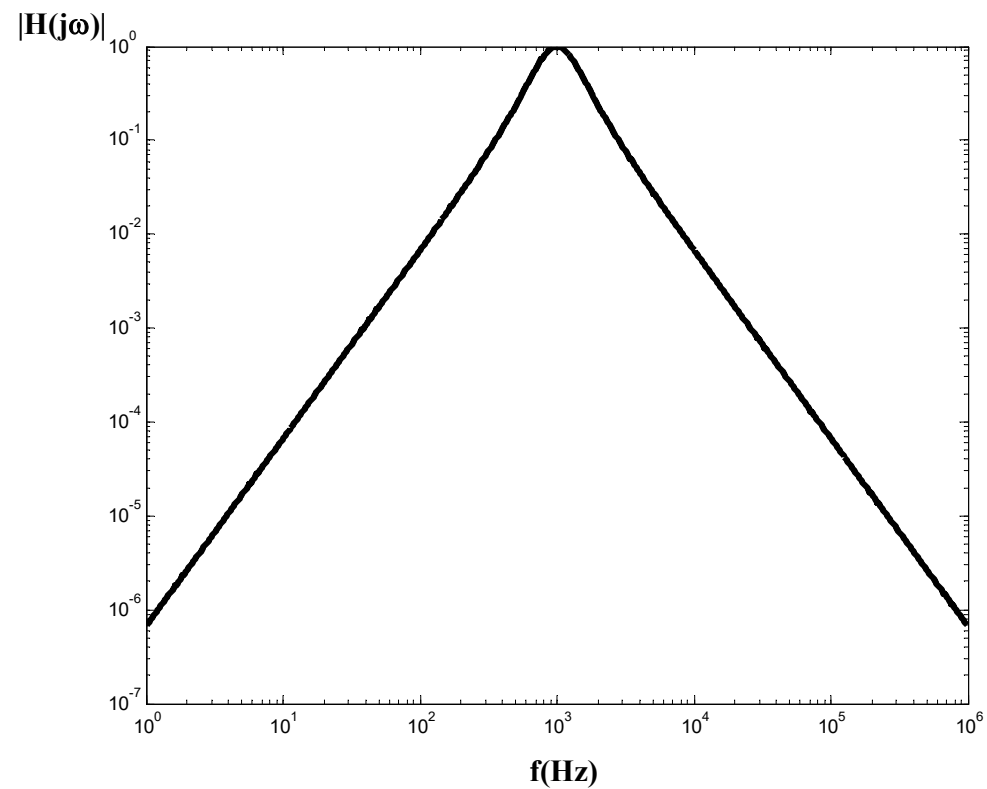
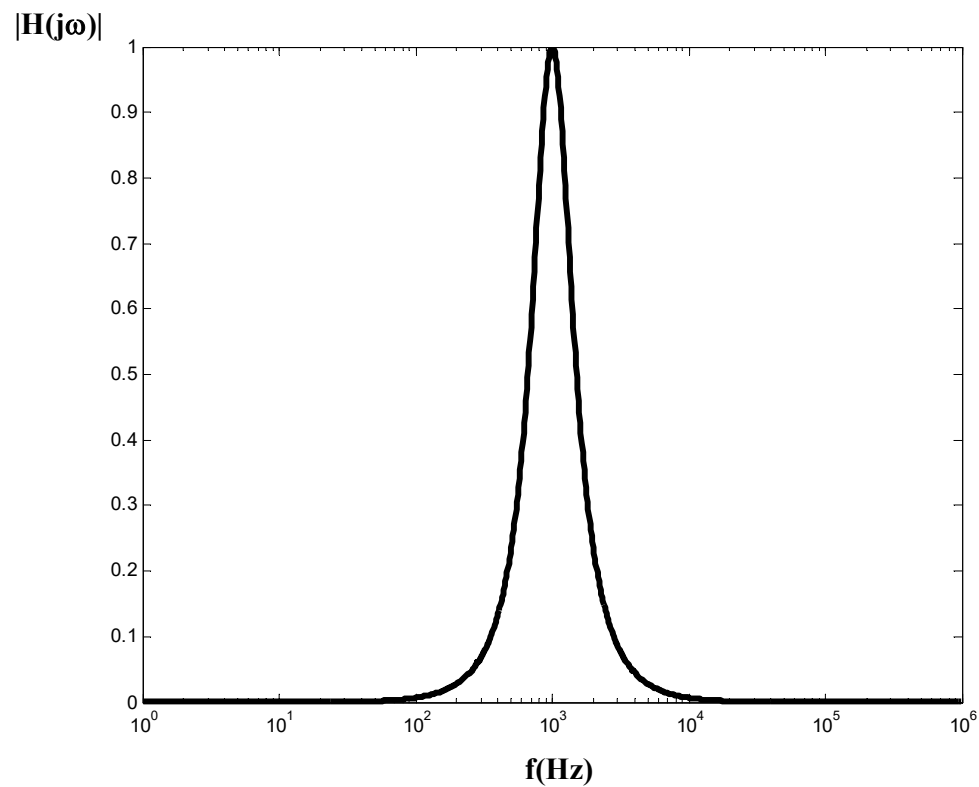


$$\mathbf{ABCD} = \begin{bmatrix} 1 & R_s + \frac{1}{sC_1} + sL_1(1-k^2) \\ 0 & 1 \end{bmatrix} \begin{bmatrix} nk & 0 \\ 0 & \frac{1}{nk} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ \frac{1}{R_L} + sC_2 + \frac{1}{sL_2} & 1 \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix}$$

$$H = 2 \sqrt{\frac{R_s}{R_L}} \frac{\dot{V}_L}{\dot{V}_s} = 2 \sqrt{\frac{R_s}{R_L}} \frac{1}{A}$$

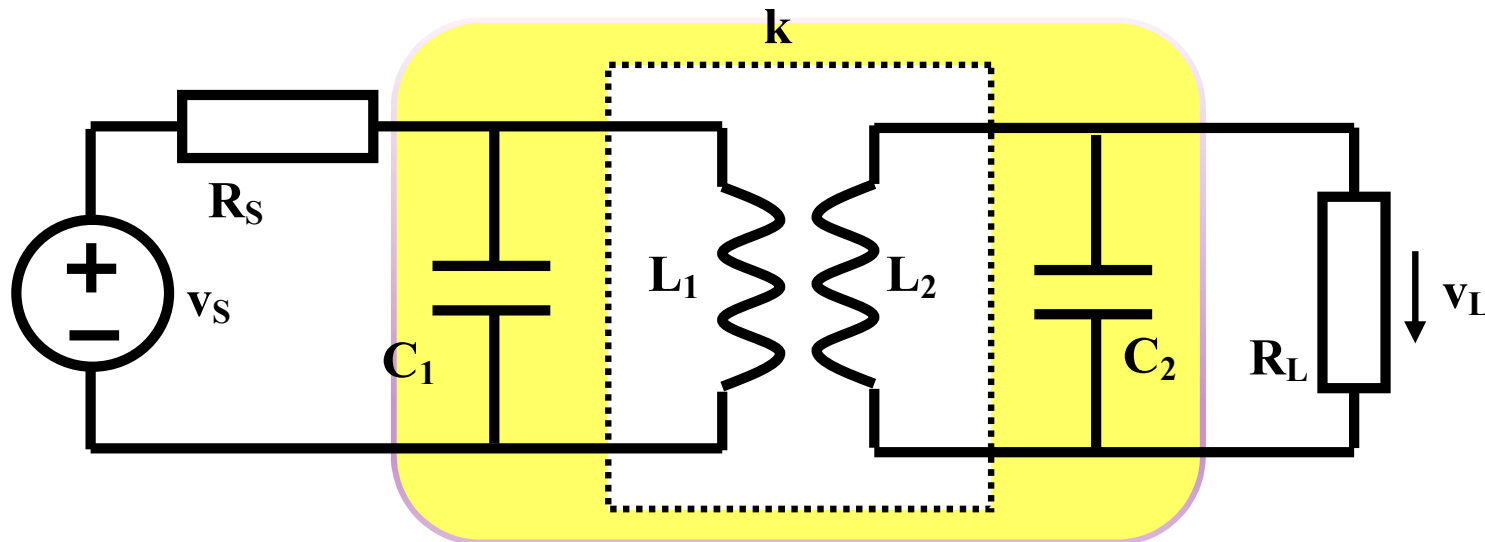
四阶系统，这里不讨论其传函形式
Matlab数值计算代码见教材

带通特性



双谐振精致设计

不是独立电感的简单谐振，而是耦合电感的复杂双谐振（耦合在一起的双谐振）
利用对特征阻抗的研究，获得设计公式



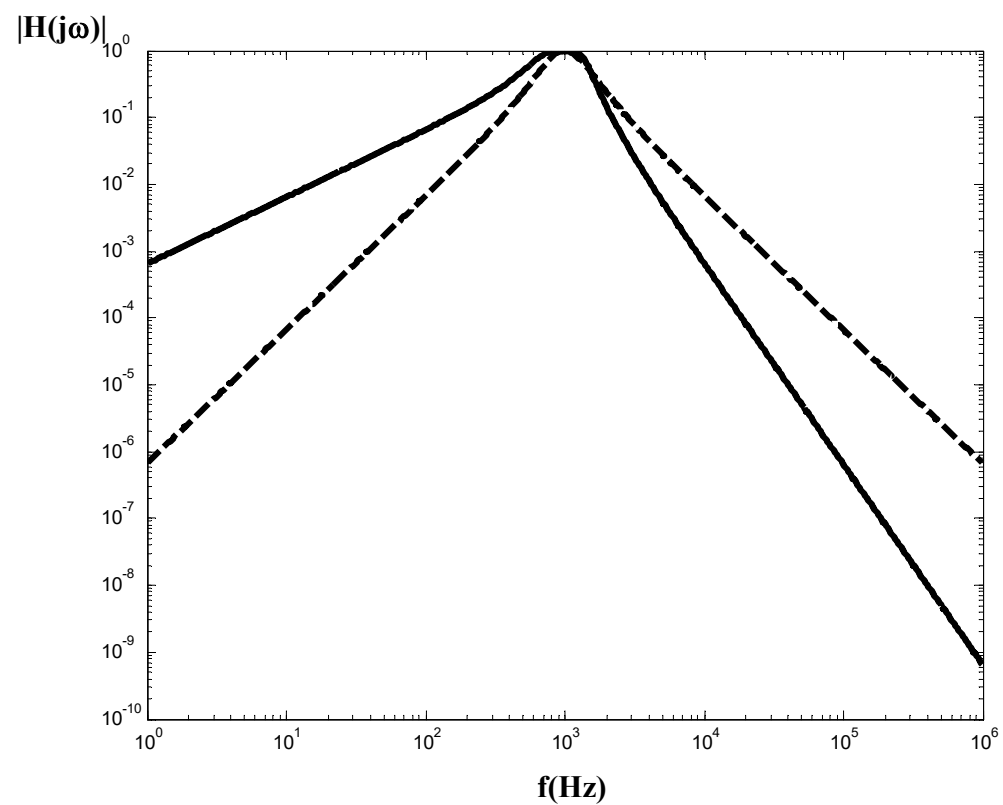
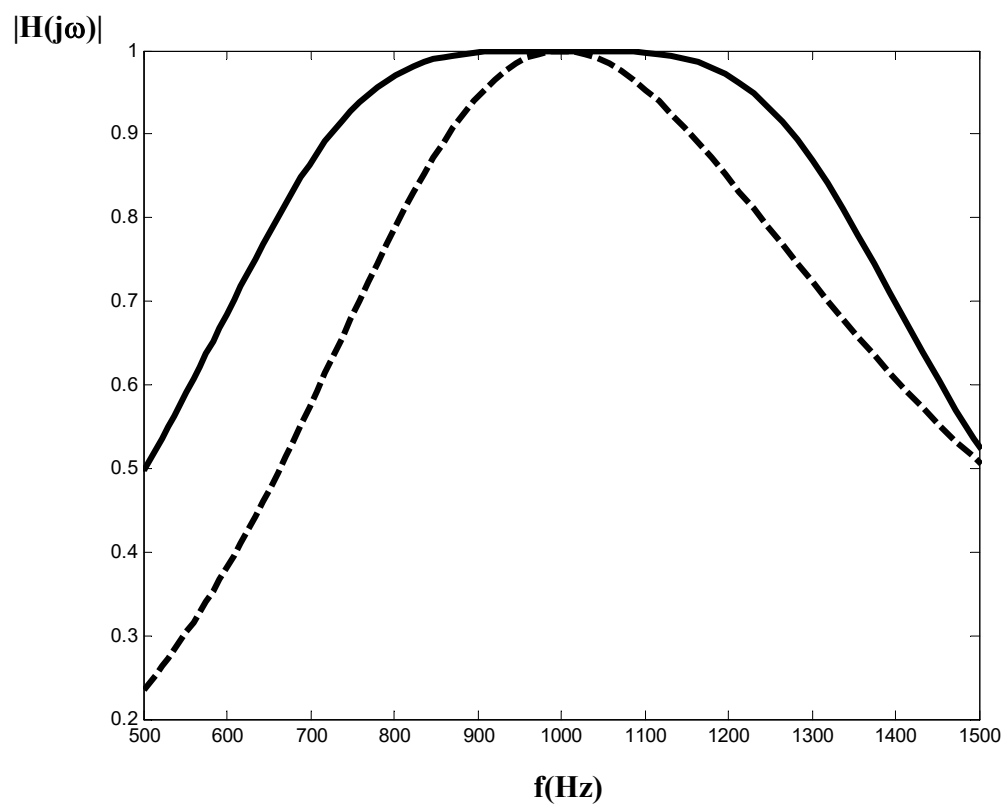
$$R_S = 200\Omega, R_L = 50\Omega, k = 0.5, n = 2 = \sqrt{R_S/R_L}$$

$$L_1 = 20.44\text{mH}, L_2 = L_1/n^2 = 5.11\text{mH}, M = k \cdot \sqrt{L_1 \cdot L_2} = 5.11\text{mH}$$

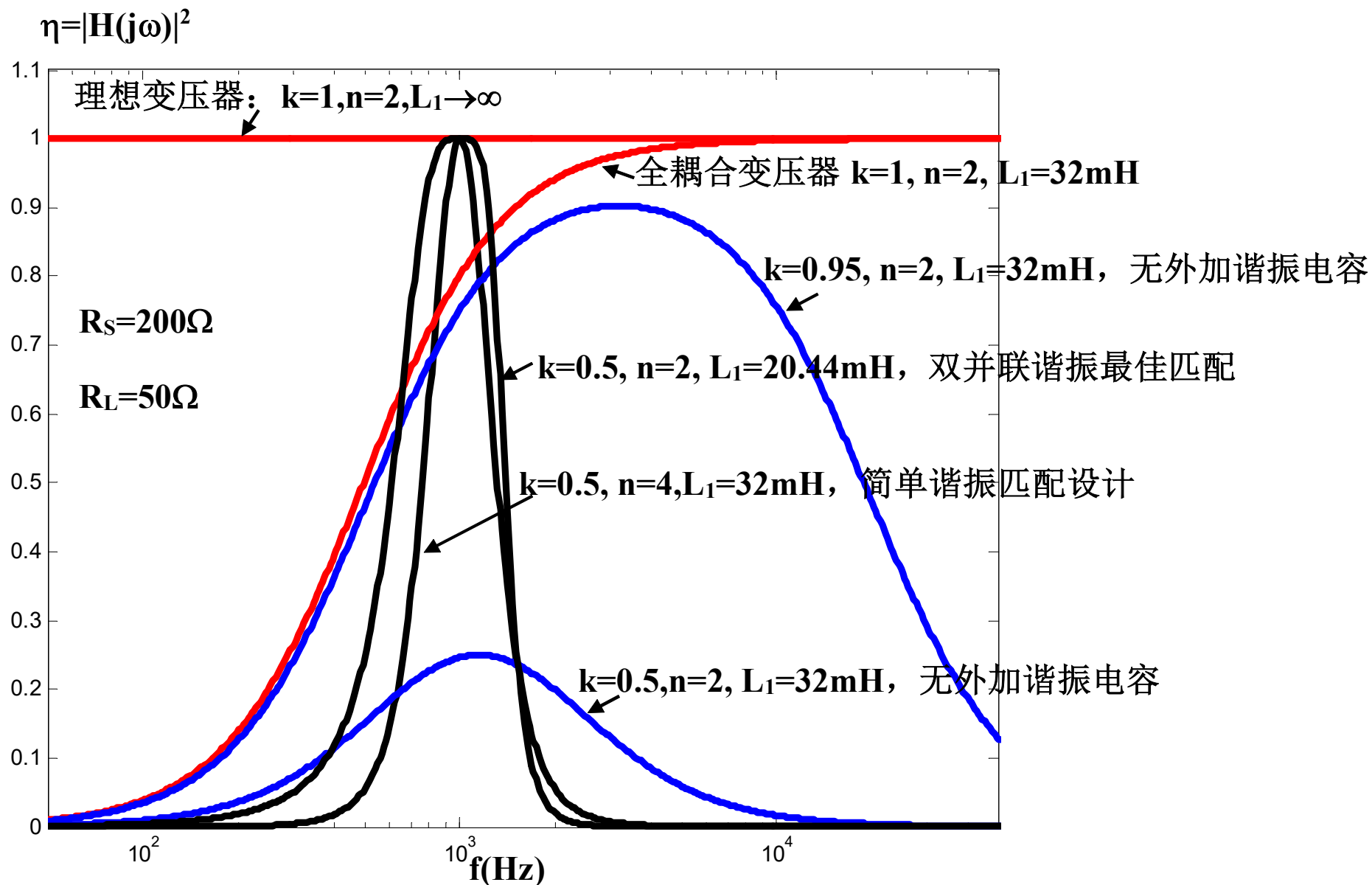
$$C_1 = 1.43\mu\text{F}, C_2 = 5.72\mu\text{F}$$

设计原理和公式自看教材，不要求
这里直接给出代入公式后给出的数值结论

通带平坦特性



变压器阻抗匹配网络



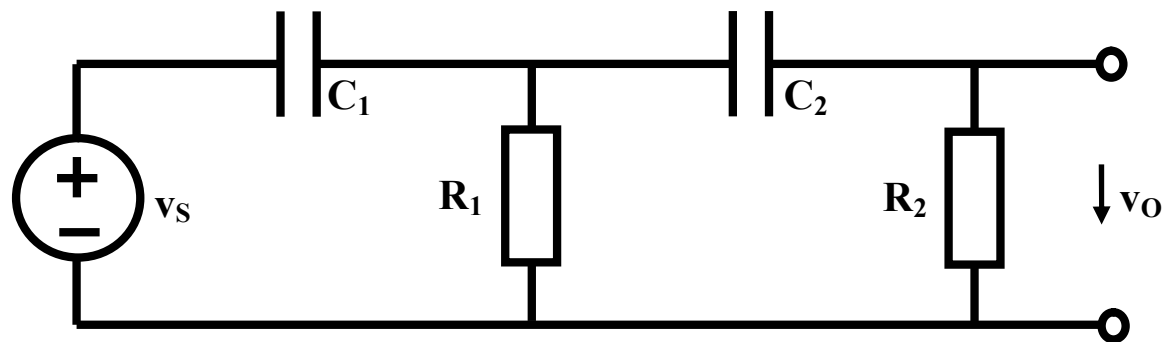
第08讲 阻抗匹配 小结

- 通过阻抗变换网络实现阻抗匹配是为了实现某种系统最优性能
 - 最大功率传输匹配：共轭匹配
 - 最小噪声系数匹配：《通信电路》课程讨论
 - 最大功率输出匹配：本课程CAD练习中会涉及到，但不做要求，
 - ...
- 阻抗匹配网络多采用双向的无损的电容、电感、变压器、传输线等实现
 - 双向才具有阻抗变换作用
 - 无损是对匹配网络一般的要求
- 利用LC的谐振特性，可以实现阻抗匹配网络
 - 要求掌握L型LC匹配网络的设计口诀：并大串小Q相等
- 互感变压器是无损的双向网络，具有阻抗变换作用，同时由于自身的感性，配合电容可实现带通滤波特性
 - 要求掌握互感变压器的z参量T模型和h参量漏磁励磁电感模型

本讲内容在教材中的章节位置

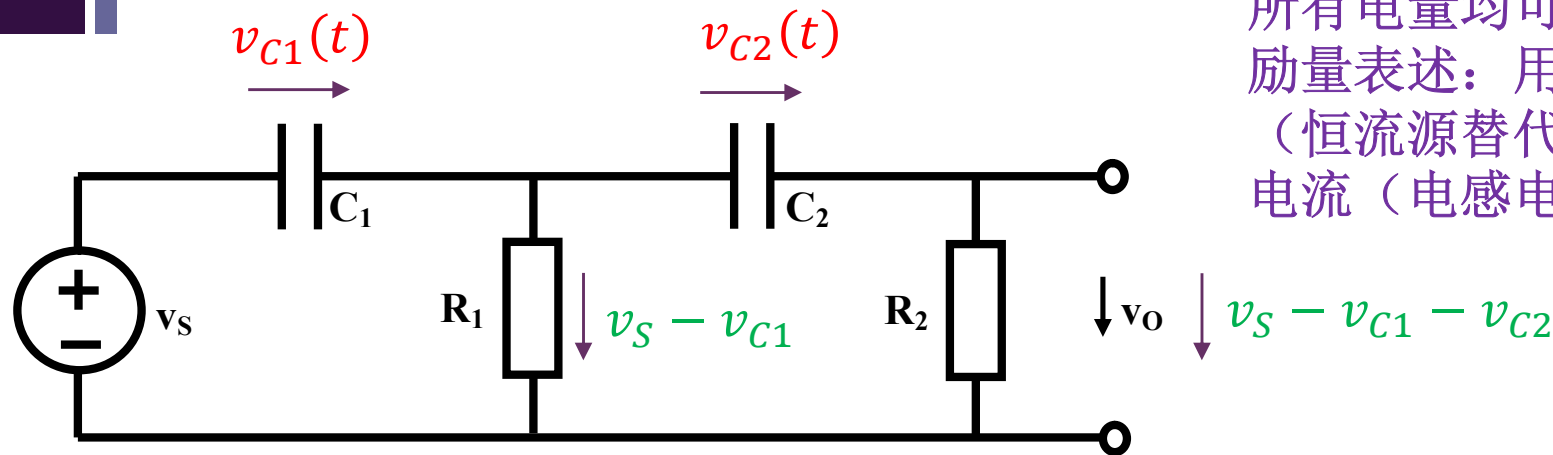
- P634: 最大功率传输匹配
- P588: 互感变压器
- P817: 互感变压器阻抗匹配网络
- P828: LC阻抗匹配网络
- P836: 阻抗变换原理

作业5.3 二阶RC高通滤波器



- 1、列写电路状态方程
- 2、列写以 v_o 为未知量的二阶微分方程
- 3、列写频域传递函数
- 4、从微分方程（或频域传递函数）说明关键参量： ξ , ω_0
- 5、假设两个电容初始电压均为0，激励源为阶跃信号源 $v_s(t)=V_0U(t)$ ，用五要素法获得输出电压表达式（考察 $R_1=R_2=R$, $C_1=C_2=C$ 的特殊情况）

状态方程：以状态变量为未知量



所有电量均可用状态变量和激励量表述：用恒压源替代电容（恒流源替代电感），求电容电流（电感电压）

$$C_2 \frac{dv_{C2}}{dt} = i_{C2} = i_{R2} = \frac{v_s - v_{C1} - v_{C2}}{R_2}$$

$$\frac{dv_{C2}}{dt} = \frac{v_s}{R_2 C_2} - \frac{v_{C1}}{R_2 C_2} - \frac{v_{C2}}{R_2 C_2}$$

$$C_1 \frac{dv_{C1}}{dt} = i_{C1} = i_{R1} + i_{C2} = \frac{v_s - v_{C1}}{R_1} + \frac{v_s - v_{C1} - v_{C2}}{R_2}$$

$$\frac{dv_{C1}}{dt} = \frac{v_s}{R_1 C_1} + \frac{v_s}{R_2 C_1} - \frac{v_{C1}}{R_1 C_1} - \frac{v_{C1}}{R_2 C_1} - \frac{v_{C2}}{R_2 C_1}$$

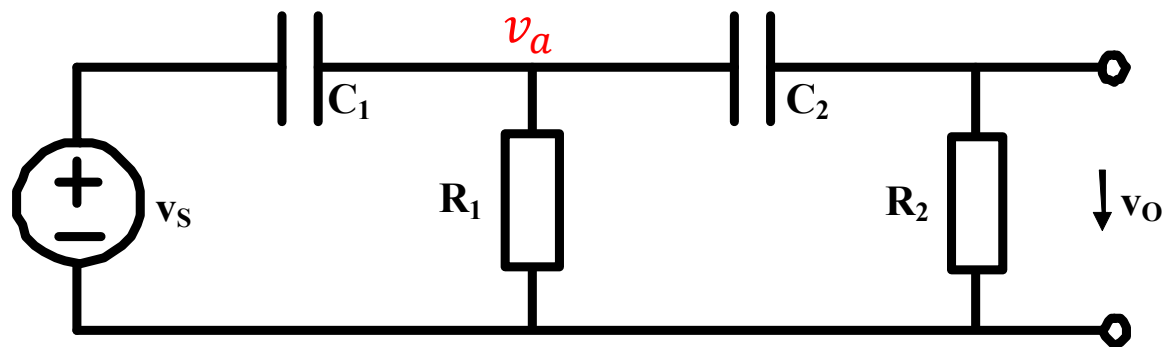
$$\frac{d}{dt} \begin{bmatrix} v_{C1} \\ v_{C2} \end{bmatrix} = \begin{bmatrix} -\frac{1}{R_1 C_1} - \frac{1}{R_2 C_1} & -\frac{1}{R_2 C_1} \\ -\frac{1}{R_2 C_2} & -\frac{1}{R_2 C_2} \end{bmatrix} \begin{bmatrix} v_{C1} \\ v_{C2} \end{bmatrix} + \begin{bmatrix} \frac{1}{R_1 C_1} + \frac{1}{R_2 C_1} \\ \frac{1}{R_2 C_2} \end{bmatrix} v_s$$

$$\frac{d}{dt} \mathbf{x} = \mathbf{A} \mathbf{x} + \mathbf{g}$$

以 v_o 为未知量的二阶微分方程

不妨从频域入手求

- 线性时不变系统相量域方法和线性电阻电路一样
- 用简单分压概念求

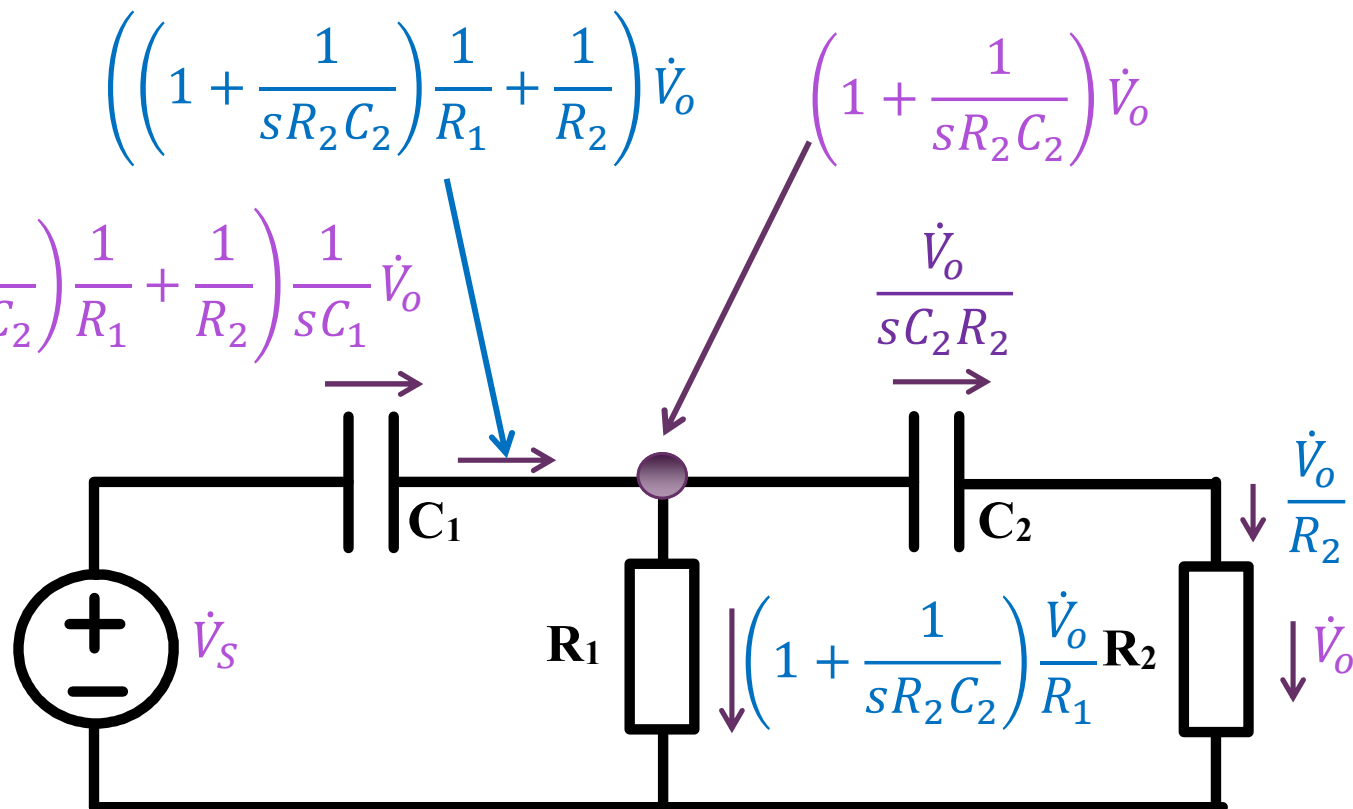


$$\begin{aligned}
 H(s) &\stackrel{s=j\omega}{\cong} \frac{\dot{V}_o}{\dot{V}_s} = \frac{\dot{V}_o}{\dot{V}_a} \cdot \frac{\dot{V}_a}{\dot{V}_s} = \frac{R_2}{R_2 + \frac{1}{sC_2}} \cdot \frac{\frac{1}{\frac{1}{R_1} + \frac{1}{R_2 + \frac{1}{sC_2}}}}{\frac{1}{sC_1} + \frac{1}{\frac{1}{R_1} + \frac{1}{R_2 + \frac{1}{sC_2}}}} \\
 &= \frac{sR_2C_2}{sR_2C_2 + 1} \cdot \frac{1}{\frac{1}{sC_1} \left(\frac{1}{R_1} + \frac{sC_2}{sR_2C_2 + 1} \right) + 1} = \frac{sR_2C_2}{sR_2C_2 + 1} \cdot \frac{sR_1C_1}{1 + \frac{sR_1C_2}{sR_2C_2 + 1} + sR_1C_1} \\
 &= \frac{s^2R_1C_1R_2C_2}{sR_2C_2 + 1 + sR_1C_2 + (sR_2C_2 + 1)sR_1C_1} = \frac{s^2R_1C_1R_2C_2}{1 + s(R_1C_1 + R_1C_2 + R_2C_2) + s^2R_1R_2C_1C_2}
 \end{aligned}$$

倒推法

利用串并联关系倒推

串联同电流，电压加
并联同电压，电流加

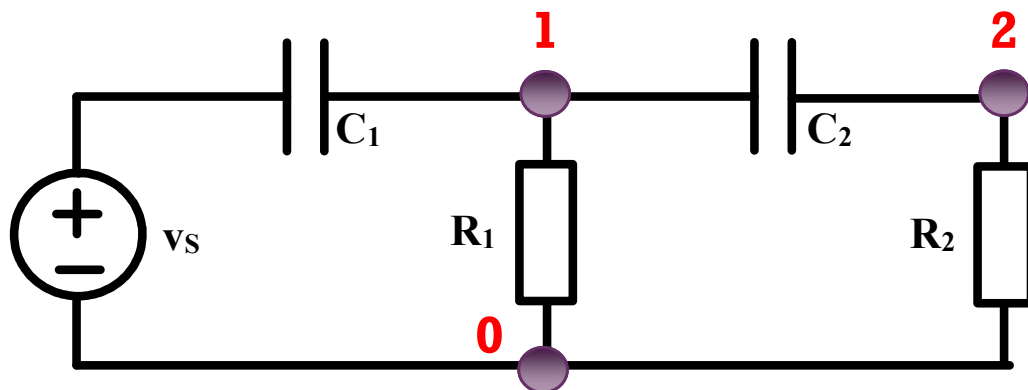


$$\dot{V}_S = \left(\left(1 + \frac{1}{sR_2C_2} \right) \frac{1}{sR_1C_1} + \frac{1}{sR_2C_1} \right) \dot{V}_o + \left(1 + \frac{1}{sR_2C_2} \right) \dot{V}_o$$

$$\left(1 + \frac{1}{sR_2C_1} + \frac{1}{sR_1C_1} + \frac{1}{sR_2C_2} + \frac{1}{s^2R_1C_1R_2C_2} \right) \dot{V}_o = \dot{V}_S$$

$$\left(s^2 + \left(\frac{1}{R_2C_1} + \frac{1}{R_1C_1} + \frac{1}{R_2C_2} \right) s + \frac{1}{R_1C_1R_2C_2} \right) \dot{V}_o = s^2 \dot{V}_S$$

$$\frac{d^2}{dt^2} v_o(t) + \left(\frac{1}{R_2C_1} + \frac{1}{R_1C_1} + \frac{1}{R_2C_2} \right) \frac{d}{dt} v_o(t) + \frac{1}{R_1R_2C_1C_2} v_o(t) = \frac{d^2}{dt^2} v_s(t)$$



$$\begin{bmatrix} sC_1 + G_1 + sC_2 & -sC_2 \\ -sC_2 & sC_2 + G_2 \end{bmatrix} \begin{bmatrix} \dot{V}_1 \\ \dot{V}_2 \end{bmatrix} = \begin{bmatrix} sC_1 \dot{V}_S \\ 0 \end{bmatrix}$$

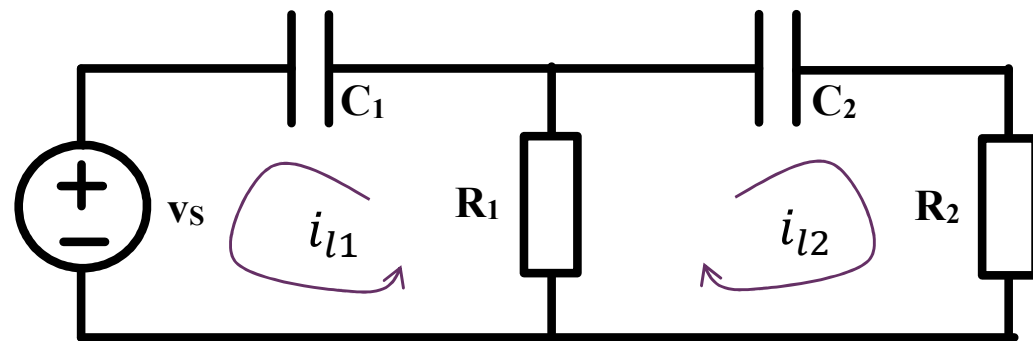
$$\begin{bmatrix} \dot{V}_1 \\ \dot{V}_2 \end{bmatrix} = \frac{\begin{bmatrix} sC_2 + G_2 & sC_2 \\ sC_2 & s(C_1 + C_2) + G_1 \end{bmatrix}}{(s(C_1 + C_2) + G_1)(sC_2 + G_2) - (-sC_2)^2} \begin{bmatrix} sC_1 \dot{V}_S \\ 0 \end{bmatrix}$$

$$\dot{V}_0 = \dot{V}_2 = \frac{sC_2 sC_1 \dot{V}_S}{(s(C_1 + C_2) + G_1)(sC_2 + G_2) - (-sC_2)^2}$$

$$= \frac{s^2 C_1 C_2}{s^2 C_1 C_2 + s(C_2 G_1 + C_1 G_2 + C_2 G_2) + G_1 G_2} \dot{V}_S$$

$$= \frac{s^2}{s^2 + s\left(\frac{1}{R_1 C_1} + \frac{1}{R_2 C_2} + \frac{1}{R_2 C_1}\right) + \frac{1}{R_1 C_1 R_2 C_2}} \dot{V}_S = \frac{s^2}{s^2 + 2\xi\omega_0 s + \omega_0^2} \dot{V}_S$$

回路电流法



$$\begin{bmatrix} R_1 + \frac{1}{sC_1} & R_1 \\ R_1 & R_1 + R_2 + \frac{1}{sC_2} \end{bmatrix} \begin{bmatrix} \dot{I}_{l1} \\ \dot{I}_{l2} \end{bmatrix} = \begin{bmatrix} -\dot{V}_S \\ 0 \end{bmatrix}$$

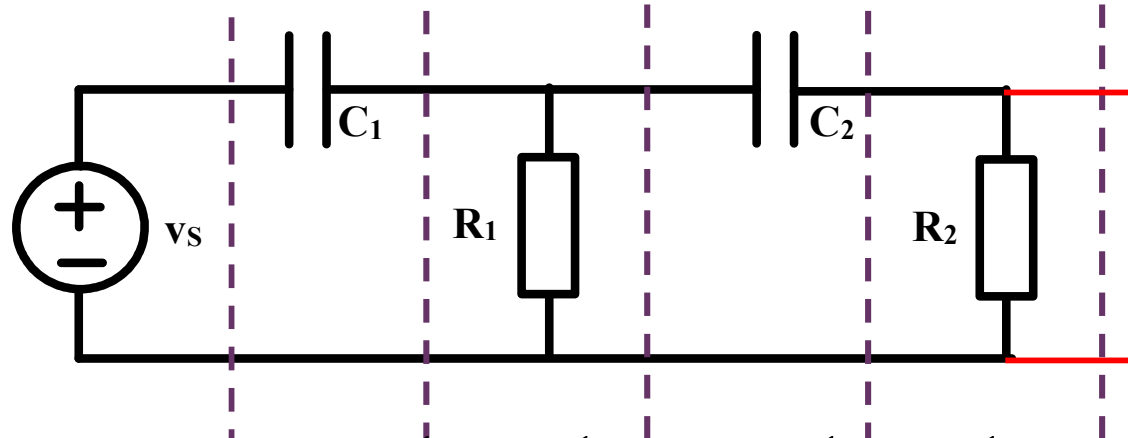
$$\begin{bmatrix} \dot{I}_{l1} \\ \dot{I}_{l2} \end{bmatrix} = \frac{\begin{bmatrix} R_1 + R_2 + \frac{1}{sC_2} & -R_1 \\ -R_1 & R_1 + \frac{1}{sC_1} \end{bmatrix}}{\left(R_1 + \frac{1}{sC_1}\right)\left(R_1 + R_2 + \frac{1}{sC_2}\right) - R_1^2} \begin{bmatrix} -\dot{V}_S \\ 0 \end{bmatrix}$$

$$\dot{V}_o = R_2 \dot{I}_{l2} = \frac{R_2 R_1 \dot{V}_S}{R_1 R_2 + R_1 \frac{1}{sC_2} + R_1 \frac{1}{sC_1} + R_2 \frac{1}{sC_1} + \frac{1}{sC_1} \frac{1}{sC_2}}$$

$$= \frac{s^2 C_1 C_2 R_1 R_2 \dot{V}_S}{s^2 C_1 C_2 R_1 R_2 + s C_1 R_1 + s C_2 R_1 + s C_2 R_2 + 1}$$

$$= \frac{s^2}{s^2 + s \left(\frac{1}{R_1 C_1} + \frac{1}{R_2 C_2} + \frac{1}{R_2 C_1} \right) + \frac{1}{R_1 C_1 R_2 C_2}} \dot{V}_S = \frac{s^2}{s^2 + 2\xi \omega_0 s + \omega_0^2} \dot{V}_S$$

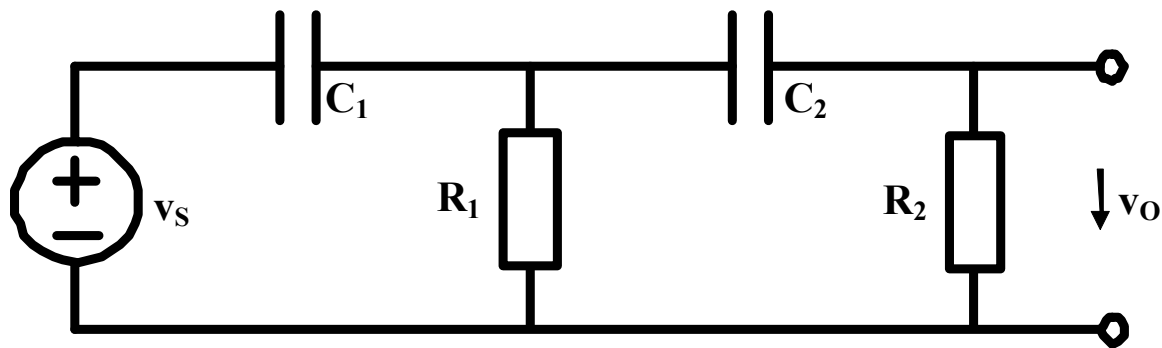
网络参量法



$$\begin{aligned}
 \begin{bmatrix} A & B \\ C & D \end{bmatrix} &= \begin{bmatrix} 1 & \frac{1}{sC_1} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{R_1} & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & \frac{1}{sC_2} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{R_2} & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 + \frac{1}{sR_1C_1} & \frac{1}{sC_1} \\ \frac{1}{R_1} & 1 \end{bmatrix} \begin{bmatrix} 1 + \frac{1}{sR_2C_2} & \frac{1}{sC_2} \\ \frac{1}{R_2} & 1 \end{bmatrix} \\
 &= \begin{bmatrix} \left(1 + \frac{1}{sR_1C_1}\right)\left(1 + \frac{1}{sR_2C_2}\right) + \frac{1}{sR_2C_1} & \dots \\ \dots & \dots \end{bmatrix}
 \end{aligned}$$

$$\begin{aligned}
 H(s) &= \frac{\dot{V}_o}{\dot{V}_s} = \frac{1}{A} \\
 &= \frac{1}{\left(1 + \frac{1}{sR_1C_1}\right)\left(1 + \frac{1}{sR_2C_2}\right) + \frac{1}{sR_2C_1}} = \frac{1}{1 + \frac{1}{s}\left(\frac{1}{R_1C_1} + \frac{1}{R_2C_2} + \frac{1}{R_2C_1}\right) + \frac{1}{s^2}\frac{1}{R_1C_1R_2C_2}} \\
 &= \frac{1}{s^2 + s\left(\frac{1}{R_1C_1} + \frac{1}{R_2C_2} + \frac{1}{R_2C_1}\right) + \frac{1}{R_1C_1R_2C_2}} = \frac{1}{s^2 + 2\xi\omega_0s + \omega_0^2}
 \end{aligned}$$

针对LTI系统，从频域转时域是简单的



$$H(s) \stackrel{s=j\omega}{\equiv} \frac{\dot{V}_o}{\dot{V}_s} = \frac{s^2 R_1 C_1 R_2 C_2}{1 + s(R_1 C_1 + R_1 C_2 + R_2 C_2) + s^2 R_1 R_2 C_1 C_2} = \frac{s^2}{s^2 + 2\xi\omega_0 s + \omega_0^2}$$

$$(1 + s(R_1 C_1 + R_1 C_2 + R_2 C_2) + s^2 R_1 R_2 C_1 C_2) \dot{V}_o = s^2 R_1 C_1 R_2 C_2 \dot{V}_s$$

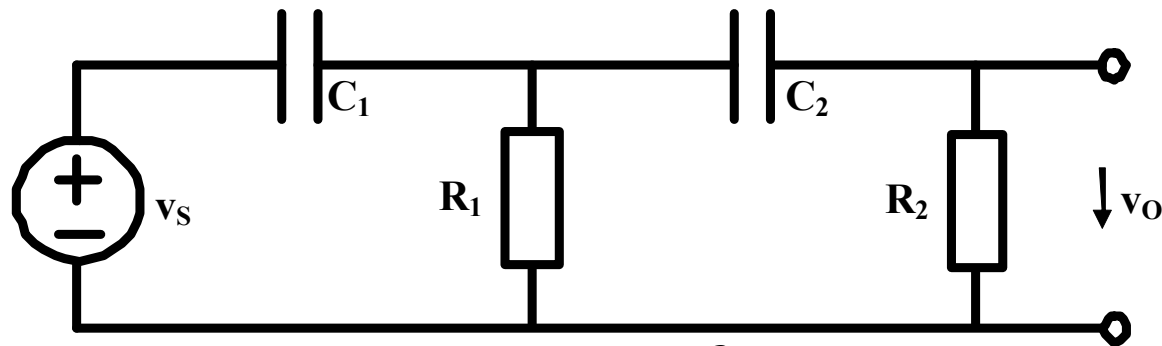
$$R_1 R_2 C_1 C_2 s^2 \dot{V}_o + (R_1 C_1 + R_1 C_2 + R_2 C_2) s \dot{V}_o + \dot{V}_o = R_1 C_1 R_2 C_2 s^2 \dot{V}_s$$

$$R_1 R_2 C_1 C_2 \frac{d^2}{dt^2} v_o(t) + (R_1 C_1 + R_1 C_2 + R_2 C_2) \frac{d}{dt} v_o(t) + v_o(t) = R_1 C_1 R_2 C_2 \frac{d^2}{dt^2} v_s(t)$$

$$\frac{d^2}{dt^2} v_o(t) + \frac{R_1 C_1 + R_1 C_2 + R_2 C_2}{R_1 R_2 C_1 C_2} \frac{d}{dt} v_o(t) + \frac{1}{R_1 R_2 C_1 C_2} v_o(t) = \frac{d^2}{dt^2} v_s(t)$$

$$\frac{d^2}{dt^2} v_o(t) + 2\xi\omega_0 \frac{d}{dt} v_o(t) + \omega_0^2 v_o(t) = \frac{d^2}{dt^2} v_s(t)$$

系统参量



$$H(s) = \frac{\dot{V}_o}{\dot{V}_s} = \frac{s^2}{s^2 + \left(\frac{1}{R_1 C_1} + \frac{1}{R_2 C_1} + \frac{1}{R_2 C_2} \right) s + \frac{1}{R_1 C_1 R_2 C_2}} = H_0 \frac{s^2}{s^2 + 2\xi\omega_0 s + \omega_0^2}$$

$$\frac{d^2}{dt^2} v_o(t) + \left(\frac{1}{R_2 C_1} + \frac{1}{R_1 C_1} + \frac{1}{R_2 C_2} \right) \frac{d}{dt} v_o(t) + \frac{1}{R_1 R_2 C_1 C_2} v_o(t) = \frac{d^2}{dt^2} v_s(t)$$

$$\frac{d^2}{dt^2} v_o(t) + 2\xi\omega_0 \frac{d}{dt} v_o(t) + \omega_0^2 v_o(t) = \frac{d^2}{dt^2} v_s(t)$$

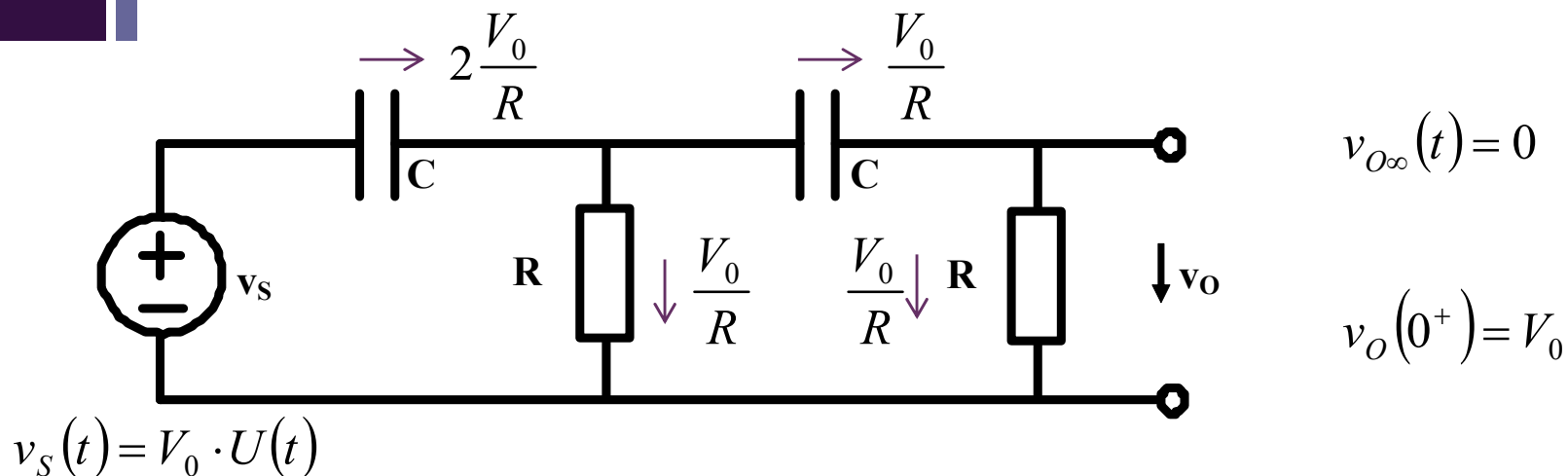
电路中只有线性**RC**（无源元件）的二阶动态电路只能是过阻尼的。要想实现欠阻尼，电路中需要**LC**：集成电路中电感不易集成，需要等效电感、等效负阻（含正反馈环路的受控源元件）以实现欠阻尼

$$H_0 = 1$$

$$\omega_0 = \frac{1}{\sqrt{R_1 C_1 R_2 C_2}}$$

$$\xi = 0.5 \left(\sqrt{\frac{R_2 C_2}{R_1 C_1}} + \sqrt{\frac{R_1 C_1}{R_2 C_2}} + \sqrt{\frac{R_1 C_2}{R_2 C_1}} \right) \geq 0.5 \left(2 + \sqrt{\frac{R_1 C_2}{R_2 C_1}} \right) = 1 + 0.5 \sqrt{\frac{R_1 C_2}{R_2 C_1}} > 1$$

五要素法求解： $R_1=R_2=R, C_1=C_2=C$



$$\omega_0 = \frac{1}{\sqrt{R_1 C_1 R_2 C_2}} = \frac{1}{RC}$$

$$\xi = 0.5 \left(\sqrt{\frac{R_2 C_2}{R_1 C_1}} + \sqrt{\frac{R_1 C_1}{R_2 C_2}} + \sqrt{\frac{R_1 C_2}{R_2 C_1}} \right) = 1.5$$

$$\begin{aligned}
 \frac{d}{dt} v_O(0^+) &= \frac{d}{dt} v_s(0^+) - \frac{d}{dt} v_{C1}(0^+) - \frac{d}{dt} v_{C2}(0^+) \\
 &= 0 - \frac{1}{C_1} i_{C1}(0^+) - \frac{1}{C_2} i_{C2}(0^+) \\
 &= -\frac{1}{C} 2 \frac{V_0}{R} - \frac{1}{C} \frac{V_0}{R} = -3 \frac{V_0}{RC} = -3 \omega_0 V_0
 \end{aligned}$$

$$\begin{aligned}
 v_O(t) &= v_{O\infty}(t) + (V_{O0} - V_{O\infty}) e^{-\xi \omega_0 t} \cosh \sqrt{\xi^2 - 1} \omega_0 t + \left(\frac{\dot{V}_{O0} - \dot{V}_{O\infty}}{\xi \omega_0} + V_{O0} - V_{O\infty} \right) \frac{\xi}{\sqrt{\xi^2 - 1}} e^{-\xi \omega_0 t} \sinh \sqrt{\xi^2 - 1} \omega_0 t \\
 &= V_0 e^{-\xi \omega_0 t} \cosh \sqrt{\xi^2 - 1} \omega_0 t - V_0 \frac{\xi}{\sqrt{\xi^2 - 1}} e^{-\xi \omega_0 t} \sinh \sqrt{\xi^2 - 1} \omega_0 t
 \end{aligned}$$

阶跃响应

过阻尼：指数衰减形态

$$\begin{aligned}
 v_o(t) &= V_0 e^{-\xi \omega_0 t} \cosh \sqrt{\xi^2 - 1} \omega_0 t - V_0 \frac{\xi}{\sqrt{\xi^2 - 1}} e^{-\xi \omega_0 t} \sinh \sqrt{\xi^2 - 1} \omega_0 t \\
 &= V_0 e^{-\xi \omega_0 t} \frac{e^{\sqrt{\xi^2 - 1} \omega_0 t} + e^{-\sqrt{\xi^2 - 1} \omega_0 t}}{2} - V_0 \frac{\xi}{\sqrt{\xi^2 - 1}} e^{-\xi \omega_0 t} \frac{e^{\sqrt{\xi^2 - 1} \omega_0 t} - e^{-\sqrt{\xi^2 - 1} \omega_0 t}}{2} \\
 &= V_0 \frac{1}{2} \left(1 - \frac{\xi}{\sqrt{\xi^2 - 1}} \right) e^{-\xi \omega_0 t + \sqrt{\xi^2 - 1} \omega_0 t} + V_0 \frac{1}{2} \left(1 + \frac{\xi}{\sqrt{\xi^2 - 1}} \right) e^{-\xi \omega_0 t - \sqrt{\xi^2 - 1} \omega_0 t} \\
 &= \frac{1}{2} \left(1 - \frac{\xi}{\sqrt{\xi^2 - 1}} \right) V_0 e^{-(\xi - \sqrt{\xi^2 - 1}) \omega_0 t} + \frac{1}{2} \left(1 + \frac{\xi}{\sqrt{\xi^2 - 1}} \right) V_0 e^{-(\xi + \sqrt{\xi^2 - 1}) \omega_0 t} \\
 &= -0.1708 V_0 e^{-\frac{t}{2.618RC}} + 1.1708 V_0 e^{-\frac{t}{0.382RC}}
 \end{aligned}$$

长寿命项

短寿命项

$$\begin{aligned}
 \omega_0 &= \frac{1}{RC} \\
 \xi &= 1.5
 \end{aligned}$$

$$\tau_1 = \frac{\xi + \sqrt{\xi^2 - 1}}{\omega_0} = 2.618RC$$

$$\tau_2 = \frac{\xi - \sqrt{\xi^2 - 1}}{\omega_0} = \frac{1}{(\xi + \sqrt{\xi^2 - 1}) \omega_0} = 0.382RC$$

时频对应

$$H(s) = \frac{s^2}{s^2 + 2\xi\omega_0 s + \omega_0^2} = \frac{s^2}{s^2 + 3\omega_0 s + \omega_0^2}$$

$$H(s) \leftrightarrow h(t)$$

$$G(s) = \frac{1}{s} H(s)$$

$$g(t) = \int_{-\infty}^t h(\gamma) d\gamma$$

$$G(s) = \frac{1}{s} H(s) = \frac{s}{s^2 + 3\omega_0 s + \omega_0^2} = \frac{s}{\left(s + \frac{3 - \sqrt{5}}{2} \omega_0\right) \left(s + \frac{3 + \sqrt{5}}{2} \omega_0\right)}$$

$$= \frac{a_1}{s + \frac{3 - \sqrt{5}}{2} \omega_0} + \frac{a_2}{s + \frac{3 + \sqrt{5}}{2} \omega_0} = \frac{a_1}{s + \frac{1}{2.618RC}} + \frac{a_2}{s + \frac{1}{0.382RC}}$$

$$a_1 + a_2 = 1$$

$$a_1 \frac{3 + \sqrt{5}}{2} \omega_0 + a_2 \frac{3 - \sqrt{5}}{2} \omega_0 = 0$$

$$a_1 = \frac{5 - 3\sqrt{5}}{10} = -0.1708$$

$$a_2 = \frac{5 + 3\sqrt{5}}{10} = 1.1708$$

$$= \frac{a_1 \left(s + \frac{3 + \sqrt{5}}{2} \omega_0\right) + a_2 \left(s + \frac{3 - \sqrt{5}}{2} \omega_0\right)}{\left(s + \frac{3 - \sqrt{5}}{2} \omega_0\right) \left(s + \frac{3 + \sqrt{5}}{2} \omega_0\right)}$$

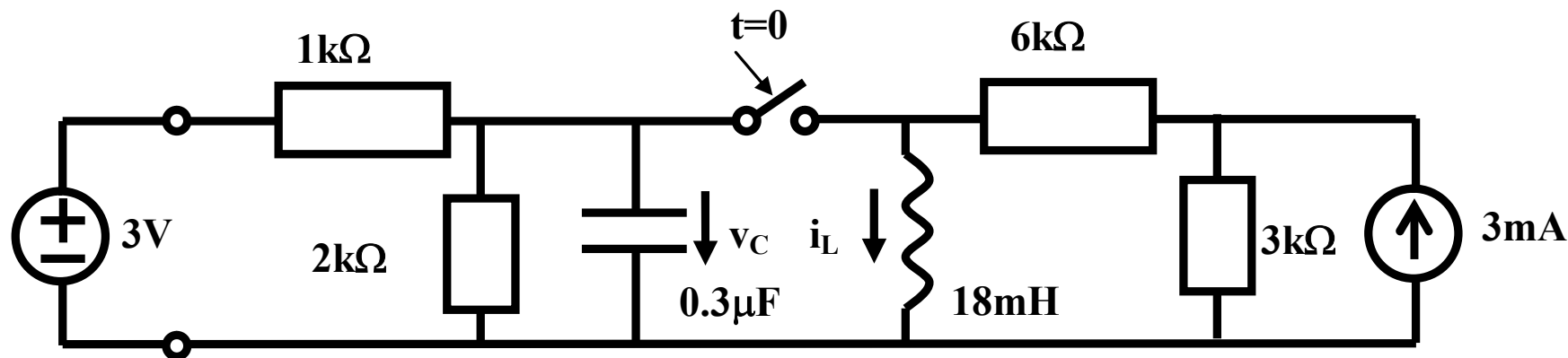
$$= \frac{(a_1 + a_2)s + a_1 \frac{3 + \sqrt{5}}{2} \omega_0 + a_2 \frac{3 - \sqrt{5}}{2} \omega_0}{\left(s + \frac{3 - \sqrt{5}}{2} \omega_0\right) \left(s + \frac{3 + \sqrt{5}}{2} \omega_0\right)}$$

$$v_o(t) = V_0 g(t)$$

$$= -0.1708 V_0 e^{-\frac{t}{2.618RC}} U(t)$$

$$+ 1.1708 V_0 e^{-\frac{t}{0.382RC}} U(t)$$

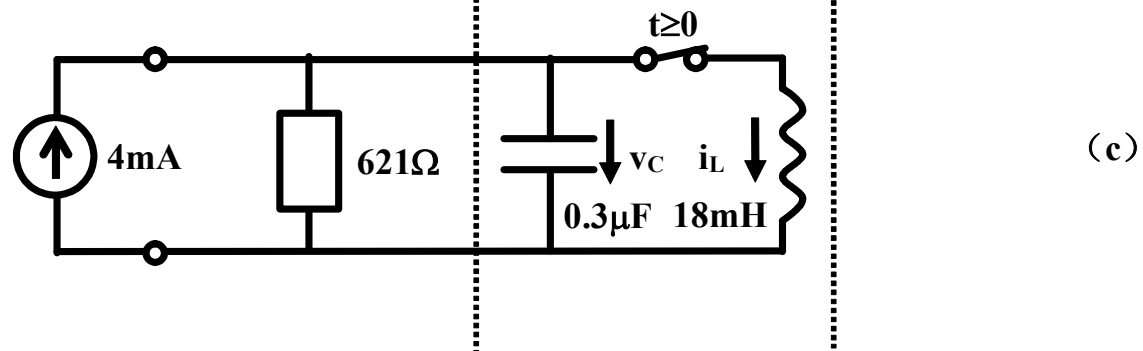
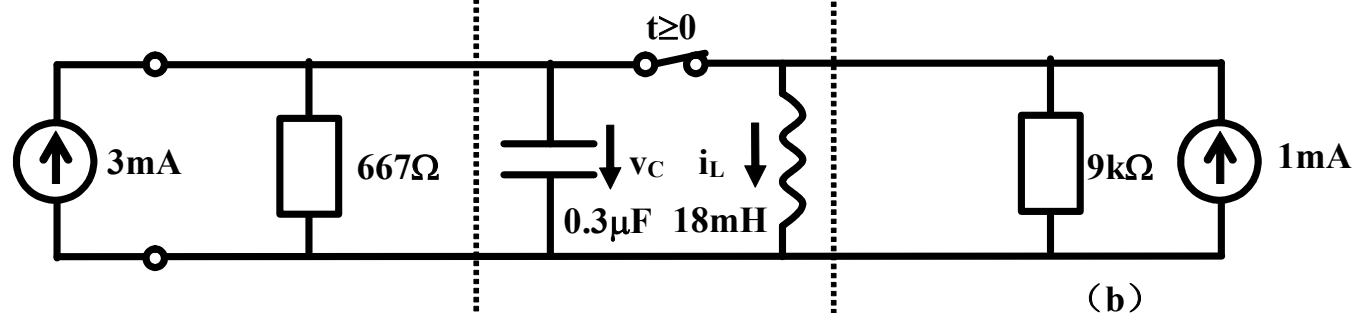
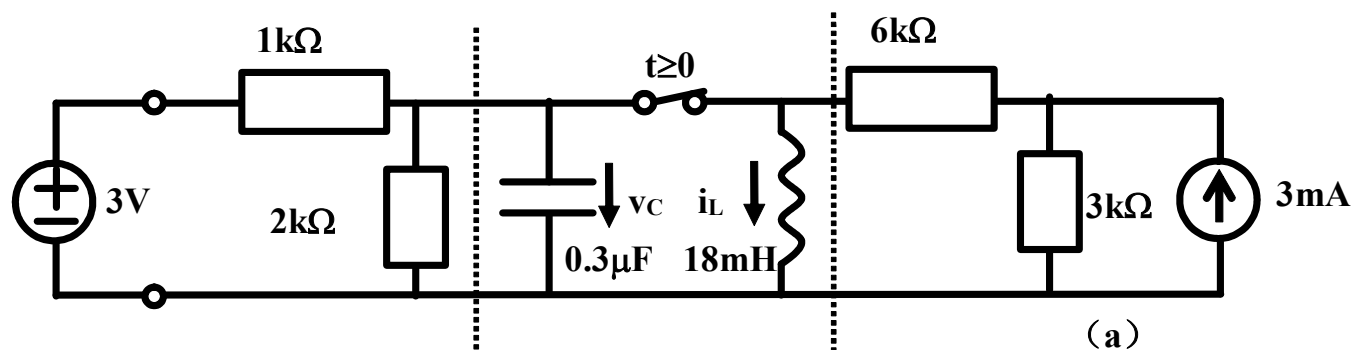
作业5.4 分析与验证



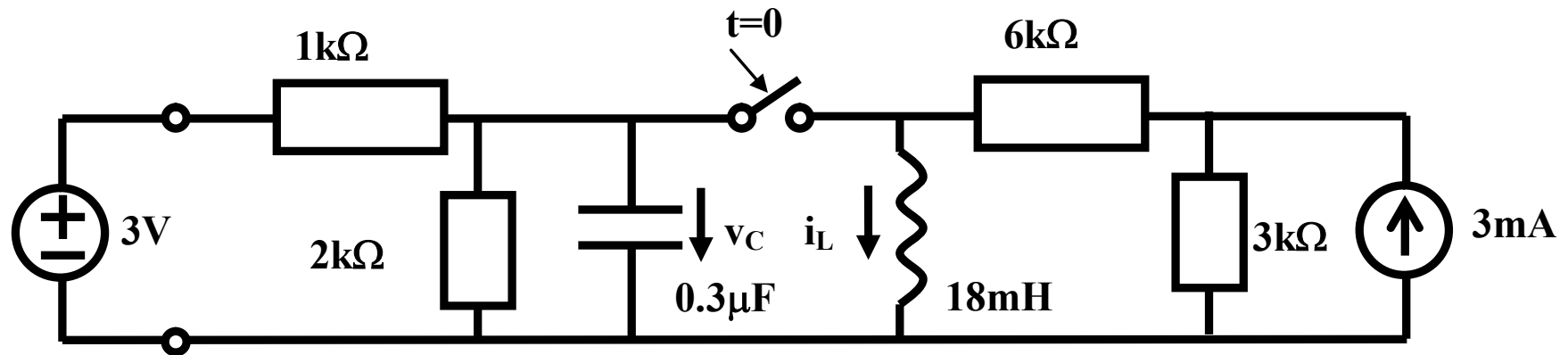
- 开关在 $t=0$ 时刻闭合。开关闭合前电路已经稳定。求开关闭合后，电容电压 $v_C(t)$ 和电感电流 $i_L(t)$ 的变化规律
 - 课件已给电容电压 $v_C(t)$ 的变化规律，求 $i_L(t)$ 的变化规律
 - 验证

$$v_C(t) = v_L(t) = L \frac{di_L(t)}{dt} \quad (t \geq 0)$$

开关闭合后，是RLC并联谐振电路

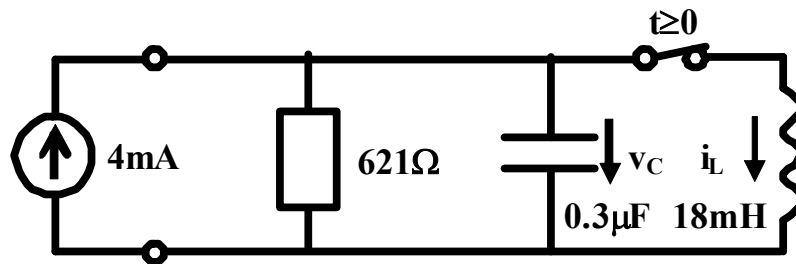


五要素法：阻尼系数，自由振荡频率

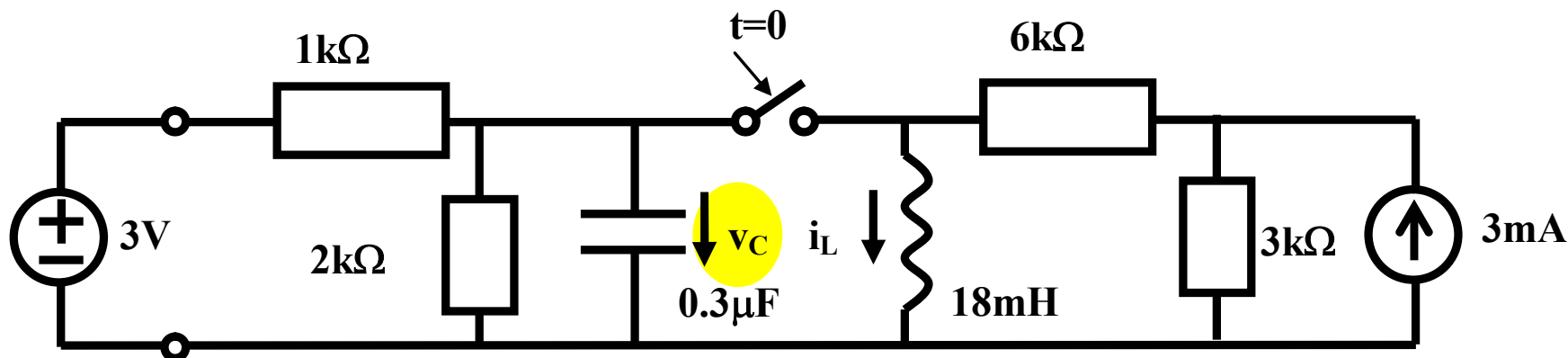
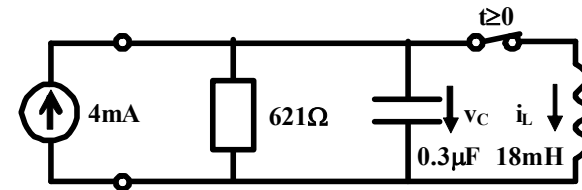


$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{18m \times 0.3\mu}} = 13.6 \times 10^3 \text{ rad/s} \quad f_0 = 2.166 \text{ kHz}$$

$$\xi = \frac{G}{2Y_0} = \frac{G}{2\sqrt{C/L}} = \frac{1}{2R} \sqrt{\frac{L}{C}} = \frac{1}{2 \times 621} \sqrt{\frac{18m}{0.3\mu}} = 0.1973$$



五要素法：两个初值



$$v_C(0^-) = \frac{2k\Omega}{1k\Omega + 2k\Omega} \times 3V = 2V$$

$$v_C(0^+) = v_C(0^-) = 2V$$

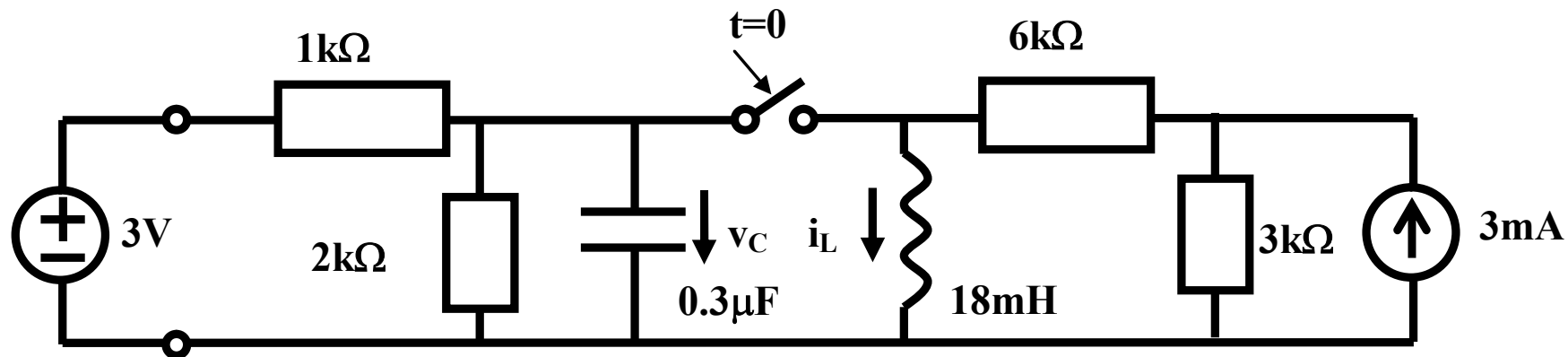
$$i_L(0^-) = \frac{3k\Omega}{6k\Omega + 3k\Omega} \times 3mA = 1mA$$

$$i_L(0^+) = i_L(0^-) = 1mA$$

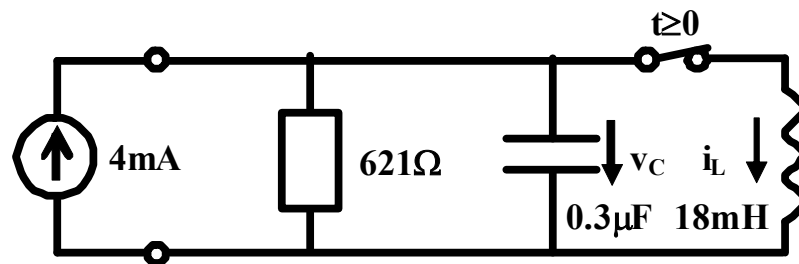
$$\frac{dv_C(0^+)}{dt} = \frac{1}{C} i_C(0^+) = \frac{1}{C} (i_S(0^+) - i_L(0^+) - i_R(0^+)) = \frac{1}{C} \left(4mA - 1mA - \frac{v_C(0^+)}{R} \right)$$

$$= \frac{1}{0.3\mu F} \left(4mA - 1mA - \frac{2V}{621\Omega} \right) = -\frac{0.2222mA}{0.3\mu F} = -0.7407V/ms$$

五要素法：稳态响应



$$v_{C,\infty}(t) = 0 \quad v_{C,\infty}(0^+) = 0 \quad \frac{dv_{C,\infty}(0^+)}{dt} = 0$$



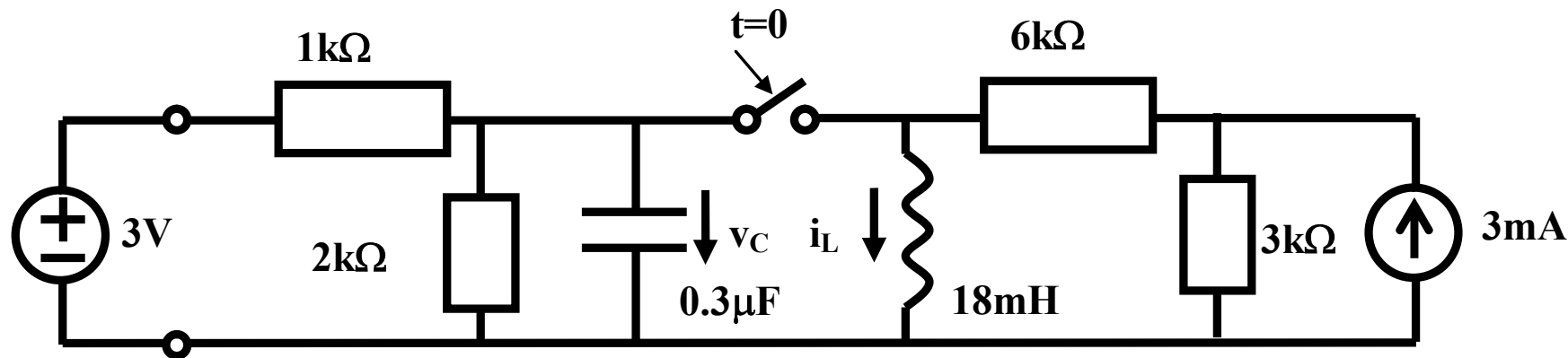
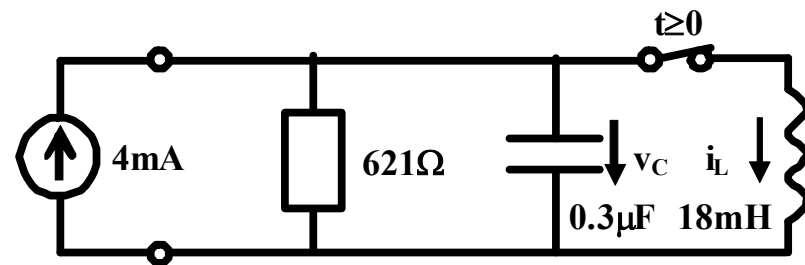
五要素表达式

幅度指数衰减的正弦振荡波形

$$\begin{aligned}
 v_C(t) &= v_{C,\infty}(t) + (V_0 - V_{\infty,0})e^{-\xi\omega_0 t} \cos(\sqrt{1-\xi^2}\omega_0 t) \\
 &+ \left(V_0 - V_{\infty,0} + \frac{\dot{V}_0 - \dot{V}_{\infty,0}}{\xi\omega_0} \right) \frac{\xi}{\sqrt{1-\xi^2}} e^{-\xi\omega_0 t} \sin(\sqrt{1-\xi^2}\omega_0 t) \\
 &= 0 + (2 - 0)e^{-\xi\omega_0 t} \cos(\sqrt{1-\xi^2}\omega_0 t) \\
 &+ \left(2 - 0 + \frac{-0.7407 \times 10^3 - 0}{0.1973 \times 13.6 \times 10^3} \right) \frac{\xi}{\sqrt{1-\xi^2}} e^{-\xi\omega_0 t} \sin(\sqrt{1-\xi^2}\omega_0 t) \\
 &= 2e^{-\xi\omega_0 t} \cos(\sqrt{1-\xi^2}\omega_0 t) + 1.7241 \frac{\xi}{\sqrt{1-\xi^2}} e^{-\xi\omega_0 t} \sin(\sqrt{1-\xi^2}\omega_0 t) \\
 &= 2e^{-\frac{t}{0.3724 \times 10^{-3}}} \cos(13.34 \times 10^3 t) + 0.347e^{-\frac{t}{0.3724 \times 10^{-3}}} \sin(13.34 \times 10^3 t) \\
 &= 2.03e^{-\frac{t}{0.3724 \times 10^{-3}}} \sin(13.34 \times 10^3 t + 1.4)
 \end{aligned}$$

单位：伏特
 $t \geq 0$

电感电流五要素



$$i_L(0^+) = i_L(0^-) = 1mA$$

$$\frac{d}{dt}i_L(0^+) = \frac{1}{L}v_L(0^+) = \frac{v_C(0^+)}{L} = \frac{2V}{18mH} = 111.1 A/s$$

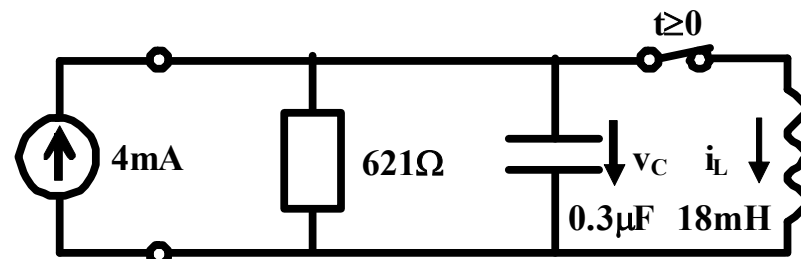
$$i_{L\infty}(t) = 4mA$$

电感电流

$$\begin{aligned}
 i_L(t) &= i_{L,\infty}(t) + (I_0 - I_{\infty,0})e^{-\xi\omega_0 t} \cos\left(\sqrt{1-\xi^2}\omega_0 t\right) \\
 &+ \left(I_0 - I_{\infty,0} + \frac{\dot{I}_0 - \dot{I}_{\infty,0}}{\xi\omega_0}\right) \frac{\xi}{\sqrt{1-\xi^2}} e^{-\xi\omega_0 t} \sin\left(\sqrt{1-\xi^2}\omega_0 t\right) \\
 &= 4 + (1-4)e^{-\xi\omega_0 t} \cos\left(\sqrt{1-\xi^2}\omega_0 t\right) \\
 &+ \left(1-4 + \frac{111.1 \times 10^3 - 0}{0.1973 \times 13.6 \times 10^3}\right) \frac{\xi}{\sqrt{1-\xi^2}} e^{-\xi\omega_0 t} \sin\left(\sqrt{1-\xi^2}\omega_0 t\right) \\
 &= 4 - 3e^{-\xi\omega_0 t} \cos\left(\sqrt{1-\xi^2}\omega_0 t\right) + 38.41 \frac{\xi}{\sqrt{1-\xi^2}} e^{-\xi\omega_0 t} \sin\left(\sqrt{1-\xi^2}\omega_0 t\right) \\
 &= 4 - 3e^{-\frac{t}{0.3724 \times 10^{-3}}} \cos(13.34 \times 10^3 t) + 7.73e^{-\frac{t}{0.3724 \times 10^{-3}}} \sin(13.34 \times 10^3 t) \\
 &= 4 + 8.29e^{-\frac{t}{0.3724 \times 10^{-3}}} \sin(13.34 \times 10^3 t - 0.37)
 \end{aligned}$$

单位：毫安
 $t \geq 0$

验证



$$v_C(t) = 2.03e^{-\frac{t}{0.3724 \times 10^{-3}}} \sin(13.34 \times 10^3 t + 1.4)$$

单位：伏特

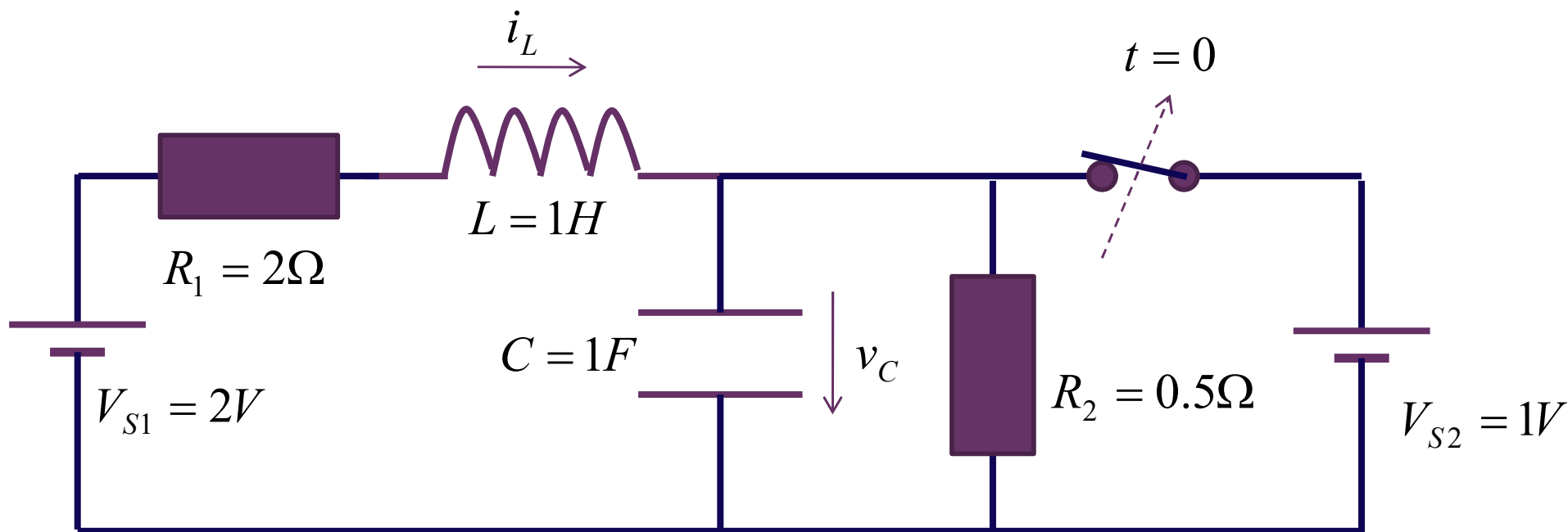
$$i_L(t) = 4 + 8.29e^{-\frac{t}{0.3724 \times 10^{-3}}} \sin(13.34 \times 10^3 t - 0.37)$$

单位：毫安

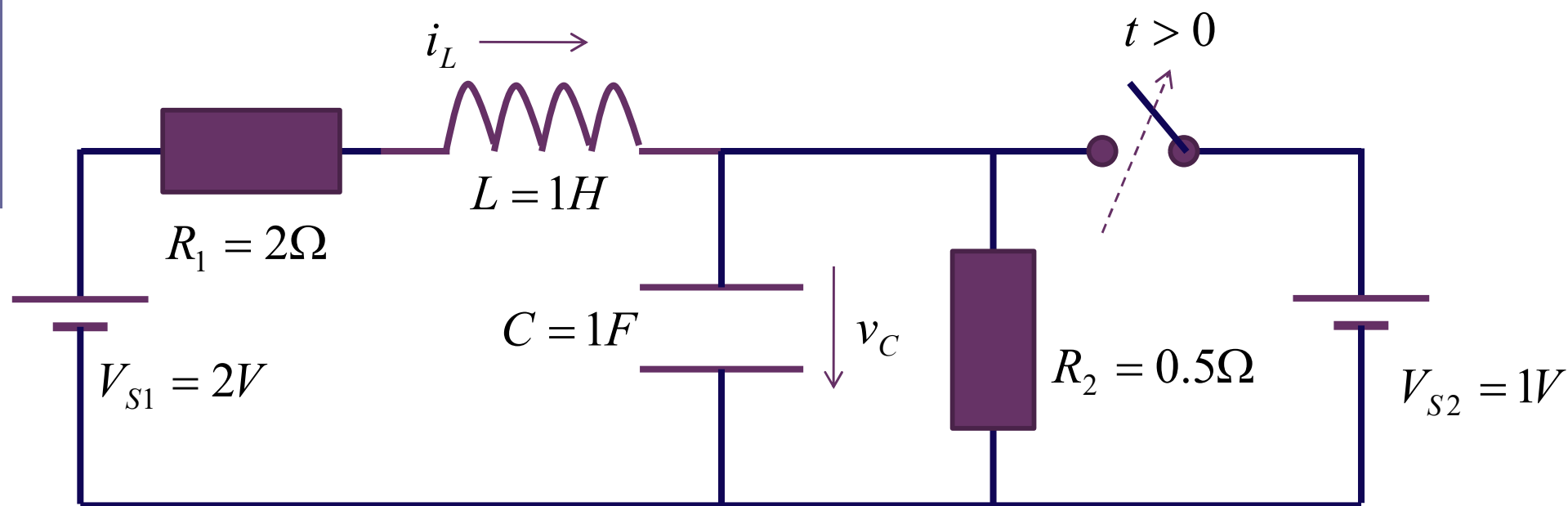
$$\begin{aligned} v_L(t) &= L \frac{di_L(t)}{dt} = 18 \times 10^{-3} \times \frac{d}{dt} \left(4 + 8.29e^{-\frac{t}{0.3724 \times 10^{-3}}} \sin(13.34 \times 10^3 t - 0.37) \right) \times 10^{-3} \\ &= 18 \times 10^{-6} \times \left(-\frac{8.29}{0.3724 \times 10^{-3}} e^{-\frac{t}{0.3724 \times 10^{-3}}} \sin(13.34 \times 10^3 t - 0.37) \right. \\ &\quad \left. + 8.29e^{-\frac{t}{0.3724 \times 10^{-3}}} \times 13.34 \times 10^3 \cos(13.34 \times 10^3 t - 0.37) \right) \\ &= 18 \times 10^{-3} \times e^{-\frac{t}{0.3724 \times 10^{-3}}} \left(-22.26 \sin(13.34 \times 10^3 t - 0.37) + 110.6 \cos(13.34 \times 10^3 t - 0.37) \right) \\ &= 18 \times 10^{-3} \times e^{-\frac{t}{0.3724 \times 10^{-3}}} \times 112.8 \times \sin \left(13.34 \times 10^3 t - 0.37 + 3.14 - \arctan \frac{110.6}{22.26} \right) \\ &= 2.03e^{-\frac{t}{0.3724 \times 10^{-3}}} \sin(13.34 \times 10^3 t - 0.37 + 3.14 - 1.37) = 2.03e^{-\frac{t}{0.3724 \times 10^{-3}}} \sin(13.34 \times 10^3 t + 1.4) = v_C(t) \end{aligned}$$

作业5.5 非简单串并联电路

需要首先确定系统参量



- 1/用五要素法或待定系数法获得电容电压和电感电流
- 2/选作：利用状态方程求解，求A矩阵特征根，用待定系数法求解



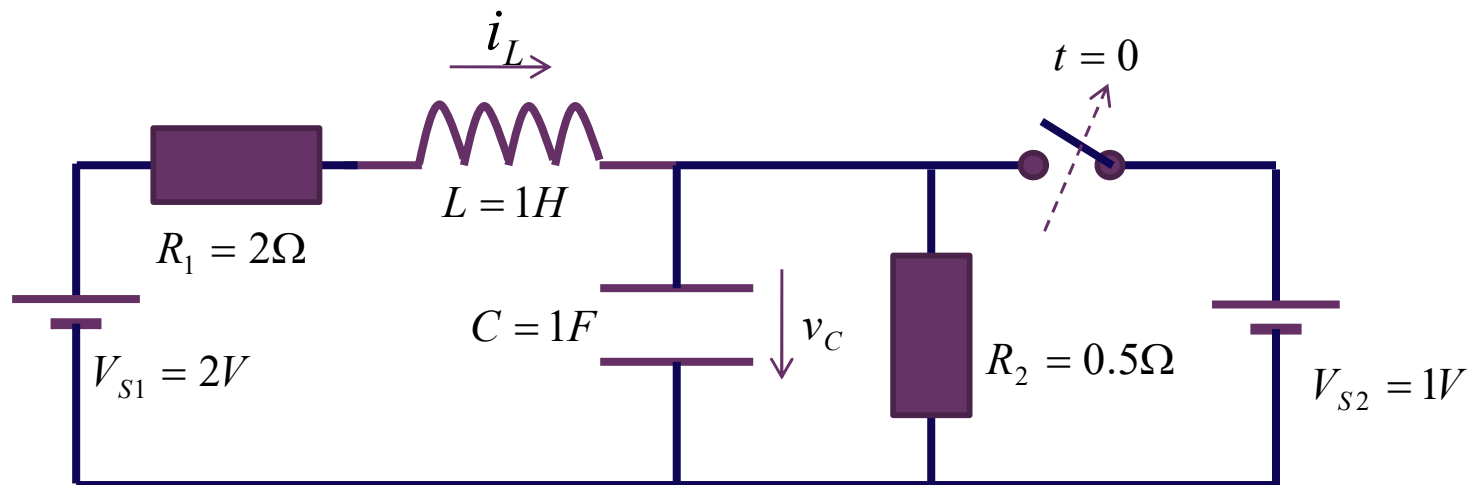
$$\frac{\dot{V}_C}{\dot{V}_{S1}} = \frac{R_2 \parallel \frac{1}{sC}}{R_1 + sL + R_2 \parallel \frac{1}{sC}} = \frac{\frac{R_2}{1 + sR_2C}}{R_1 + sL + \frac{R_2}{1 + sR_2C}} = \frac{1}{1 + \frac{R_1}{R_2} + s\left(\frac{L}{R_2} + R_1C\right) + s^2LC}$$

$$= \frac{R_2}{R_2 + R_1} \frac{1}{1 + s\left(\frac{L}{R_2 + R_1} + \frac{R_2R_1}{R_2 + R_1}C\right) + s^2LC} \frac{R_2}{R_2 + R_1} = H_0 \frac{1}{1 + 2\xi \frac{s}{\omega_n} + \left(\frac{s}{\omega_n}\right)^2}$$

$$H_0 = \frac{R_2}{R_2 + R_1} = \frac{0.5}{2 + 0.5} = 0.2 \quad \omega_n = \frac{1}{\sqrt{LC \frac{R_2}{R_2 + R_1}}} = \frac{1}{\sqrt{1 \times 1 \times 0.2}} = \sqrt{5}$$

$$\xi = 0.5 \sqrt{\frac{R_2}{R_2 + R_1} \left(\frac{Z_0}{R_2} + \frac{R_1}{Z_0}\right)} = 0.5\sqrt{0.2} \left(\frac{1}{0.5} + \frac{2}{1}\right) = \frac{2}{\sqrt{5}}$$

初值

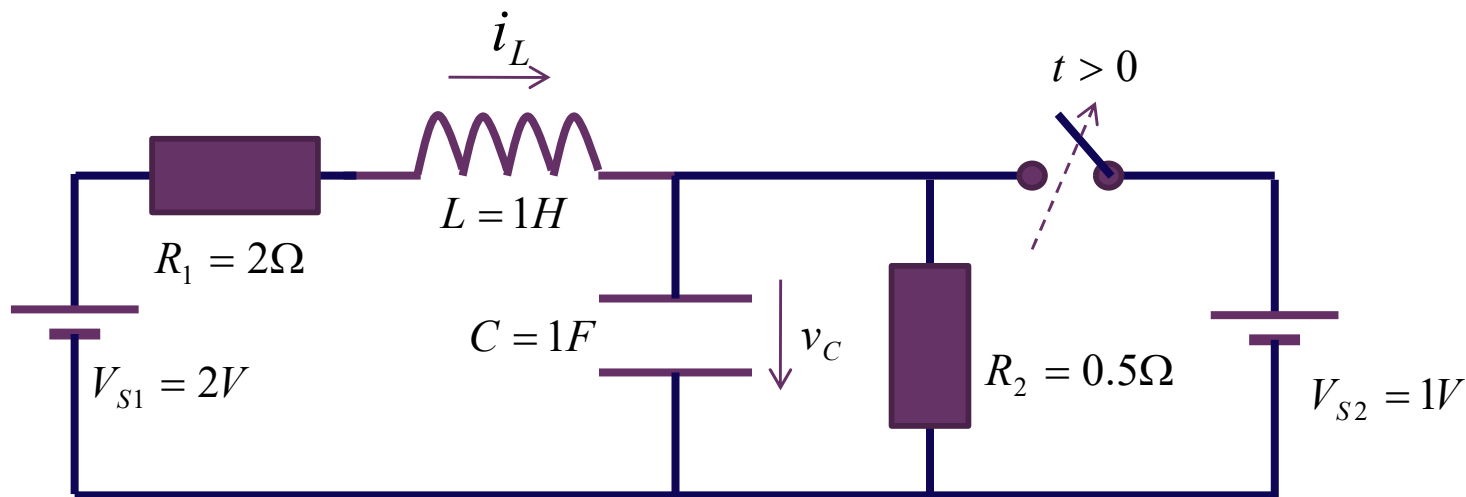


$$v_C(0^+) = v_C(0^-) = V_{S2} = 1V \quad i_L(0^+) = i_L(0^-) = \frac{V_{S1} - V_{S2}}{R_1} = \frac{2-1}{2} = 0.5A$$

$$\frac{d}{dt}v_C(0^+) = \frac{i_C(0^+)}{C} = \frac{i_L(0^+) - i_{R2}(0^+)}{C} = \frac{i_L(0^+)}{C} - \frac{v_C(0^+)}{R_2 C} = \frac{0.5}{1} - \frac{1}{0.5 \times 1} = -1.5V/s$$

$$\begin{aligned} \frac{d}{dt}i_L(0^+) &= \frac{v_L(0^+)}{L} = \frac{v_S(0^+) - v_{R1}(0^+) - v_C(0^+)}{L} = \frac{v_S(0^+) - R_1 i_L(0^+) - v_C(0^+)}{L} \\ &= \frac{2 - 2 \times 0.5 - 1}{1} = 0 \end{aligned}$$

稳态响应



$$v_{C\infty}(t) = \frac{R_2}{R_1 + R_2} V_{S1} = \frac{0.5}{2 + 0.5} \times 2 = 0.4V$$

$$i_{L\infty}(t) = \frac{V_{S1}}{R_1 + R_2} = \frac{2}{2 + 0.5} = 0.8A$$

电容电压

$$\omega_n = \sqrt{5} \text{ s}^{-1}$$

$$\xi = \frac{2}{\sqrt{5}}$$

$$v_{C\infty}(t) = 0.4V$$

$$v_C(0^+) = 1V$$

$$\frac{d}{dt}v_C(0^+) = -1.5V/s$$

$$\begin{aligned} v_C(t) &= v_{C\infty}(t) + (V_0 - V_{\infty,0})e^{-\xi\omega_0 t} \cos(\sqrt{1-\xi^2}\omega_0 t) \\ &\quad + \left(V_0 - V_{\infty,0} + \frac{\dot{V}_0 - \dot{V}_{\infty,0}}{\xi\omega_0} \right) \frac{\xi}{\sqrt{1-\xi^2}} e^{-\xi\omega_0 t} \sin(\sqrt{1-\xi^2}\omega_0 t) \\ &= 0.4 + (1 - 0.4)e^{-\xi\omega_0 t} \cos(\sqrt{1-\xi^2}\omega_0 t) \\ &\quad + \left(1 - 0.4 + \frac{-1.5 - 0}{\frac{2}{\sqrt{5}} \times \sqrt{5}} \right) \frac{\frac{2}{\sqrt{5}}}{\sqrt{1-\frac{4}{5}}} e^{-\xi\omega_0 t} \sin(\sqrt{1-\xi^2}\omega_0 t) \\ &= 0.4 + 0.6e^{-\xi\omega_0 t} \cos(\sqrt{1-\xi^2}\omega_0 t) - 0.3e^{-\xi\omega_0 t} \sin(\sqrt{1-\xi^2}\omega_0 t) \\ &= 0.4 + 0.6e^{-2t} \cos t - 0.3e^{-2t} \sin t \quad (t \geq 0) \end{aligned}$$

电感电流

$$\omega_n = \sqrt{5} \text{ s}^{-1}$$

$$\xi = \frac{2}{\sqrt{5}}$$

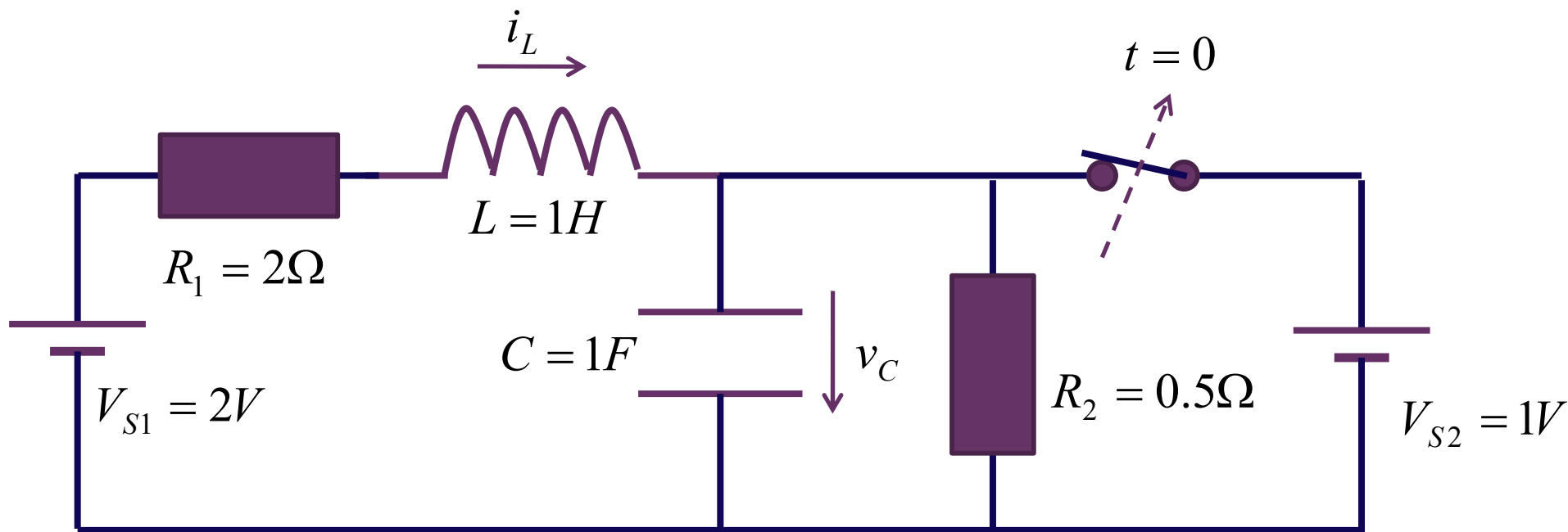
$$i_{L\infty}(t) = 0.8 \text{ A}$$

$$i_L(0^+) = 0.5 \text{ A}$$

$$\frac{d}{dt}i_L(0^+) = 0$$

$$\begin{aligned} i_L(t) &= i_{L,\infty}(t) + (I_0 - I_{\infty,0})e^{-\xi\omega_0 t} \cos\left(\sqrt{1-\xi^2}\omega_0 t\right) \\ &\quad + \left(I_0 - I_{\infty,0} + \frac{\dot{I}_0 - \dot{I}_{\infty,0}}{\xi\omega_0}\right) \frac{\xi}{\sqrt{1-\xi^2}} e^{-\xi\omega_0 t} \sin\left(\sqrt{1-\xi^2}\omega_0 t\right) \\ &= 0.8 + (0.5 - 0.8)e^{-\xi\omega_0 t} \cos\left(\sqrt{1-\xi^2}\omega_0 t\right) \\ &\quad + \left(0.5 - 0.8 + \frac{-0 - 0}{\frac{2}{\sqrt{5}} \times \sqrt{5}}\right) \frac{\frac{2}{\sqrt{5}}}{\sqrt{1-\frac{4}{5}}} e^{-\xi\omega_0 t} \sin\left(\sqrt{1-\xi^2}\omega_0 t\right) \\ &= 0.8 - 0.3e^{-\xi\omega_0 t} \cos\left(\sqrt{1-\xi^2}\omega_0 t\right) - 0.6e^{-\xi\omega_0 t} \sin\left(\sqrt{1-\xi^2}\omega_0 t\right) \\ &= 0.8 - 0.3e^{-2t} \cos t - 0.6e^{-2t} \sin t \quad (t \geq 0) \end{aligned}$$

方法2：利用状态方程求解



$$\mathbf{x}(t) = \begin{bmatrix} v_C(t) \\ i_L(t) \end{bmatrix}$$

状态变量

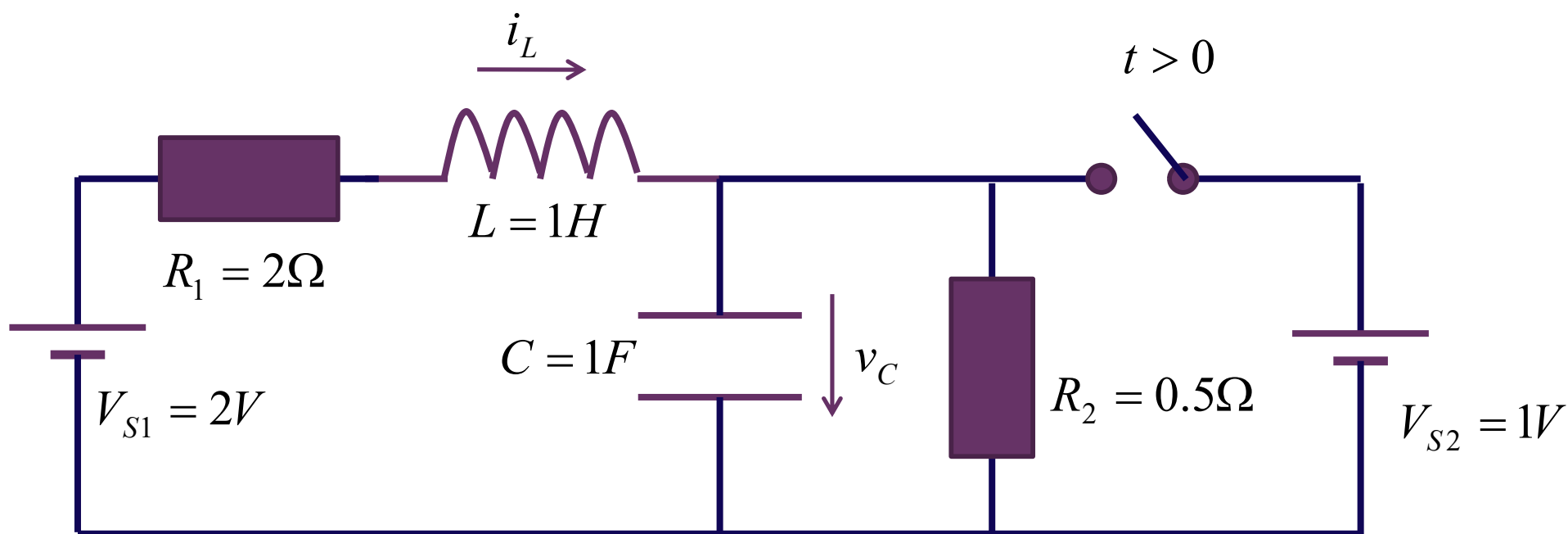
$$\mathbf{x}_0 = \begin{bmatrix} v_C(0^+) \\ i_L(0^+) \end{bmatrix} = \begin{bmatrix} 1V \\ 0.5A \end{bmatrix}$$

状态初值

$$\mathbf{x}_\infty(t) = \begin{bmatrix} v_C(t) \\ i_L(t) \end{bmatrix} = \begin{bmatrix} 0.4V \\ 0.8A \end{bmatrix}$$

稳态响应

状态方程



$$C \frac{dv_C}{dt} = i_L - \frac{v_C}{R_2}$$

$$L \frac{di_L}{dt} = v_{S1} - i_L R_1 - v_C$$

$$\frac{dv_C}{dt} = -\frac{v_C}{R_2 C} + \frac{i_L}{C}$$

$$\frac{di_L}{dt} = -\frac{v_C}{L} - i_L \frac{R_1}{L} + \frac{1}{L} v_{S1}$$

$$\frac{d}{dt} \begin{bmatrix} v_C \\ i_L \end{bmatrix} = \begin{bmatrix} -\frac{1}{R_2 C} & \frac{1}{C} \\ -\frac{1}{L} & -\frac{R_1}{L} \end{bmatrix} \begin{bmatrix} v_C \\ i_L \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{L} \end{bmatrix} v_{S1} = \begin{bmatrix} -2 & 1 \\ -1 & -2 \end{bmatrix} \begin{bmatrix} v_C \\ i_L \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \end{bmatrix} U(t)$$

特征根

$$\frac{d}{dt} \begin{bmatrix} v_C \\ i_L \end{bmatrix} = \begin{bmatrix} -2 & 1 \\ -1 & -2 \end{bmatrix} \begin{bmatrix} v_C \\ i_L \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \end{bmatrix} U(t)$$

$$\frac{d}{dt} \mathbf{x} = \mathbf{A} \mathbf{x} + \mathbf{g}$$

$$\det(\lambda \mathbf{I} - \mathbf{A}) = 0 \quad \begin{vmatrix} \lambda + 2 & -1 \\ 1 & \lambda + 2 \end{vmatrix} = 0$$

$$(\lambda + 2)^2 + 1 = 0$$

$$\lambda^2 + 4\lambda + 5 = 0$$

$$\lambda_1 = -2 + j$$

$$\lambda_2 = -2 - j$$

$$\lambda^2 + 2\xi\omega_n\lambda + \omega_n^2 = 0$$

$$\omega_n = \sqrt{5}$$

$$\xi = \frac{2}{\sqrt{5}}$$

用任意方法，获得完全相同的系统参量

待定系数法

$$\lambda_{1,2} = -2 \pm j = -\frac{1}{\tau_d} \pm j\omega_d$$

$$\begin{bmatrix} v_C(t) \\ i_L(t) \end{bmatrix} = \begin{bmatrix} v_{C\infty}(t) \\ i_{L\infty}(t) \end{bmatrix} + e^{-2t} \begin{bmatrix} A \cos t + B \sin t \\ C \cos t + D \sin t \end{bmatrix} = \begin{bmatrix} 0.4 \\ 0.8 \end{bmatrix} + e^{-2t} \begin{bmatrix} A \cos t + B \sin t \\ C \cos t + D \sin t \end{bmatrix}$$

$$\mathbf{x}_0 = \begin{bmatrix} v_C(0) \\ i_L(0) \end{bmatrix} = \begin{bmatrix} 1 \\ 0.5 \end{bmatrix} = \begin{bmatrix} 0.4 \\ 0.8 \end{bmatrix} + \begin{bmatrix} A \\ C \end{bmatrix} \quad \begin{bmatrix} A \\ C \end{bmatrix} = \begin{bmatrix} 0.6 \\ -0.3 \end{bmatrix}$$

$$\frac{d}{dt} \begin{bmatrix} v_C(t) \\ i_L(t) \end{bmatrix}_{t=0^+} = \left(-2e^{-2t} \begin{bmatrix} A \cos t + B \sin t \\ C \cos t + D \sin t \end{bmatrix} + e^{-2t} \begin{bmatrix} -A \sin t + B \cos t \\ -C \sin t + D \cos t \end{bmatrix} \right)_{t=0^+} = -2 \begin{bmatrix} A \\ C \end{bmatrix} + \begin{bmatrix} B \\ D \end{bmatrix}$$

$$\frac{d}{dt} \begin{bmatrix} v_C(0^+) \\ i_L(0^+) \end{bmatrix} = \begin{bmatrix} -2 & 1 \\ -1 & -2 \end{bmatrix} \begin{bmatrix} v_C(0^+) \\ i_L(0^+) \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \end{bmatrix} U(0^+) = \begin{bmatrix} -2 & 1 \\ -1 & -2 \end{bmatrix} \begin{bmatrix} 1 \\ 0.5 \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \end{bmatrix} = \begin{bmatrix} -1.5 \\ -2 \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \end{bmatrix} = \begin{bmatrix} -1.5 \\ 0 \end{bmatrix}$$

$$-2 \begin{bmatrix} A \\ C \end{bmatrix} + \begin{bmatrix} B \\ D \end{bmatrix} = \begin{bmatrix} B \\ D \end{bmatrix} - 2 \begin{bmatrix} 0.6 \\ -0.3 \end{bmatrix} = \begin{bmatrix} -1.5 \\ 0 \end{bmatrix} \quad \begin{bmatrix} B \\ D \end{bmatrix} = \begin{bmatrix} -0.3 \\ -0.6 \end{bmatrix}$$

最终结果

$$\begin{bmatrix} v_C(t) \\ i_L(t) \end{bmatrix} = \begin{bmatrix} 0.4 \\ 0.8 \end{bmatrix} + e^{-2t} \begin{bmatrix} A \cos t + B \sin t \\ C \cos t + D \sin t \end{bmatrix}$$

$$\begin{bmatrix} A \\ C \end{bmatrix} = \begin{bmatrix} 0.6 \\ -0.3 \end{bmatrix}$$

从初值获得待定系数

$$\begin{bmatrix} B \\ D \end{bmatrix} = \begin{bmatrix} -0.3 \\ -0.6 \end{bmatrix}$$

$$\begin{bmatrix} v_C(t) \\ i_L(t) \end{bmatrix} = \begin{bmatrix} 0.4 \\ 0.8 \end{bmatrix} + \begin{bmatrix} +0.6e^{-2t} \cos t - 0.3e^{-2t} \sin t \\ -0.3e^{-2t} \cos t - 0.6e^{-2t} \sin t \end{bmatrix}$$

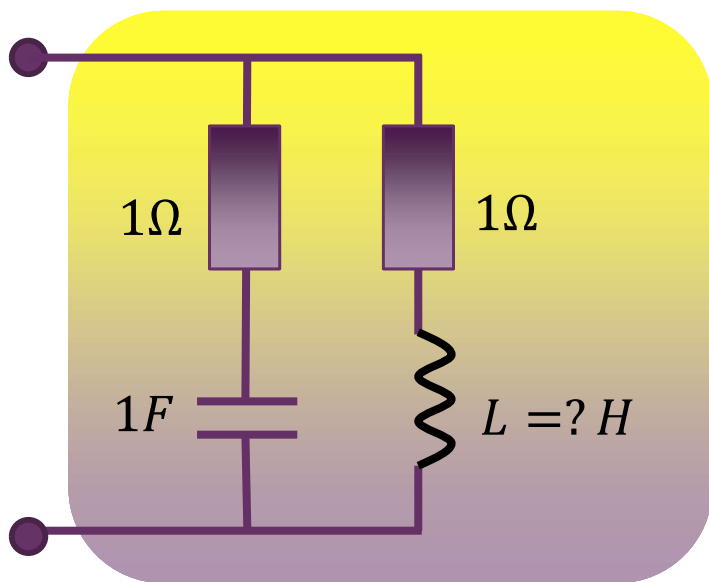
和五要素法完全一致

$(t \geq 0)$

《信号与系统》课程中，可以用拉普拉斯变换，先频域后时域获得同样的解析解

作业3.4 电感和电容的阻抗具有抵消补偿作用

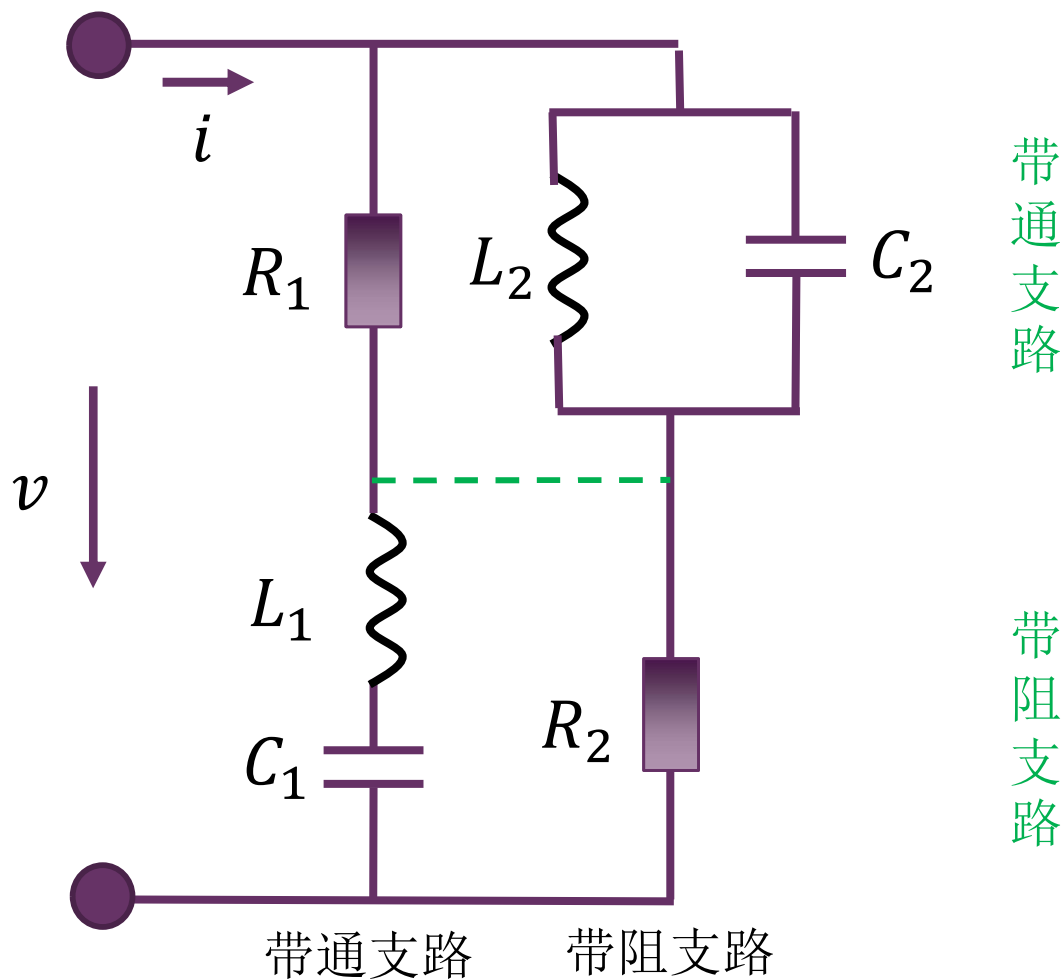
- 对于图示网络
 - (1) 说明通过选择适当的电感感值 L ，端口阻抗可以和频率 ω 无关
 - (2) L 取该值时，端口阻抗为多少 Ω ?



$$\begin{aligned}
 Y &= \frac{1}{R_1 + \frac{1}{j\omega C_2}} + \frac{1}{R_3 + j\omega L_4} \\
 &= \frac{1}{1 + \frac{1}{j\omega}} + \frac{1}{1 + j\omega L_4} = \frac{j\omega}{1 + j\omega} + \frac{1}{1 + j\omega L_4} \\
 &= \frac{j\omega(1 + j\omega L_4) + (1 + j\omega)}{(1 + j\omega)(1 + j\omega L_4)} \\
 &= \frac{1 - L_4\omega^2 + j2\omega}{1 - L_4\omega^2 + j(L_4 + 1)\omega} \\
 &= G = \frac{1 - L_4\omega^2}{1 - L_4\omega^2} = 1 \\
 &= \frac{j2\omega}{j(L_4 + 1)\omega} = \frac{2}{L_4 + 1}
 \end{aligned}$$

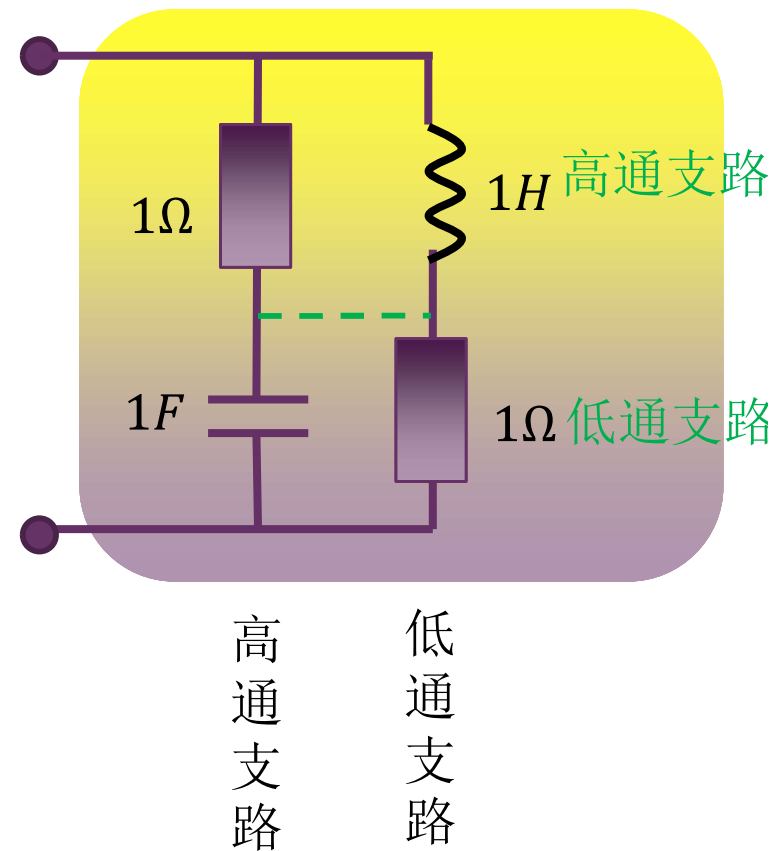
$$L_4 = 1H, \quad G = 1S, \quad R = 1\Omega$$

作业7.1 带通带阻互补为直通



带通支路

带阻支路



高通支路

低通支路

低通+高通=直通

$$R \frac{\omega_0}{s + \omega_0} + R \frac{s}{s + \omega_0} = R$$

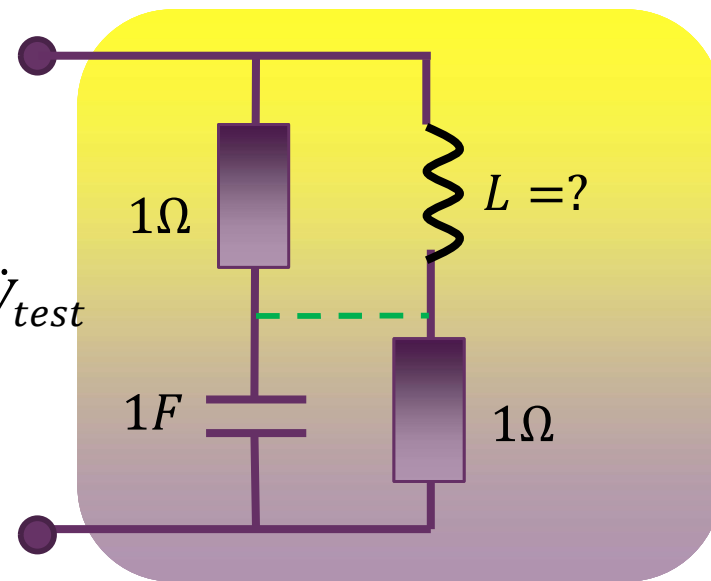
一阶低通和一阶高通互补

$$R \frac{2\xi\omega_0 s}{s^2 + 2\xi\omega_0 s + \omega_0^2} + R \frac{s^2 + \omega_0^2}{s^2 + 2\xi\omega_0 s + \omega_0^2} = R$$

二阶带通和二阶带阻互补

- 求电路中6个元件值满足什么关系时，总端口看入阻抗为纯阻？

能够抵偿时恰好为平衡电桥



$$\begin{aligned} \dot{I}_{test} &= Y\dot{V}_{test} = (Y_{12} + Y_{34})\dot{V}_{test} = \left(\frac{1}{R_1 + \frac{1}{sC_2}} + \frac{1}{R_4 + sL_3} \right) \dot{V}_{test} \\ &= \left(\frac{1}{R_1} \frac{sR_1C_2}{1 + sR_1C_2} + \frac{1}{R_4} \frac{1}{1 + sG_4L_3} \right) \dot{V}_{test} \end{aligned}$$

$$\stackrel{R_1=R_2=R}{\cong} \left(\frac{1}{R} \frac{s\tau_{12}}{1 + s\tau_{12}} + \frac{1}{R} \frac{1}{1 + s\tau_{34}} \right) \dot{V}_{test}$$

12串支高通，34串支低通

$$\stackrel{\tau_{12}=\tau_{34}}{\cong} \frac{1}{R} \left(\frac{s\tau + 1}{1 + s\tau} \right) \dot{V}_{test} = \frac{1}{R} \dot{V}_{test}$$

整体效果是直通

$$R_1 = R_2 = R$$

$$R_1C_2 = \tau_{12} = \tau_{34} = G_4L_3$$

$$L_3 = R_4R_1C_2 = 1\Omega \times 1\Omega \times 1C = 1H$$

$$Z_1Z_4 = R_1R_4 = 1 \times 1 = 1$$

$$Z_2Z_3 = \frac{1}{sC_2} sL_3 = \frac{L_3}{C_2} = \frac{1}{1} = 1$$

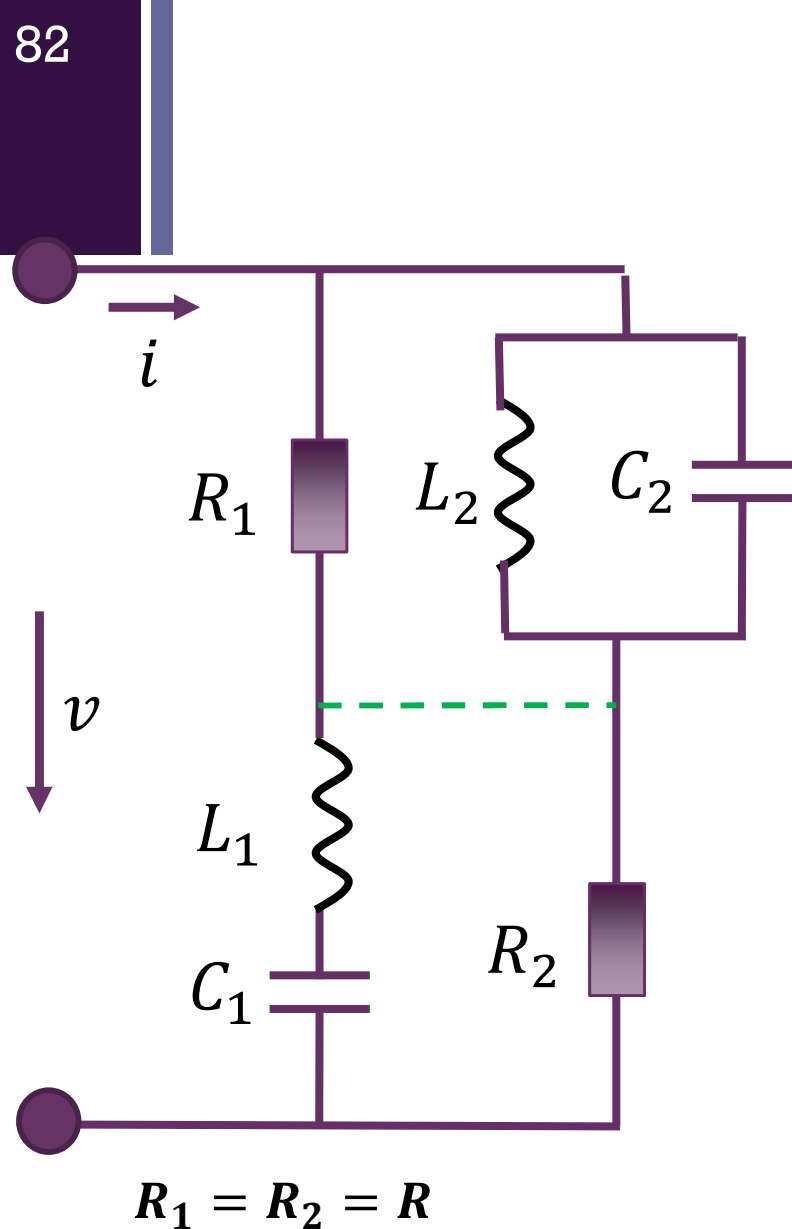
平衡电桥桥中开路、短路不会影响总端口阻抗

令桥中短路，端口加流测试

$$\dot{V}_{test} = Z\dot{I}_{test} = (Z_{13} + Z_{24})\dot{I}_{test} = \dots$$

$$= \left(R \frac{s\tau_{13}}{1 + s\tau_{13}} + R \frac{1}{1 + s\tau_{24}} \right) \dot{I}_{test} = \dots = R\dot{I}_{test}$$

13并支高通，24并路低通，整体效果是直通



$$L_1 = Q \frac{R}{\omega_0} \quad L_2 = \frac{1}{Q} \frac{R}{\omega_0}$$

$$C_1 = \frac{1}{Q} \frac{1}{\omega_0 R} \quad C_2 = Q \frac{1}{\omega_0 R}$$

$$\dot{I}_{test} = Y \dot{V}_{test} = (Y_{12} + Y_{34}) \dot{V}_{test}$$

$$= \left(\frac{1}{R_1 + j\omega L_1 + \frac{1}{j\omega C_1}} + \frac{1}{R_2 + \frac{1}{j\omega C_2 + \frac{1}{j\omega L_2}}} \right) \dot{V}_{test}$$

$$= \left(\frac{1}{R_1} \frac{1}{1 + \frac{1}{R_1} \left(j\omega L_1 + \frac{1}{j\omega C_1} \right)} + \frac{1}{R_2} \frac{R_2 \left(j\omega C_2 + \frac{1}{j\omega L_2} \right)}{1 + R_2 \left(j\omega C_2 + \frac{1}{j\omega L_2} \right)} \right) \dot{V}_{test}$$

$$\stackrel{R_1=R_2=R}{=} \left(\frac{1}{R} \frac{1}{1 + jQ_1 \left(\frac{\omega}{\omega_{01}} - \frac{\omega_{01}}{\omega} \right)} + \frac{1}{R} \frac{jQ_2 \left(\frac{\omega}{\omega_{02}} - \frac{\omega_{02}}{\omega} \right)}{1 + jQ_2 \left(\frac{\omega}{\omega_{02}} - \frac{\omega_{02}}{\omega} \right)} \right) \dot{V}_{test}$$

12串支带通, 34串支带阻

$$\stackrel{Q_1=Q_2=Q}{\omega_{01}=\omega_{02}=\omega_0}{=} \frac{1}{R} \dot{V}_{test}$$

整体效果是直通

$$R_1 = R_2 = R \quad \frac{1}{R_1} \sqrt{\frac{L_1}{C_1}} = Q_1 = Q = Q_2 = R_2 \sqrt{\frac{C_2}{L_2}}$$

$$\frac{1}{\sqrt{L_1 C_1}} = \omega_{01} = \omega_0 = \omega_{02} = \frac{1}{\sqrt{L_2 C_2}}$$

平衡电桥, 桥中短路:

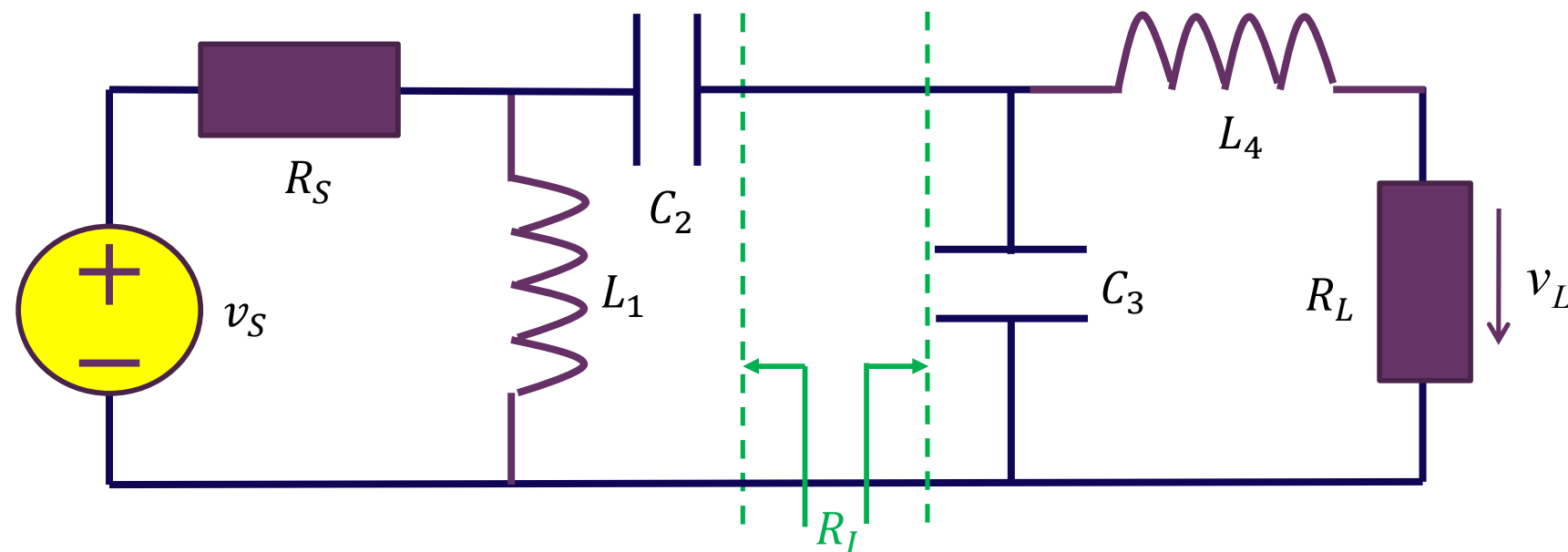
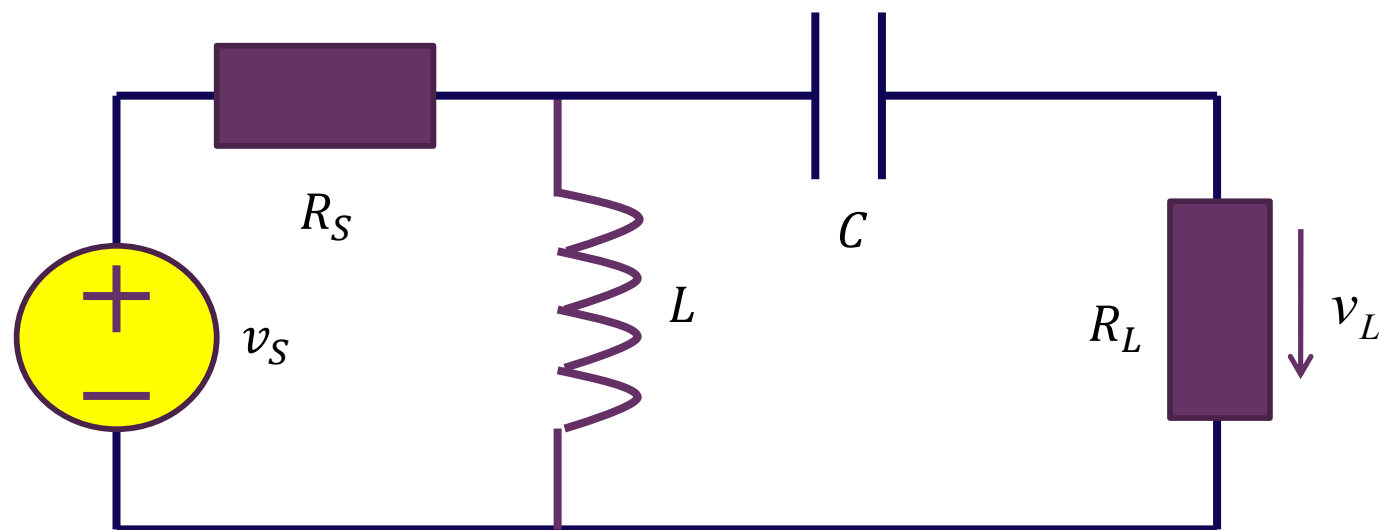
13并支带通, 24并支带阻, 整体效果是直通

第08讲 匹配网络 作业

作业1 L型匹配网络

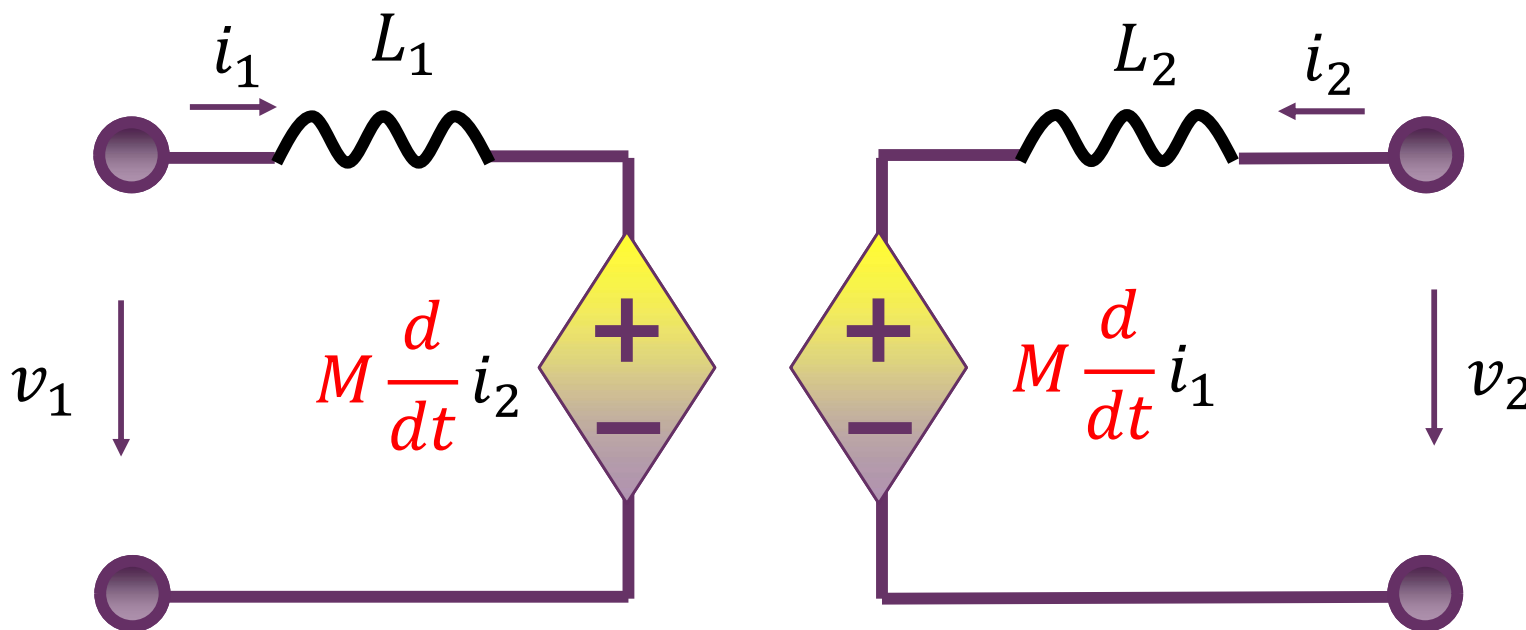
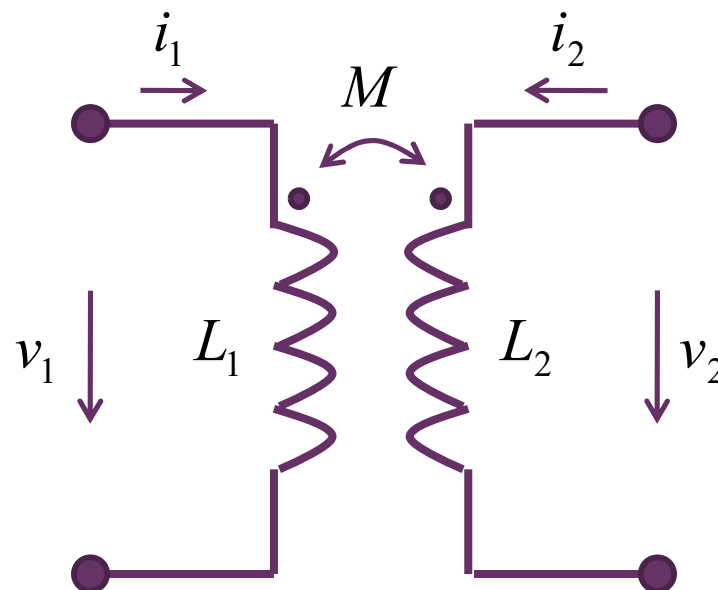
■ 如果希望在10MHz频点上实现实现1300Ω和50Ω阻抗之间的最大功率传输匹配，即希望负载获得信源的额定输出功率，高通型的L型匹配网络中的电感L、电容C如何取值？

■ 如果先把 $R_S=1300\Omega$ 通过高通L型匹配网络变换为 $R_I=255\Omega$ ，再通过低通L型匹配网络将 $R_I=255\Omega$ 变换为 $R_L=50\Omega$ ，请给出匹配网络中的电容、电感取值结果。



作业2 互感变压器h参量模型

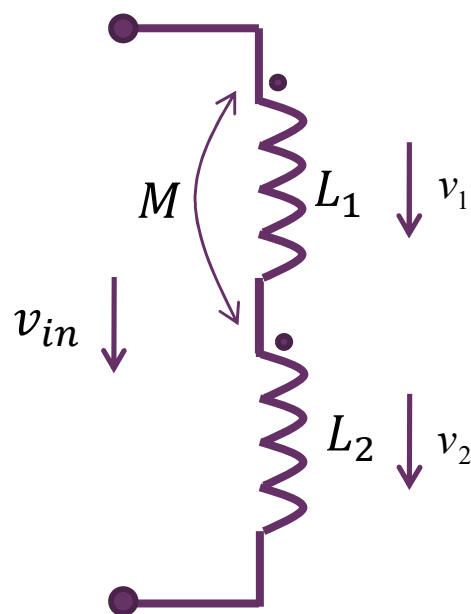
- 互感变压器的T型等效电路本质上是Z参量电路模型，请给出互感变压器的h参量电路模型
- 提示：理想变压器有h参量等效电路，把理想变压器抽象条件代入，互感变压器h参量电路模型应退化为理想变压器电路模型



没有初始电流的等效电路

作业3：有互感的电感分压系数

- 对于图示具有互感的串联电感，求其分压系数



作业4

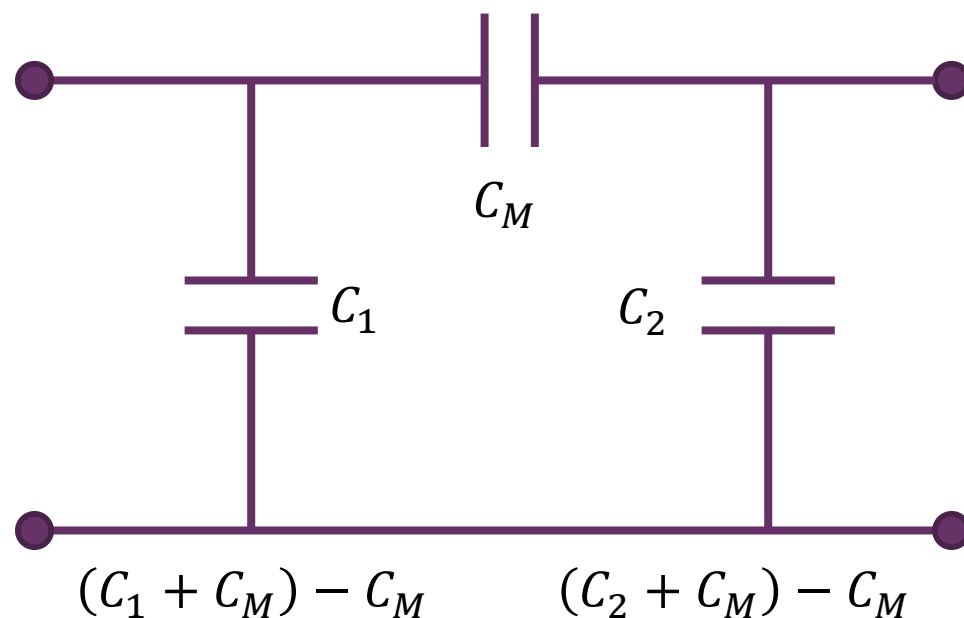
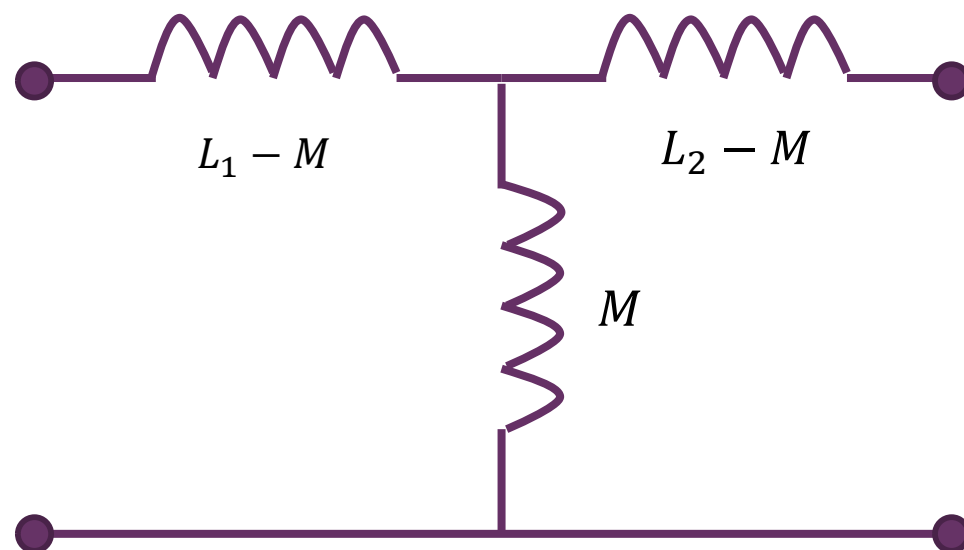
二端口电容

- T型纯电感网络是二端口电感，其元件约束方程为

$$\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} L_1 & M \\ M & L_2 \end{bmatrix} \frac{d}{dt} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix}$$

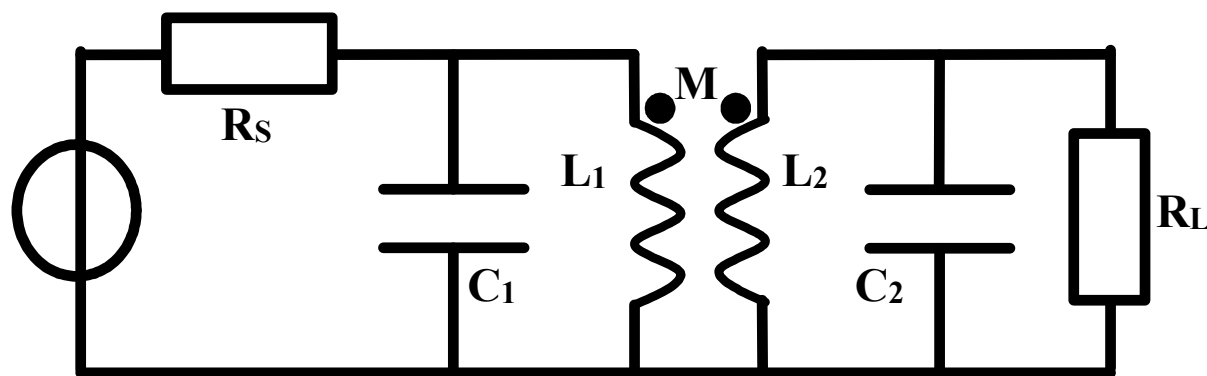
$$v = L \frac{di}{dt}$$

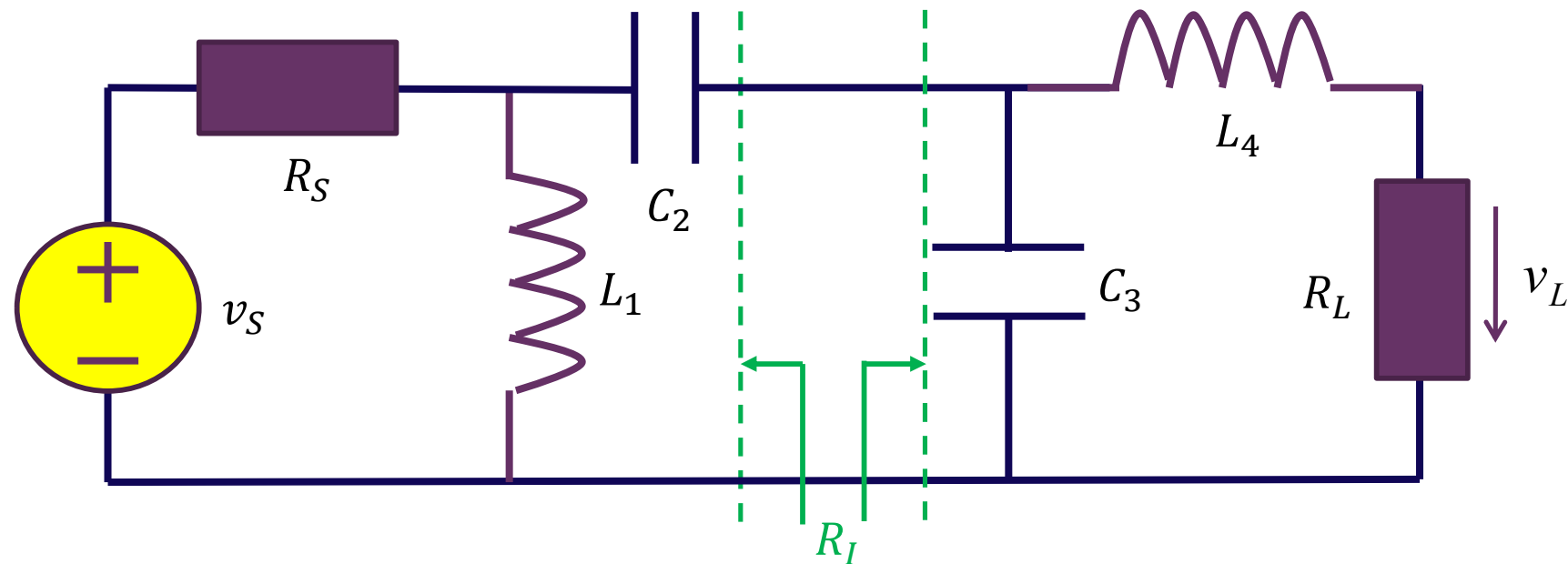
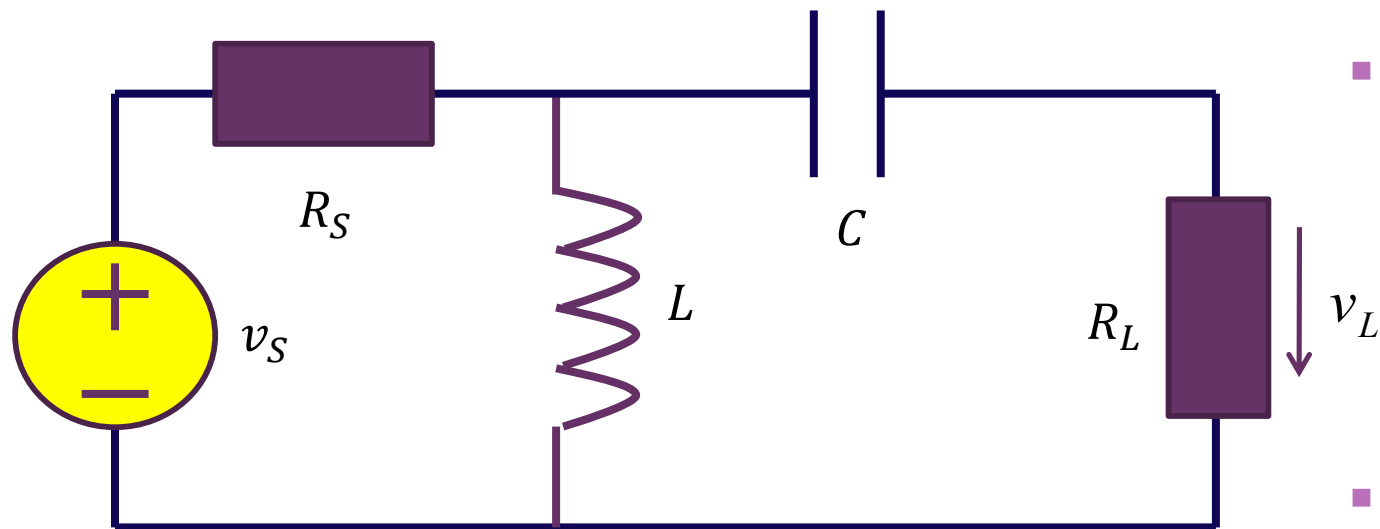
- 其对偶电路为Π型纯电容网络，它是二端口电容
 - 给出其元件约束方程及对应的y参量电路模型
 - 给出其h参量电路模型
 - 考察什么情况下有理想变压功能？



作业5 设计使得三次谐波不得通过（选作）

- 设计流程和设计案例见教材P823-826页
- 请设计一个互感耦合双并联谐振电路，要求在 $f_0=900\text{MHz}$ 上获具有平坦特性的得60MHz的匹配带宽，其中信源内阻 50Ω ，负载电阻 $1\text{k}\Omega$
- 通带设计完成后，突然发现系统还有对 $3f_0$ 进行完全抑制的需求才能确保系统性能，请在已设计好的图示电路上进行修正以实现 $3f_0$ 的抑制，并仿真确认你的修正设计，在实现对 $3f_0$ 抑制的同时，对通带带宽及通带内平坦特性的影响应该很小，设计才被认为是成功的，并确认通带特性（通带最大衰减）和 $[2610, 2790]\text{MHz}$ 频段的阻带特性（ $3f_0$ 阻带最小衰减）。





- 作业1的设定数值，对设计进行验证
 - 频域验证10MHz频点上负载获得信源额定功率，仿真确认两个网络的1dB匹配带宽，说明匹配带通提高了。
 - 研究：如果期望匹配带宽是原先的一半，如何操作？
- 时域验证负载获得信源额定功率
 - 假设电容、电感初值为0，观察瞬态过程
 - 确认稳态后，负载获得信源的额定功率（频域分析属正弦稳态分析）
 - 稳态后，电容电感平均功率为0，瞬态过程是电容、电感吸收能量储存能量的过程
- 频率取值偏离匹配频点，取1dB频点，重新做上述仿真，负载获得功率比信源额定功率低多少dB？

下节内容在教材中的章节 预习用

- P654-657: 习题8.2 器件的寄生效应
- P657-658: 习题8.3 空芯电感电路模型
- P473-523: 第六章 电路抽象
 - P488: 电阻; P491: 电容; P495: 电感; P513: 寄生效应
- P725-731: 数字非门延时/动态功耗分析